2_EDA

2025-06-11

Data Cleaning and Exploratory Data Analysis

This R markdown document will be used to perform Exploratory Data Analysis (EDA) on the merged_df_raw that was developed in the previous workbook `1_data_extraction.Rmd`.

Data Importing

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readr)
library(ggplot2)
library(grid)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(here)
```

```
## here() starts at /home/rstudio/Documents/ads503_project_g1
```

```r
library(naniar)
library(VIM)
```

```
## Loading required package: colorspace
## VIM is ready to use.
##
```

```
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```r
library(skimr)
```

```
##
## Attaching package: 'skimr'
##
## The following object is masked from 'package:naniar':
##
##     n_complete
```

```r
library(stringr)
library(rcompanion)
```

```r
merged_df <- read_csv(here('Data/Raw', 'merged_df_raw.csv'))
```

```
## Rows: 899 Columns: 77
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (2): name, location
## dbl (75): id, ccf, age, sex, painloc, painexer, relrest, pncaden, cp, trestb...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(merged_df)
```

```
## # A tibble: 6 x 77
##      id   ccf   age   sex painloc painexer relrest pncaden    cp trestbps   htn
##   <dbl> <dbl> <dbl> <dbl>   <dbl>    <dbl>   <dbl>   <dbl> <dbl>    <dbl> <dbl>
## 1     1     0    63     1      -9       -9      -9      -9     1      145     1
## 2     2     0    67     1      -9       -9      -9      -9     4      160     1
## 3     3     0    67     1      -9       -9      -9      -9     4      120     1
## 4     4     0    37     1      -9       -9      -9      -9     3      130     0
## 5     6     0    41     0      -9       -9      -9      -9     2      130     1
## 6     7     0    56     1      -9       -9      -9      -9     2      120     1
## # i 66 more variables: chol <dbl>, smoke <dbl>, cigs <dbl>, years <dbl>,
## #   fbs <dbl>, dm <dbl>, famhist <dbl>, restecg <dbl>, ekgmo <dbl>,
## #   ekgday <dbl>, ekgyr <dbl>, dig <dbl>, prop <dbl>, nitr <dbl>, pro <dbl>,
## #   diuretic <dbl>, proto <dbl>, thaldur <dbl>, thaltime <dbl>, met <dbl>,
## #   thalach <dbl>, thalrest <dbl>, tpeakbps <dbl>, tpeakbpd <dbl>, dummy <dbl>,
## #   trestbpd <dbl>, exang <dbl>, xhypo <dbl>, oldpeak <dbl>, slope <dbl>,
## #   rldv5 <dbl>, rldv5e <dbl>, ca <dbl>, restckm <dbl>, exerckm <dbl>, ...
```

Categorical Conversions

```r
## categorical features and their levels and labels
#sex
merged_df$sex <- factor(merged_df$sex,
                        levels = c(0, 1),
```

```r
                     labels = c("Female", "Male")
                     )
#painloc
#unique(merged_df$painloc)
merged_df$painloc <- factor(merged_df$painloc,
                         levels = c(0, 1),
                         labels = c("Otherwise", "Substernal")
                         )
#painexer
#unique(merged_df$painexer)
merged_df$painexer <- factor(merged_df$painexer,
                         levels = c(0, 1),
                         labels = c("Otherwise", "Provoked by Exertion")
                         )
#relrest
#unique(merged_df$relrest)
merged_df$relrest <- factor(merged_df$relrest,
                         levels = c(0, 1),
                         labels = c("Otherwise", "Relieved after rest")
                         )
#cp
#unique(merged_df$cp)
merged_df$cp <- factor(merged_df$cp,
                         levels = c(1, 2, 3, 4),
                         labels = c("Typical Angina", "Atypical Angina",
                                    "Non-anginal Pain", "Asymptomatic"),
                    ordered = TRUE)
#smoke
#unique(merged_df$smoke)
merged_df$smoke <- factor(merged_df$smoke,
                         levels = c(0, 1),
                         labels = c("No", "Yes")
                         )
#fbs
#unique(merged_df$fbs)
merged_df$fbs <- factor(merged_df$fbs,
                         levels = c(0, 1),
                         labels = c("No", "Yes")
                         )
#dm
#unique(merged_df$dm)
merged_df$dm <- factor(merged_df$dm,
                         levels = c(0, 1),
                         labels = c("No", "Yes")
                         )
#famhist
#unique(merged_df$famhist)
merged_df$famhist <- factor(merged_df$famhist,
                         levels = c(0, 1),
                         labels = c("No", "Yes")
                         )
#restecg
#unique(merged_df$restecg)
```

```r
merged_df$restecg <- factor(merged_df$restecg,
                        levels = c(0, 1, 2),
                        labels = c("Normal", "Abnormal", "Hypertrophy"),
                   ordered = TRUE)
#dig
#unique(merged_df$dig)
merged_df$dig <- factor(merged_df$dig,
                        levels = c(0, 1),
                        labels = c("No", "Yes")
                        )
#prop
#unique(merged_df$prop)
# find the number of 22's a feature outside the range/
#sum(merged_df$prop == 22, na.rm = TRUE)
# convert the 22 to NA
merged_df <- merged_df %>%
  mutate(prop = ifelse(prop == 22, NA, prop))
#create factor
merged_df$prop <- factor(merged_df$prop,
                        levels = c(0, 1),
                        labels = c("No", "Yes")
                        )
#nitr
#unique(merged_df$nitr)
merged_df$nitr <- factor(merged_df$nitr,
                     levels = c(0, 1),
                     labels = c("No", "Yes")
                     )
#pro
#unique(merged_df$pro)
merged_df$pro <- factor(merged_df$pro,
                     levels = c(0, 1),
                     labels = c("No", "Yes")
                     )
#diuretic
#unique(merged_df$diuretic)
merged_df$diuretic <- factor(merged_df$diuretic,
                     levels = c(0, 1),
                     labels = c("No", "Yes")
                     )
#proto
#unique(merged_df$proto)
merged_df$proto <- factor(merged_df$proto,
                     levels = c(0, 1),
                     labels = c("No", "Yes")
                     )
#exang
#unique(merged_df$exang)
merged_df$exang <- factor(merged_df$exang,
                     levels = c(0, 1),
                     labels = c("No", "Yes")
                     )
#xhypo
```

```r
#unique(merged_df$xhypo)
merged_df$xhypo <- factor(merged_df$xhypo,
                    levels = c(0, 1),
                    labels = c("No", "Yes")
                    )
#slope
#unique(merged_df$slope)
# check count of 0 valuyes
#sum(merged_df$slope == 0, na.rm = TRUE)
# replace 0 with NA
merged_df <- merged_df %>%
  mutate(slope = ifelse(slope == 0, NA, slope))
# factor
merged_df$slope <- factor(merged_df$slope,
                    levels = c(1, 2, 3),
                    labels = c("Upsloping", "Flat", "Downsloping"),
                    ordered = TRUE
                    )
#restwm
#unique(merged_df$restwm)
# check count of 0 valuyes
# factor
merged_df$restwm <- factor(merged_df$restwm,
                    levels = c(0, 1, 2, 3),
                    labels = c("None", "Mild/Moderate", "Moderate/Severe",
                               "Akinesis"),
                    ordered = TRUE
                    )
#thal
#unique(merged_df$thal)
# examine 1, 2, 5 values which are not in the documentation
# check count of 1,2,5 values
#sum(merged_df$thal == 1, na.rm = TRUE)
#sum(merged_df$thal == 2, na.rm = TRUE)
#sum(merged_df$thal == 5, na.rm = TRUE)
# replace 1, 2, 5 with na
merged_df <- merged_df %>%
  mutate(thal = ifelse(thal == 1, NA, thal)) %>%
  mutate(thal = ifelse(thal == 2, NA, thal)) %>%
  mutate(thal = ifelse(thal == 4, NA, thal)) %>%
  mutate(thal = ifelse(thal == 5, NA, thal))
```

Data Cleaning

This below section of code starts by removing columns that are unused as indicated by the hear-disease.names text file which serves as a data dictionary and source for information on all the data sets. Once the explicitly unnecessary columns are removed, we convert the -9's and -9.0's into NA values and set a threshold for feature removal of if the feature is missing more than 60% of the observations it is immediately dropped. Once these features have been removed we focus on removing near zero variance features.

```r
# remove unnecessary columns as indicated in heart-disease.names
# cols marked as not used or dummy variables
not_used_dummy_var <- c(
  "id", "ccf", "dummy", "restckm", "exerckm", "thalsev", "thalpul",
```

```r
  "earlobe", "lvx1", "lvx2", "lvx3", "lvx4", "lvf", "cathef", "junk", "name"
)

df_cleaning <- merged_df %>%
  select(-all_of(not_used_dummy_var))

# convert -9 to NA
df_cleaning <- df_cleaning %>%
  mutate(across(where(is.character), ~ na_if(.x, "-9"))) %>%   # character columns
  mutate(across(where(is.numeric), ~ na_if(.x, -9)))           # numeric columns
#Get NA count
sum(is.na(df_cleaning))
```

```
## [1] 15881
```

```r
#get dimensions of the data
df_cleaning %>% dim
```

```
## [1] 899  61
```

```r
#reexamine the head
df_cleaning %>% head()
```

```
## # A tibble: 6 x 61
##     age sex    painloc painexer relrest pncaden cp    trestbps   htn  chol smoke
##   <dbl> <fct>  <fct>   <fct>    <fct>     <dbl> <ord>    <dbl> <dbl> <dbl> <fct>
## 1    63 Male   <NA>    <NA>     <NA>         NA Typi~      145     1   233 <NA>
## 2    67 Male   <NA>    <NA>     <NA>         NA Asym~      160     1   286 <NA>
## 3    67 Male   <NA>    <NA>     <NA>         NA Asym~      120     1   229 <NA>
## 4    37 Male   <NA>    <NA>     <NA>         NA Non-~      130     0   250 <NA>
## 5    41 Female <NA>    <NA>     <NA>         NA Atyp~      130     1   204 <NA>
## 6    56 Male   <NA>    <NA>     <NA>         NA Atyp~      120     1   236 <NA>
## # i 50 more variables: cigs <dbl>, years <dbl>, fbs <fct>, dm <fct>,
## #   famhist <fct>, restecg <ord>, ekgmo <dbl>, ekgday <dbl>, ekgyr <dbl>,
## #   dig <fct>, prop <fct>, nitr <fct>, pro <fct>, diuretic <fct>, proto <fct>,
## #   thaldur <dbl>, thaltime <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## #   tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, xhypo <fct>,
## #   oldpeak <dbl>, slope <ord>, rldv5 <dbl>, rldv5e <dbl>, ca <dbl>,
## #   restef <dbl>, restwm <ord>, exeref <dbl>, exerwm <dbl>, thal <dbl>, ...
```

```r
# calculate percentage missing
missing_threshold <- 60
missing_pct <- df_cleaning %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(everything(), names_to = "column", values_to = "missing_percent") %>%
  arrange(desc(missing_percent))

# filter by col missing
missing_cols <- missing_pct %>%
  filter(missing_percent > missing_threshold)
var_missing_gt_60 <- missing_cols$column

missing_cols
```

```
## # A tibble: 12 x 2
##    column  missing_percent
```

```
##    <chr>              <dbl>
##  1 pncaden            100
##  2 exeref              99.8
##  3 exerwm              99.4
##  4 restef              96.9
##  5 restwm              96.7
##  6 dm                  89.4
##  7 smoke               74.4
##  8 ca                  67.6
##  9 proto               64.0
## 10 om2                 63.6
## 11 ramus               63.1
## 12 diag                62.1
```

```r
# drop cols missing greater than threshold
df_cleaning <- df_cleaning %>%
  select(-all_of(var_missing_gt_60))
dim(df_cleaning)
```

```
## [1] 899  49
```

```r
# check near zero variance vars
nzv_cols <- nearZeroVar(df_cleaning %>%
    select(-num), saveMetrics = TRUE)
nzv_cols %>% filter(nzv == TRUE)
```

```
##       freqRatio percentUnique zeroVar  nzv
## dig    27.65517     0.2224694   FALSE TRUE
## xhypo  37.22727     0.2224694   FALSE TRUE
```

```r
# get column names of nzv
nzv_names <- nzv_cols %>%
  filter(nzv ==TRUE) %>%
  rownames()
nzv_names
```

```
## [1] "dig"   "xhypo"
```

```r
# remove nzv variables
df_cleaning <- df_cleaning %>%
  select(-all_of(nzv_names))

# add back ca column for EDA
df_cleaning$ca <- merged_df$ca
df_cleaning$ca <- na_if(df_cleaning$ca, -9)

dim(df_cleaning)
```

```
## [1] 899  48
```

In the process of removing variables with >60% missing data, the `ca` variable, which represents *Major vessels via fluoroscopy* was removed due to having 67.6% missing data. This variable was part of the *cleveland-14* set. It has re-added to the final data set after *nzv* variables were removed.

Export cleaned dataset

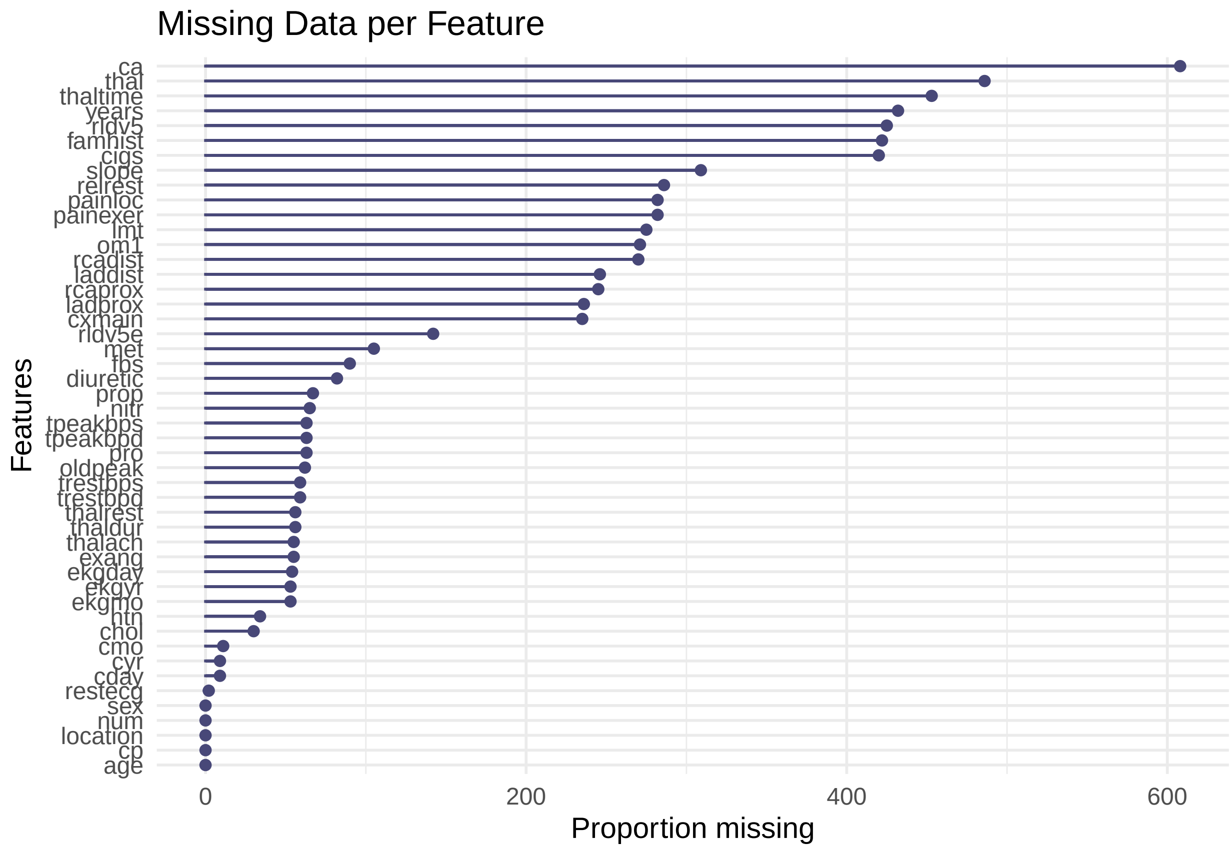
```r
# export to file
write.csv(df_cleaning, here("Data/Clean", "heart_disease_cleaned.csv"), row.names = FALSE)
saveRDS(df_cleaning, here("Data/Clean", "heart_disease_cleaned.rds"))
```

Exploratory Data Analysis

```r
df <- readRDS(here("Data/Clean", "heart_disease_cleaned.rds"))
# head(df)

missing_var <- df %>%
  summarise(across(everything(), ~ mean(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "missing_pct")

missing_var %>%
  filter(missing_pct > 0) %>%
  arrange(desc(missing_pct))
```

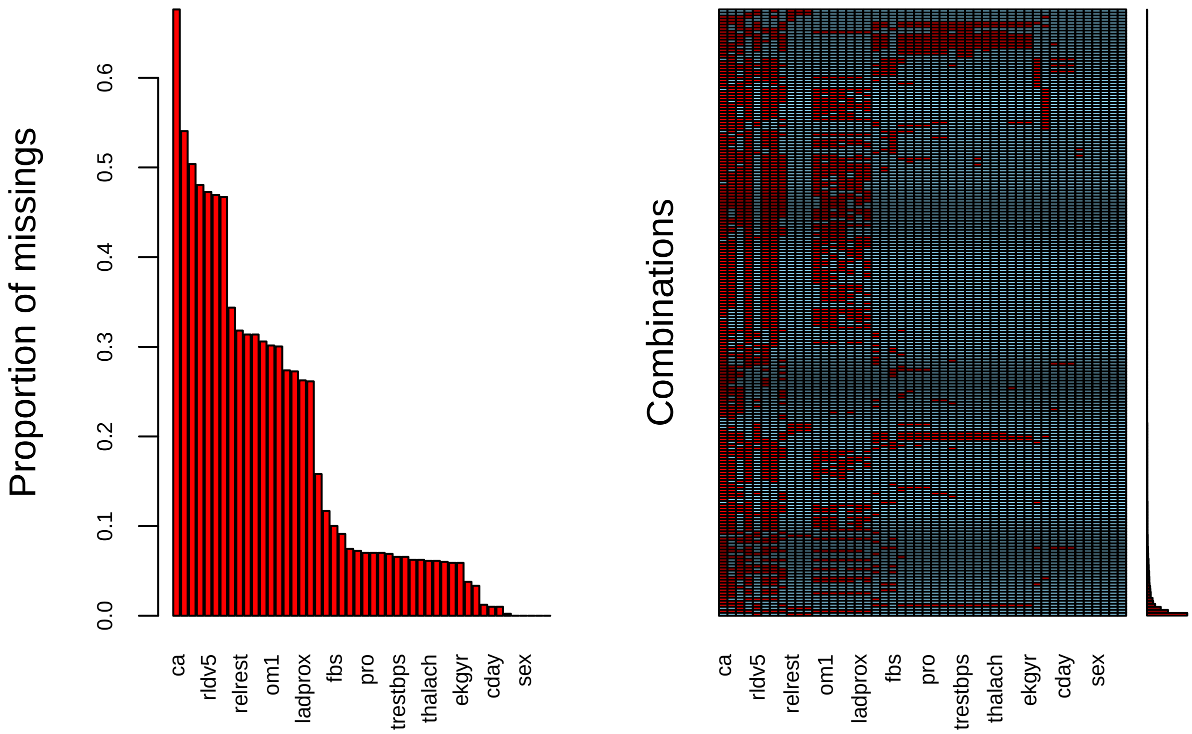
```
## # A tibble: 43 x 2
##    variable missing_pct
##    <chr>          <dbl>
##  1 ca             0.676
##  2 thal           0.541
##  3 thaltime       0.504
##  4 years          0.481
##  5 rldv5          0.473
##  6 famhist        0.469
##  7 cigs           0.467
##  8 slope          0.344
##  9 relrest        0.318
## 10 painloc        0.314
## # i 33 more rows
```

```r
# bar chart of missingness per feature
gg_miss_var(df) +
  labs(
    title = "Missing Data per Feature",
    x = "Features",
    y = "Proportion missing"
  )
```




```r
# missingness matrix
aggr(df, numbers = TRUE, sortVars = TRUE, cex.axis = 0.7,
     main = "Missingness Matrix",
     sub = "Red = missing, Blue = observed")
```

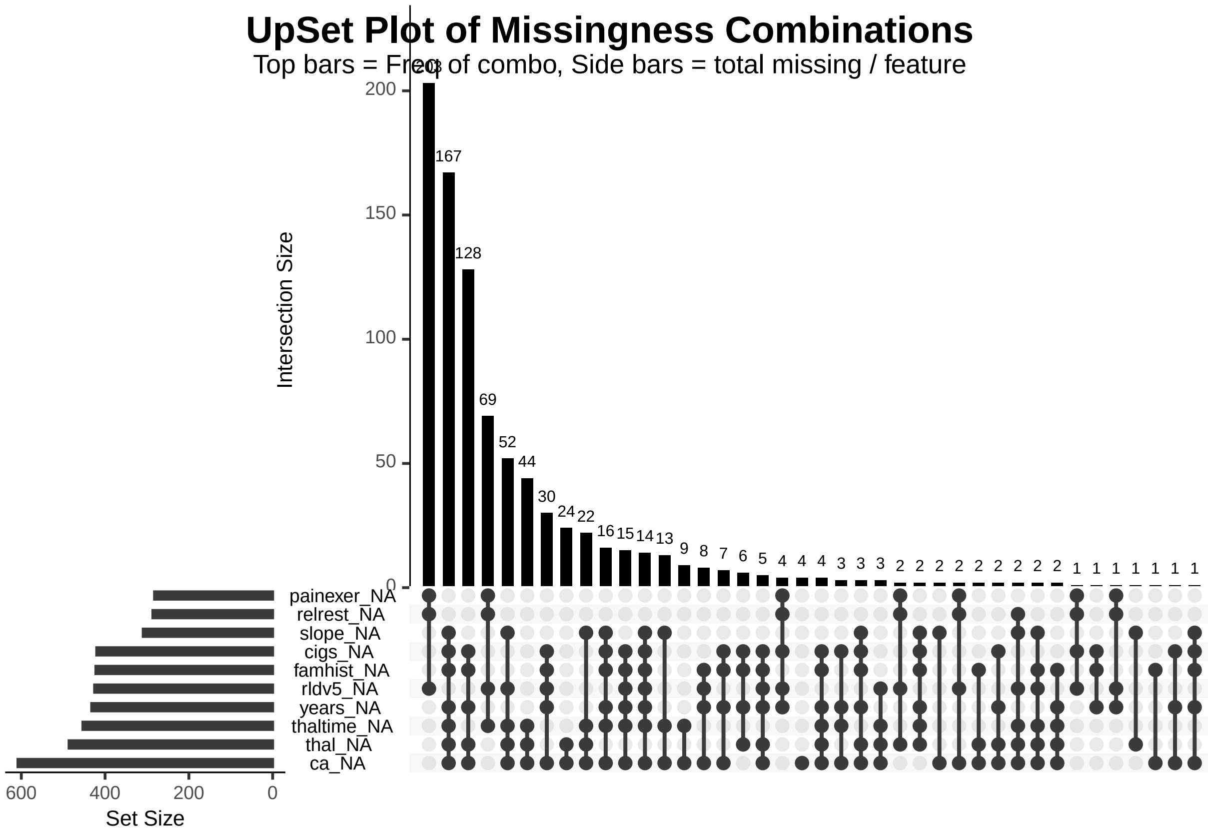
```
## Warning in plot.aggr(res, ...): not enough vertical space to display
## frequencies (too many combinations)
```




```
##
## Variables sorted by number of missings:
## Variable       Count
##        ca 0.676307008
##      thal 0.540600667
##  thaltime 0.503893215
##     years 0.480533927
##     rldv5 0.472747497
##   famhist 0.469410456
##      cigs 0.467185762
##     slope 0.343715239
##   relrest 0.318131257
##   painloc 0.313681869
##  painexer 0.313681869
##       lmt 0.305895439
##       om1 0.301446051
##   rcadist 0.300333704
##   laddist 0.273637375
##   rcaprox 0.272525028
##   ladprox 0.262513904
##    cxmain 0.261401557
##    rldv5e 0.157953281
##       met 0.116796440
##       fbs 0.100111235
##  diuretic 0.091212458
##      prop 0.074527253
```

```
##      nitr 0.072302558
##       pro 0.070077864
##  tpeakbps 0.070077864
##  tpeakbpd 0.070077864
##   oldpeak 0.068965517
##  trestbps 0.065628476
##  trestbpd 0.065628476
##   thaldur 0.062291435
##  thalrest 0.062291435
##   thalach 0.061179088
##     exang 0.061179088
##    ekgday 0.060066741
##     ekgmo 0.058954394
##     ekgyr 0.058954394
##       htn 0.037819800
##      chol 0.033370412
##       cmo 0.012235818
##      cday 0.010011123
##       cyr 0.010011123
##   restecg 0.002224694
##       age 0.000000000
##       sex 0.000000000
##        cp 0.000000000
##       num 0.000000000
##  location 0.000000000
```

```r
# missingness correlation / upset plot
gg_miss_upset(df, nsets = 10, main.bar.color = "black")
# Add a title manually
grid.text("UpSet Plot of Missingness Combinations", y = 0.95, gp = gpar(fontsize = 14, fontface = "bold
grid.text("Top bars = Freq of combo, Side bars = total missing / feature", y = 0.91, gp = gpar(fontsize
```



Variables Missing > 40%

- ca: cleveland-14 variable - represents # of major vessels (0 - 3) colored by fluoroscopy.
- thal: cleveland-14 variable - results from thallium stress test
- thaltime: time duration for thallium stress test
- years: number of years as a smoker
- rldv5: height at rest
- famhist: family history of coronary artery disease
- cigs: cigarettes smoked per day

Upset

The UpSet plot shows the intersections of variable sets, similar to a Venn Diagram. This plot shows the combinations of variables that have missing values and the total of thos combinations.

- based on upset table, `ca`, `thal`, and `thaltime` are missing together
- missingness might not be random

Custom Functions

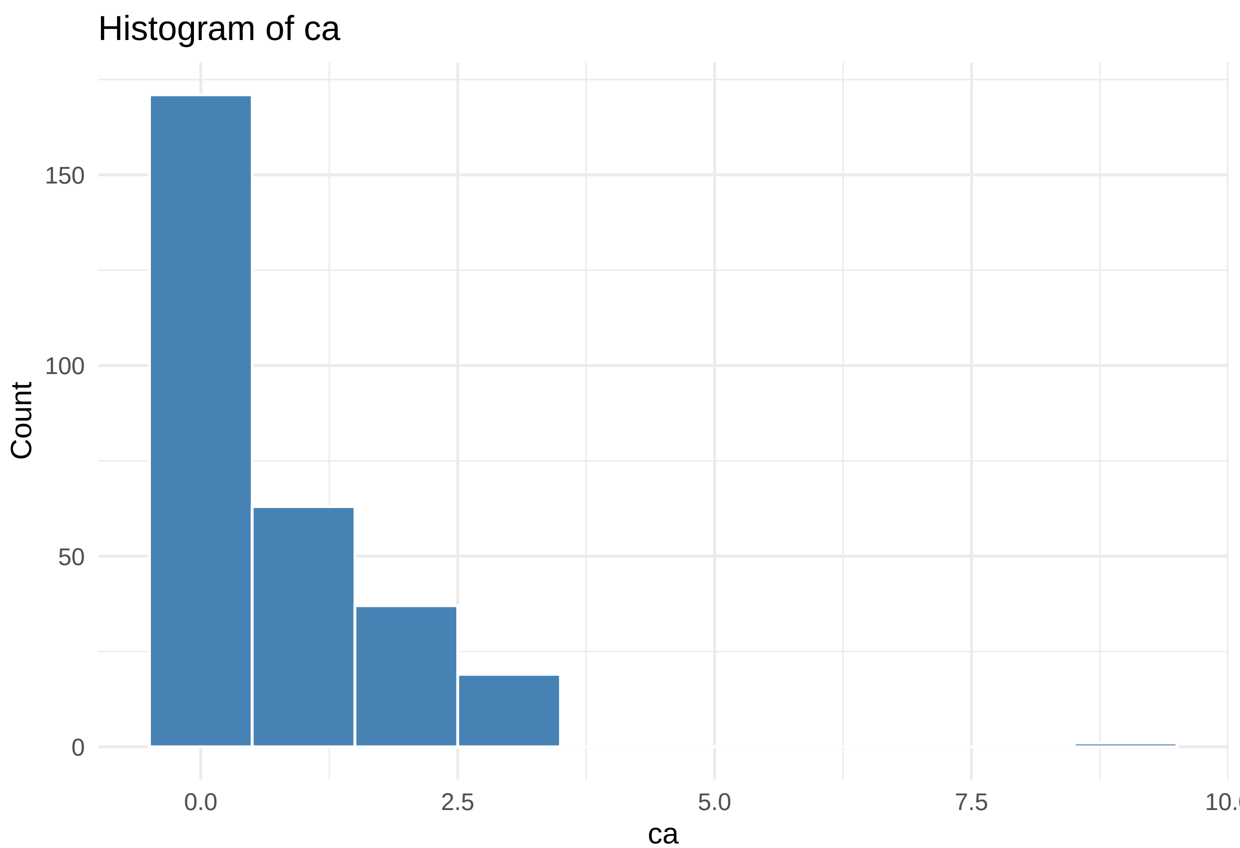
```
# function for showing histogram
show_histogram <- function(dataframe, feature) {
  ggplot(dataframe, aes(x = {{feature}})) +
    geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(
    title = paste("Histogram of", rlang::as_label(enquo(feature))),
```

```r
    x = rlang::as_label(enquo(feature)),
    y = "Count"
  ) +
  theme_minimal()
}
# function for showing desnsity plot
show_densityplot <- function(dataframe, feature) {
  ggplot(df, aes(x = {{ feature }})) +
    geom_density(fill = "darkorange", alpha = 0.5) +
    labs(
      title = paste("Density Plot of", rlang::as_label(enquo(feature))),
      x = rlang::as_label(enquo(feature)),
      y = "Count"
    ) +
    theme_minimal()
}

show_barplot <- function(dataframe, feature) {
  ggplot(dataframe, aes(x = {{feature}})) +
  geom_bar(fill = "steelblue") +
  labs(
    title = paste("Bar Plot of", rlang::as_label(enquo(feature))),
      x = rlang::as_label(enquo(feature)),
    y = "Count") +
  theme_minimal()
}
```

Feature: ca

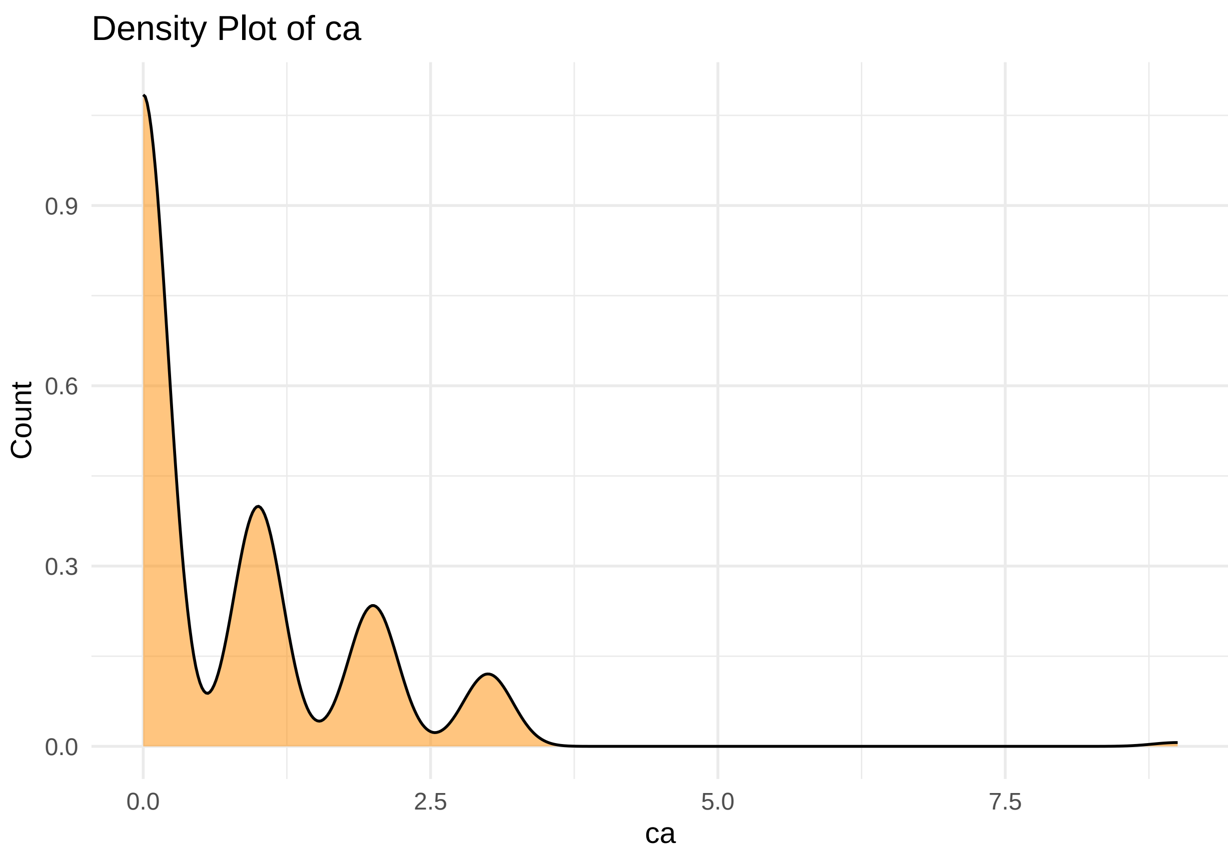
```r
show_histogram(df, ca)
```

```
## Warning: Removed 608 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
show_densityplot(df, ca)
```

```
## Warning: Removed 608 rows containing non-finite outside the scale range
## (`stat_density()`).
```



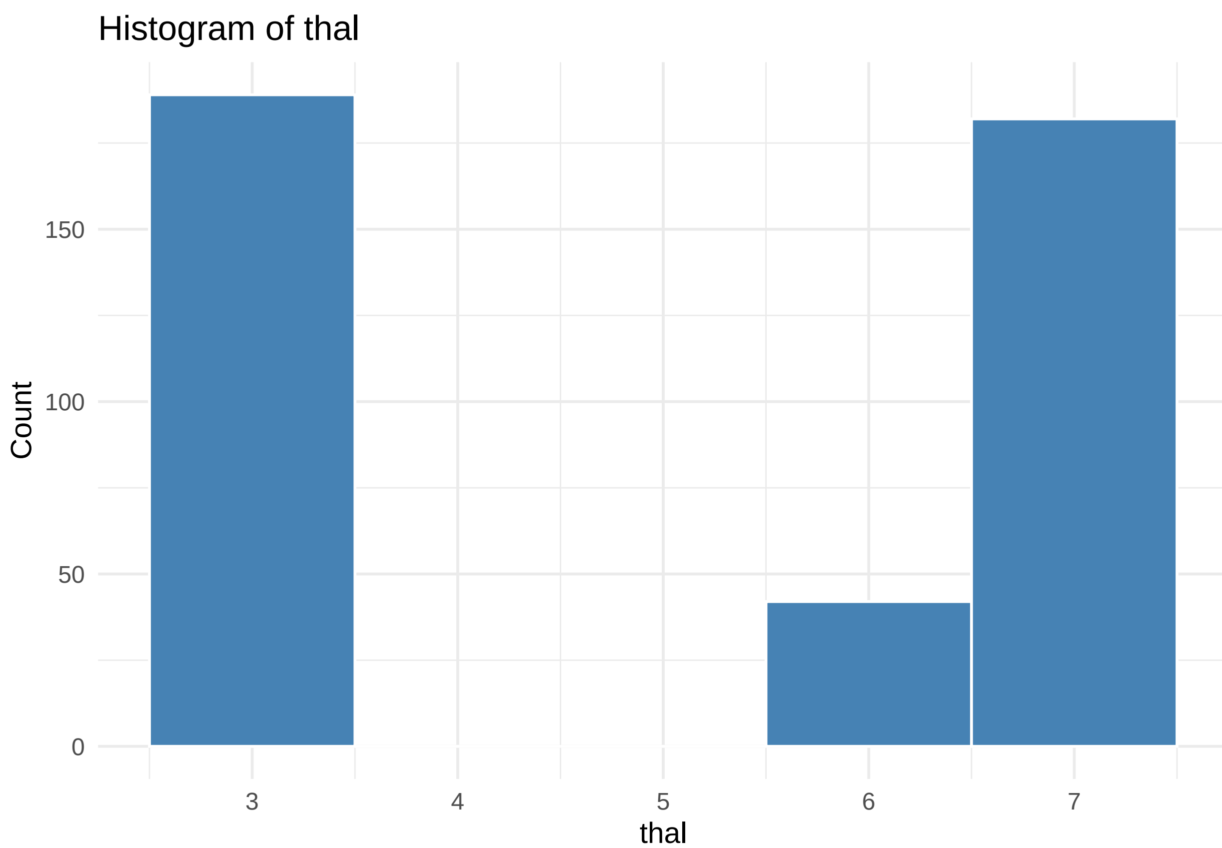

```r
# impute with median and missingness flag
df <- df %>%
  mutate(
    ca_missing = ifelse(is.na(ca), 1, 0),
    ca_imputed = ifelse(is.na(ca), median(ca, na.rm = TRUE), ca)
  )

# drop imputed column
df <- df %>%
  select(-ca)
```

Feature `ca` has a right skew therefore median was used for imputation.

Feature: `thal`

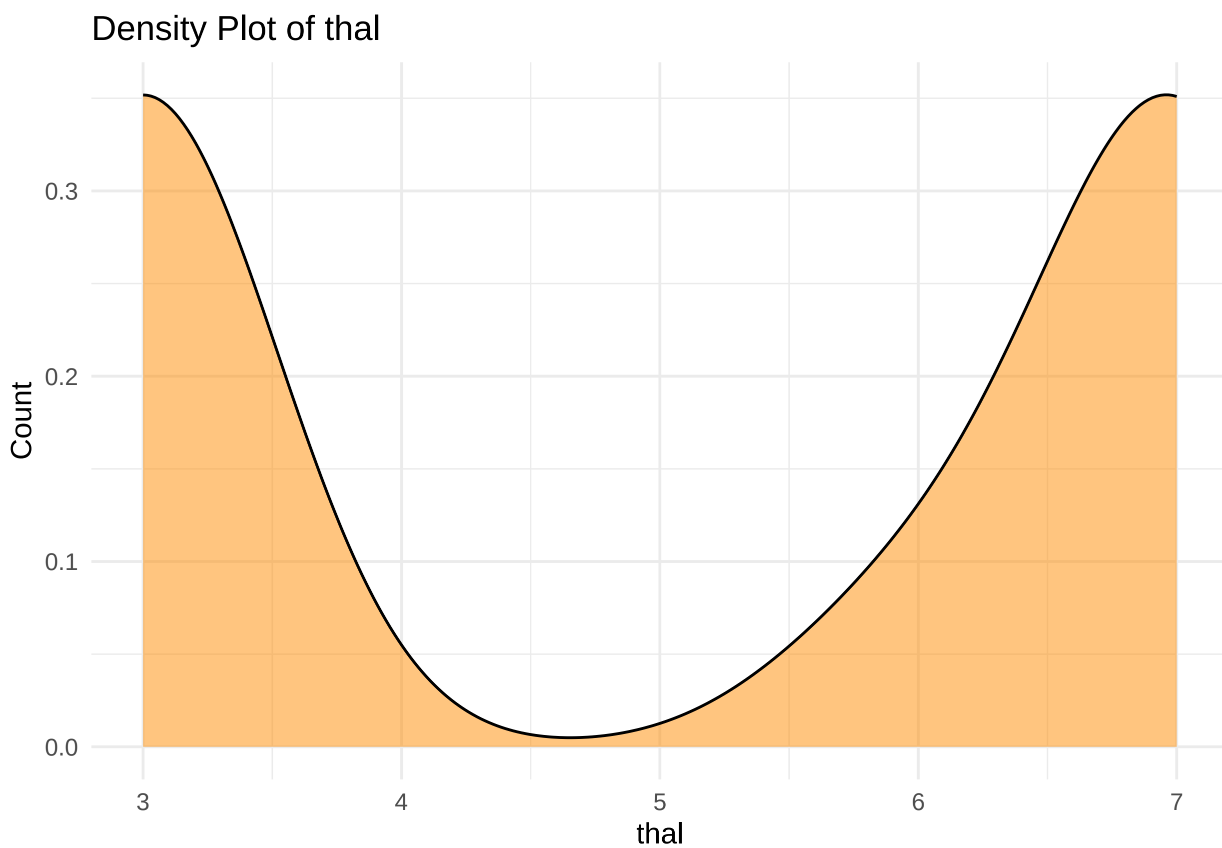
```r
show_histogram(df, thal)
```

```
## Warning: Removed 486 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```r
show_densityplot(df, thal)
```

```
## Warning: Removed 486 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```
table(df$thal)
```

```
##
##   3   6   7
## 189  42 182
```

```
# impute with mode and missingness flag
df <- df %>%
  mutate(
    thal_missing = ifelse(is.na(df$thal), 1, 0),
    thal_imputed = ifelse(is.na(df$thal), 3, df$thal) # mode imputation
  )

# drop imputed column
df <- df %>%
  select(-thal)
```

Feature: `thaltime`

```
summary(df$thaltime)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00    3.00    6.00    5.69    8.00   20.00     453
```

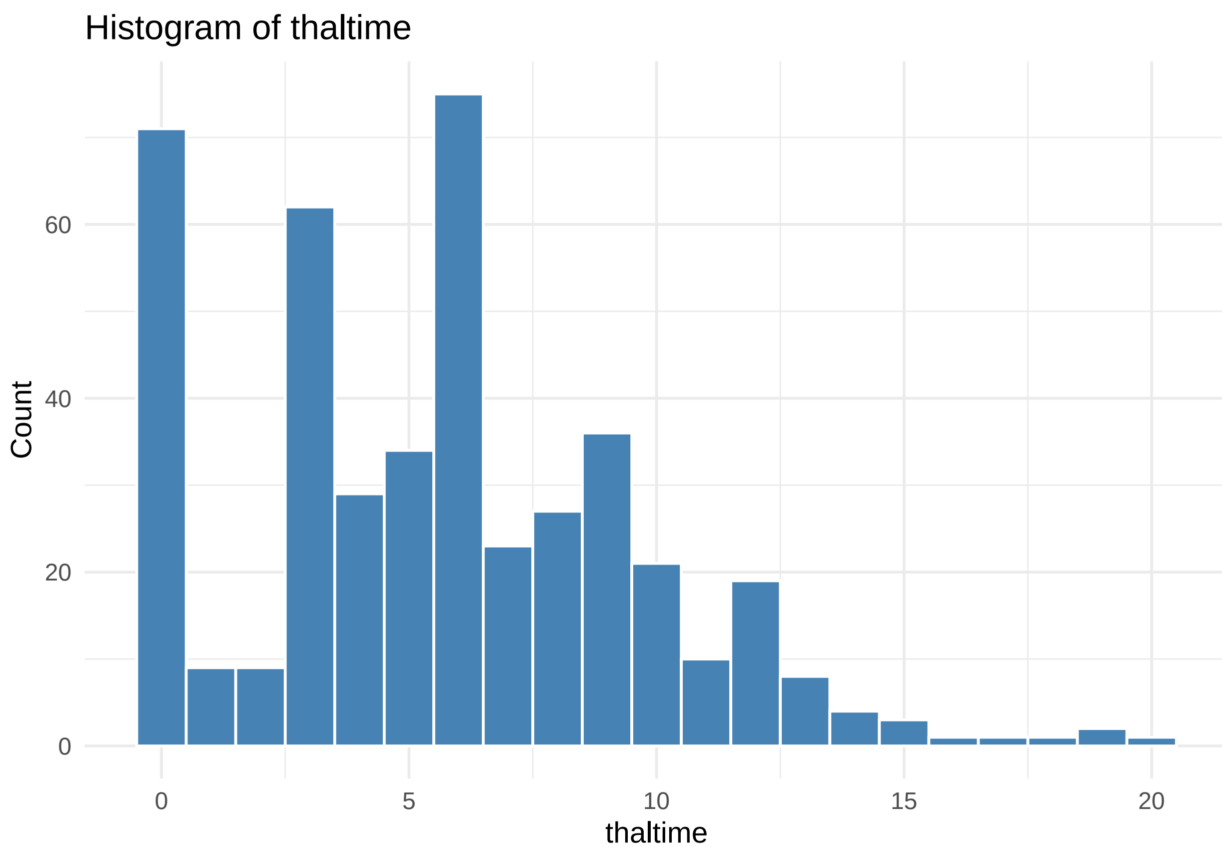
```
sort(table(df$thaltime), decreasing = TRUE)
```

```
##
##    0    6    3    9    8   10    4   12    5  7.5  5.5    7  4.5    2   11    1
```

```
##   70   65   57   28   20   19   17   17   16   11   10    9    8    7    6    5
##  3.5  6.5  8.5   13  1.5  8.6   15  2.5  3.6  4.8  5.1  5.8  6.3  7.4  7.8  9.3
##    5    5    4    4    3    3    3    2    2    2    2    2    2    2    2    2
##  9.5 13.2   14   19  0.5  0.7  3.8  4.2  4.6  4.7  5.2  5.4  6.4  6.8  7.9  9.4
##    2    2    2    2    1    1    1    1    1    1    1    1    1    1    1    1
## 10.2 10.5 10.6 10.7 11.1 11.3 11.7 11.8 12.6 13.5 14.3 14.5 15.8 17.5 17.8   20
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
```

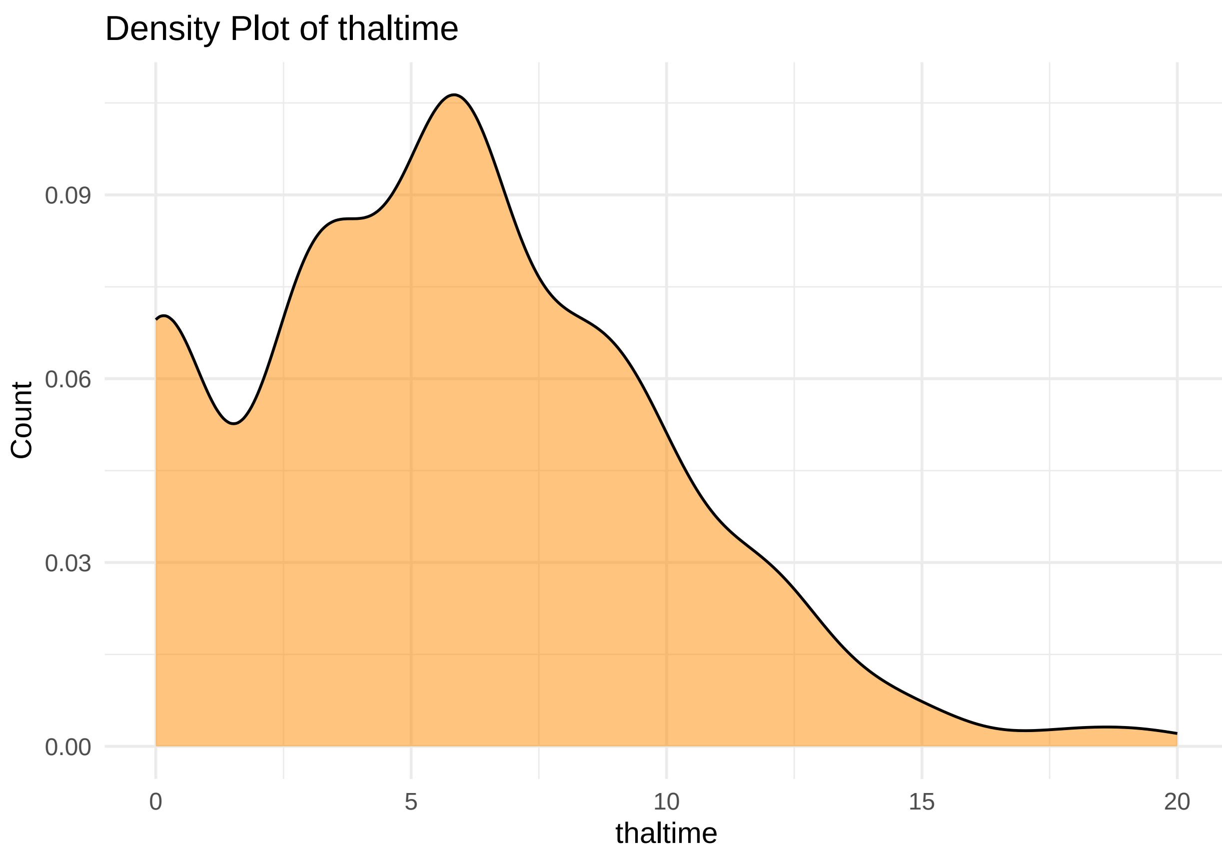
```r
# histogram
show_histogram(df, thaltime)
```

```
## Warning: Removed 453 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```r
show_densityplot(df, thaltime)
```

```
## Warning: Removed 453 rows containing non-finite outside the scale range
## (`stat_density()`).
```



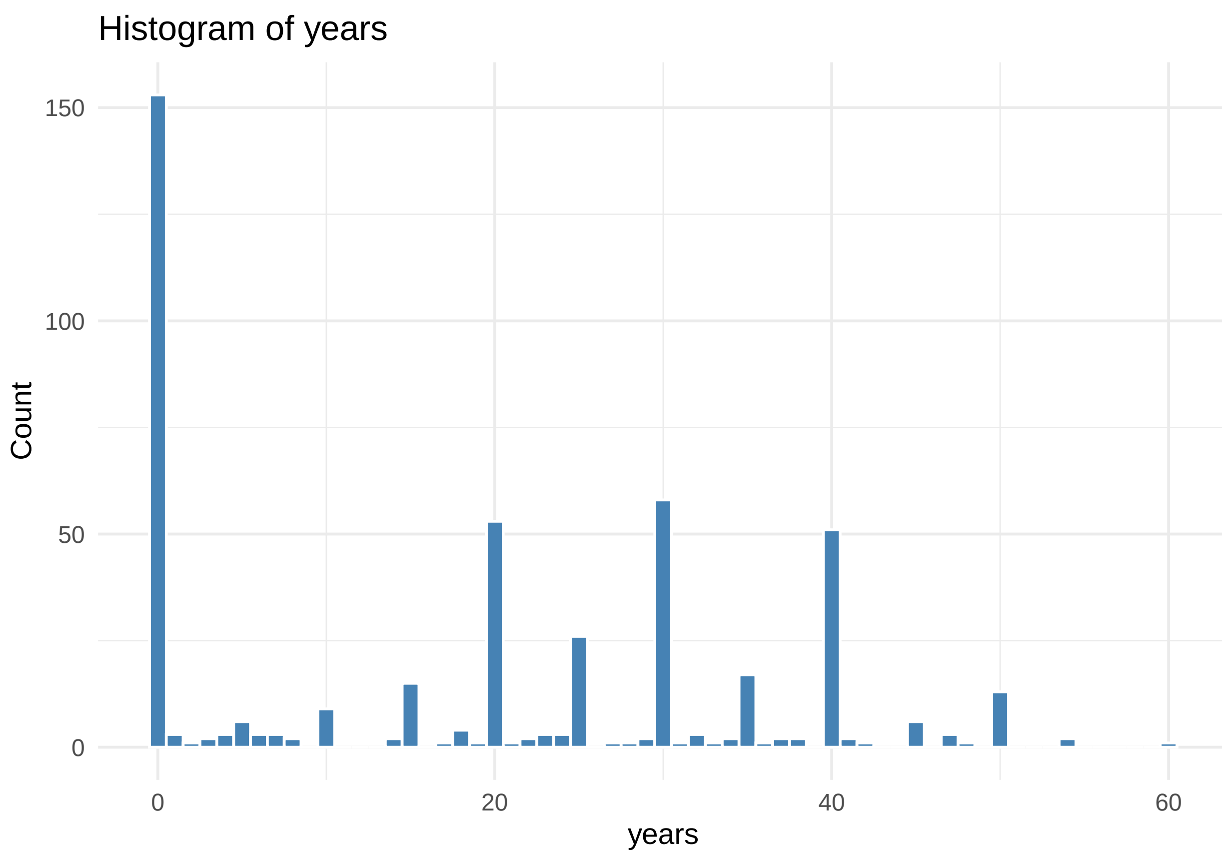

```r
# impute with median + missingness flag
df <- df %>%
  mutate(
    thaltime_missing = ifelse(is.na(thaltime), 1, 0),
    thaltime_imputed = ifelse(is.na(thaltime), median(thaltime, na.rm = TRUE), thaltime)
  )

# drop imputed column
df <- df %>%
  select(-thaltime)
```

Feature `thaltime` showed a right skew. Imputed with median.

Features: `years` and `cigs`

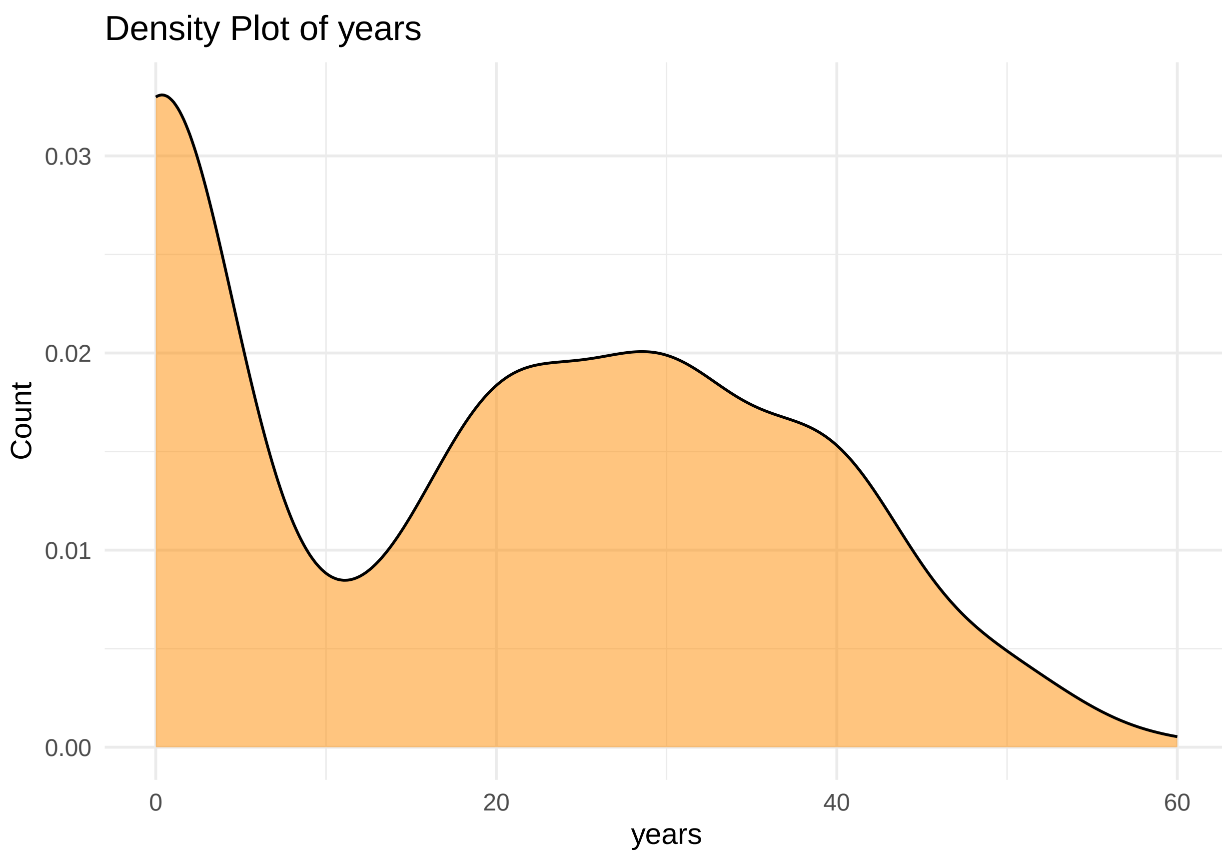
```r
show_histogram(df, years)
```

```
## Warning: Removed 432 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



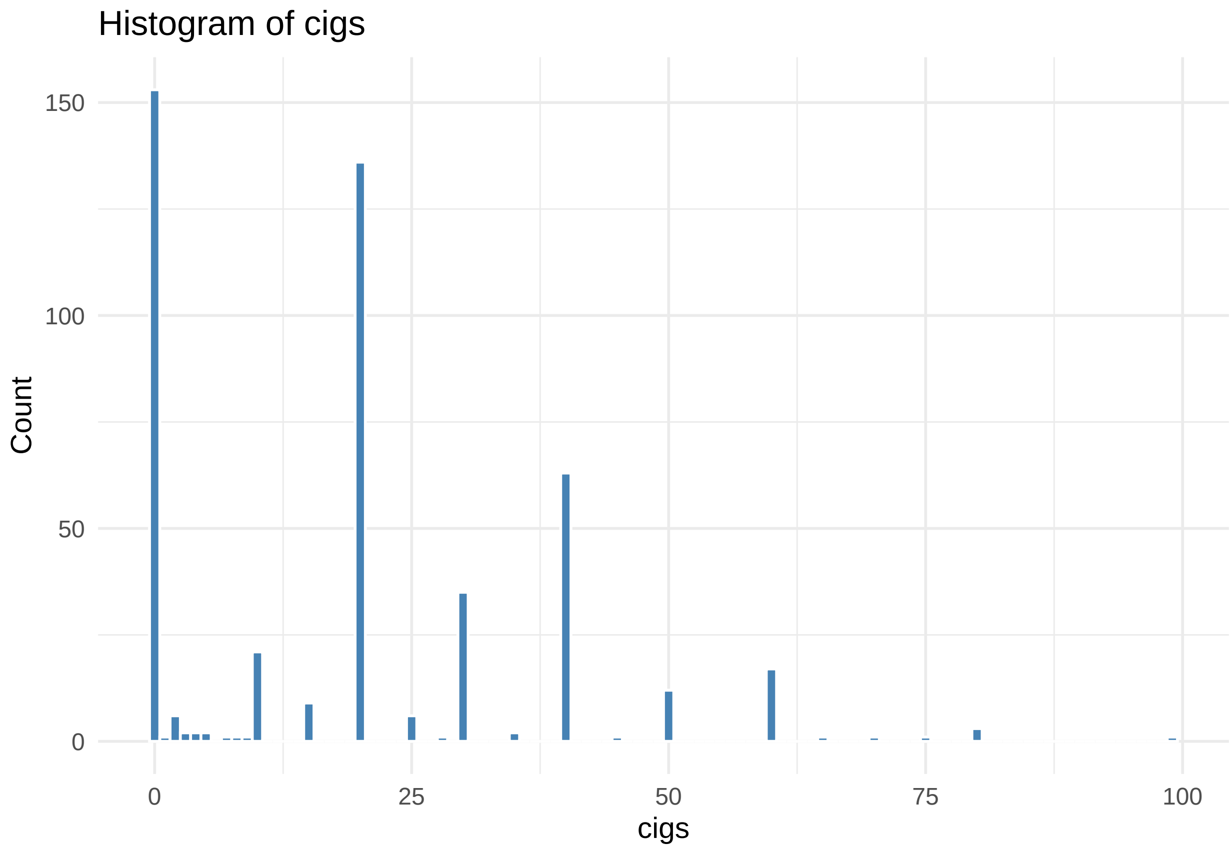

```
show_densityplot(df, years)
```

```
## Warning: Removed 432 rows containing non-finite outside the scale range
## (`stat_density()`).
```



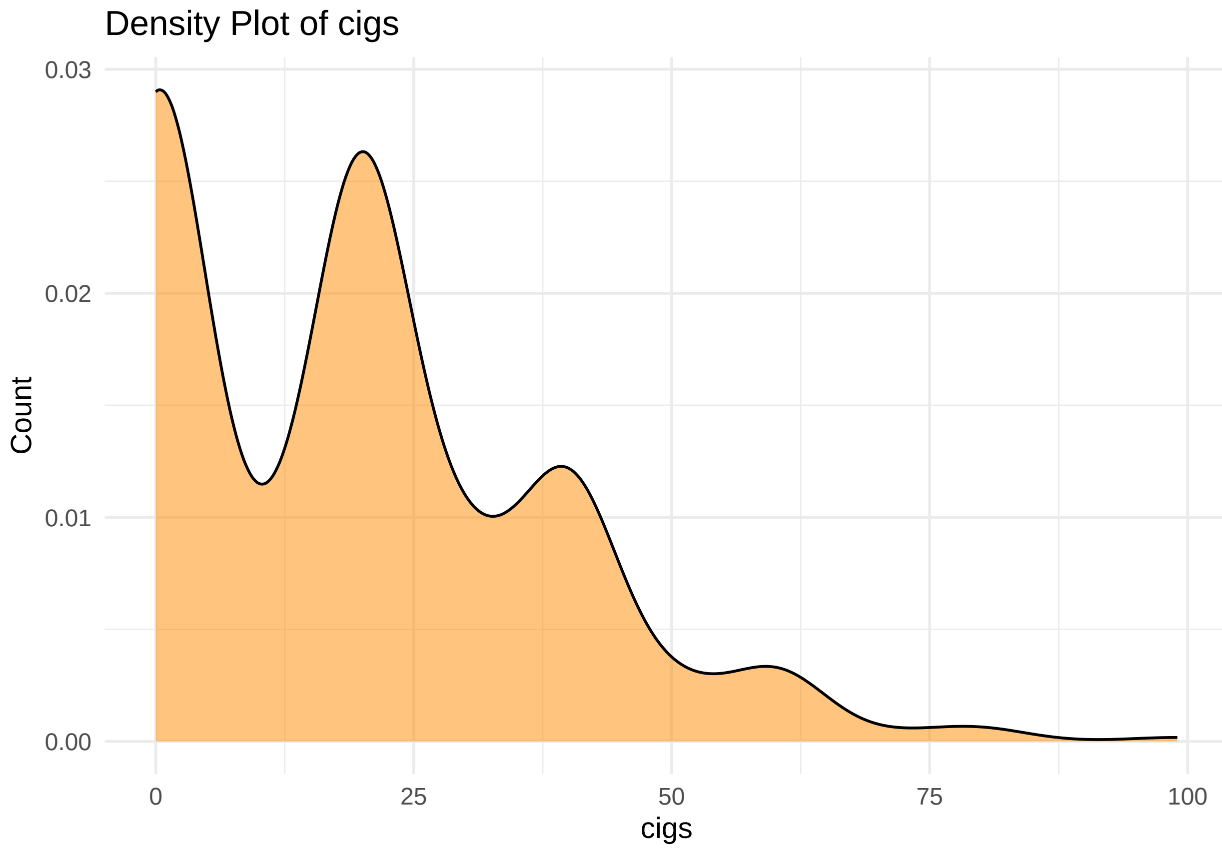

```
show_histogram(df, cigs)
```

```
## Warning: Removed 420 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



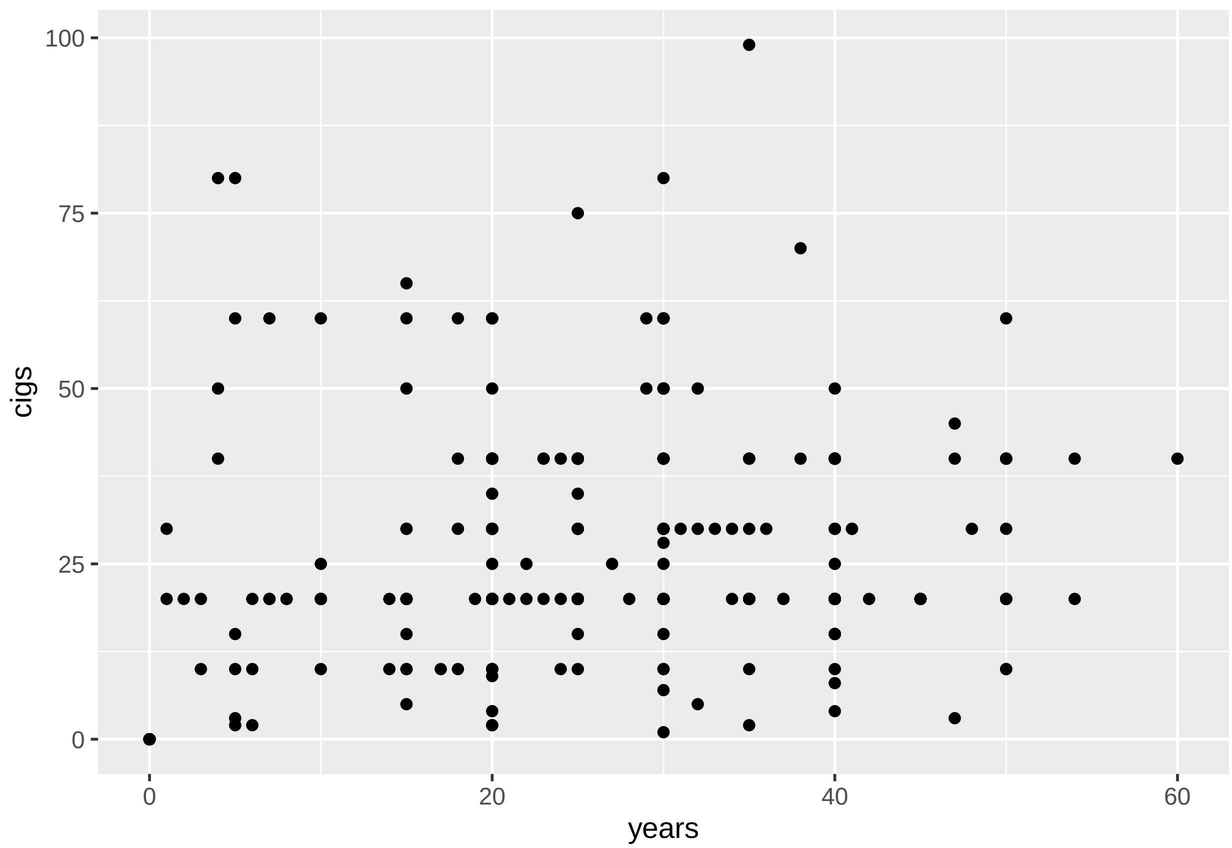

```
show_densityplot(df, cigs)
```

```
## Warning: Removed 420 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```
ggplot(df, aes(x = years, y = cigs)) + geom_point()
```

```
## Warning: Removed 435 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
corr_val <- cor(df$years, df$cigs, use = "complete.obs", method = "pearson")
cat("Correlation value:", corr_val)
```

```
## Correlation value: 0.5890923
```

```r
# create smoke_profile feature
# generative AI was used to create this code to create the smoke_profile feature
df <- df %>%
  mutate(
    # Categorize smoking duration
    years_cat = case_when(
      is.na(years)          ~ "missing",
      years == 0            ~ "non",
      years <= 10           ~ "short",
      years <= 20           ~ "medium",
      years <= 40           ~ "long",
      TRUE                  ~ "very_long"
    ),

    # Categorize cigarette intensity
    cigs_cat = case_when(
      is.na(cigs)           ~ "missing",
      cigs == 0             ~ "non",
      cigs <= 10            ~ "light",
      cigs <= 20            ~ "moderate",
      cigs <= 40            ~ "heavy",
      TRUE                  ~ "very_heavy"
```

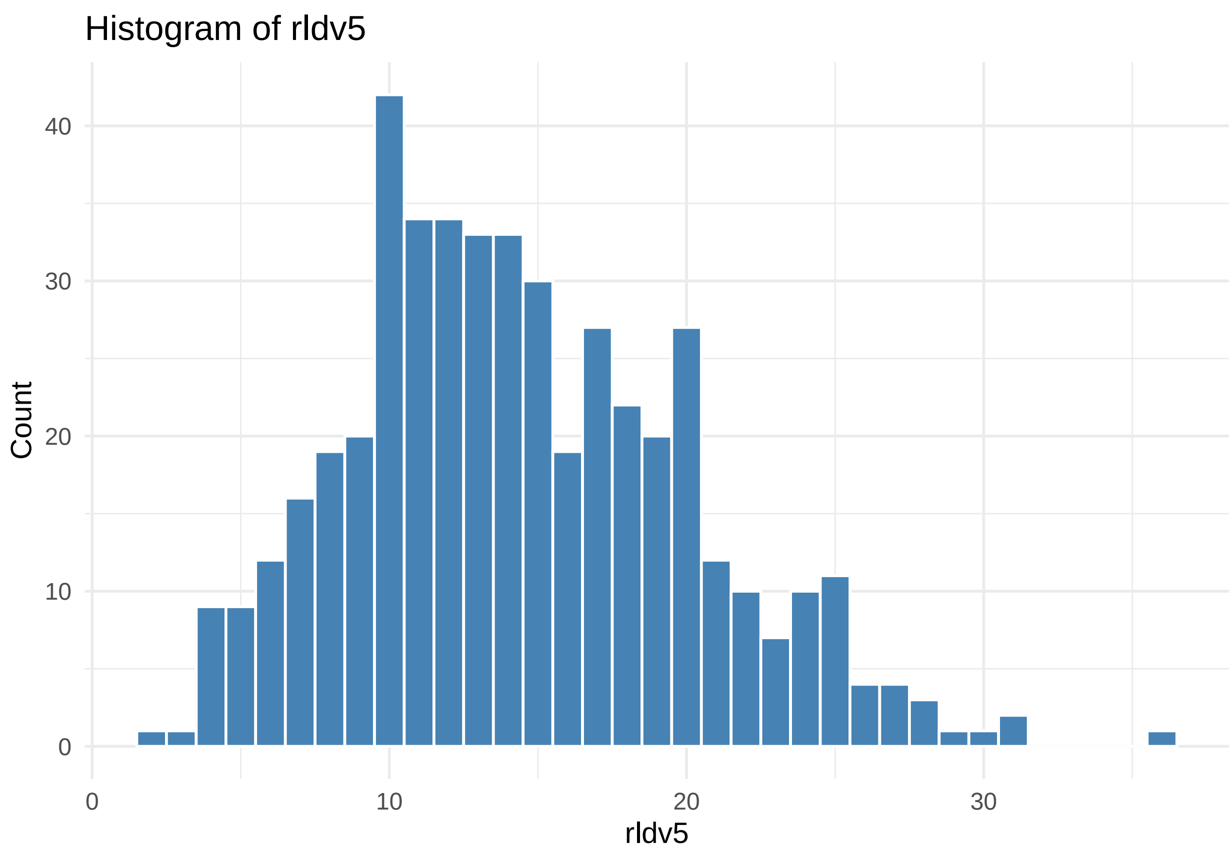
```r
    ),

    # Combine into single profile
    smoke_profile = case_when(
      years_cat == "non" & cigs_cat == "non"               ~ "non",
      years_cat == "missing" | cigs_cat == "missing"       ~ "missing",
      TRUE                                                  ~ paste0(cigs_cat, "_", years_cat)
    ),

    # Optional: convert to factor
    smoke_profile = factor(smoke_profile)
  )

# drop helper columns
df <- df %>%
  select(-c(cigs_cat, years_cat))

# drop imputed column
df <- df %>%
  select(-c(years, cigs))
```

Features `years` and `cigs` show moderate collinearity (Pearson correlation = 0.59). Due to missing data, a `smoke_profile` was engineered to combine the two columns.

Feature: `rldv5`

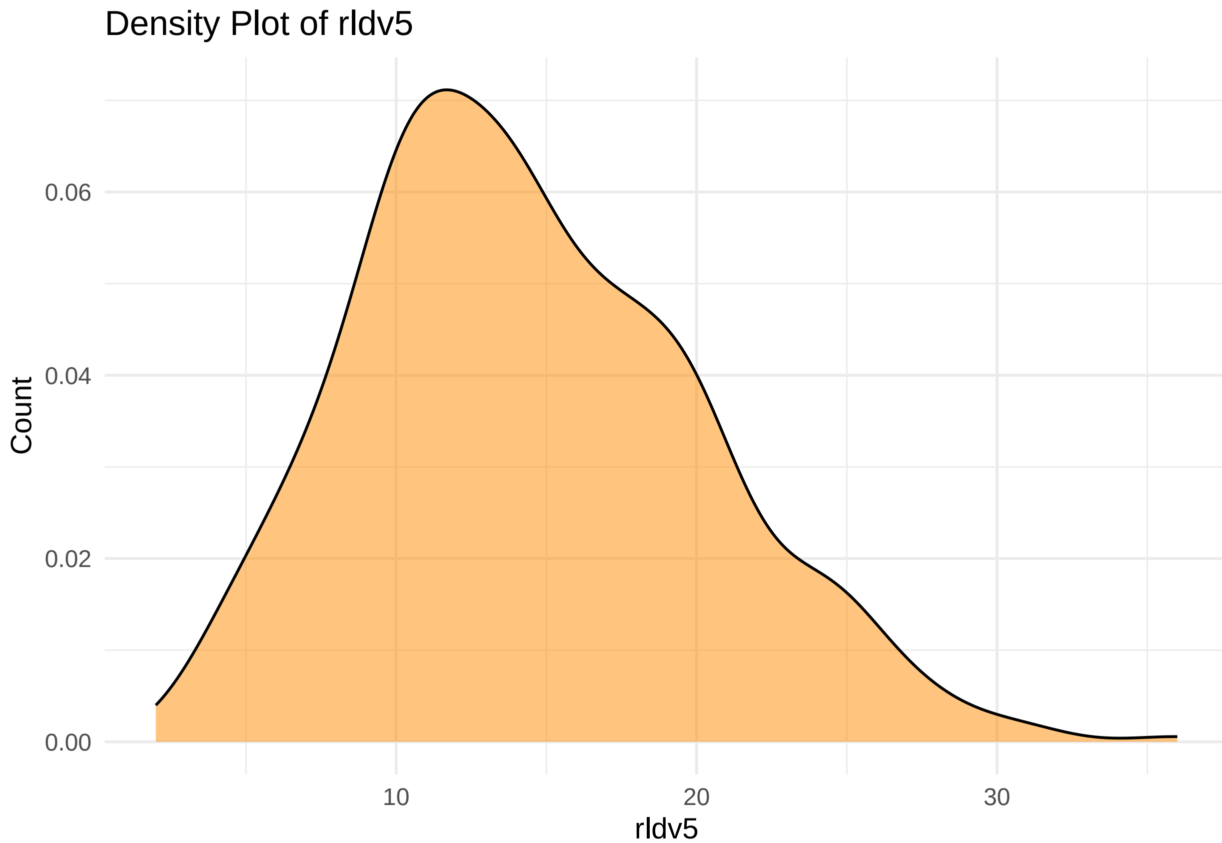
```r
show_histogram(df, rldv5)
```

```
## Warning: Removed 425 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```r
show_densityplot(df, rldv5)
```

```
## Warning: Removed 425 rows containing non-finite outside the scale range
## (`stat_density()`).
```

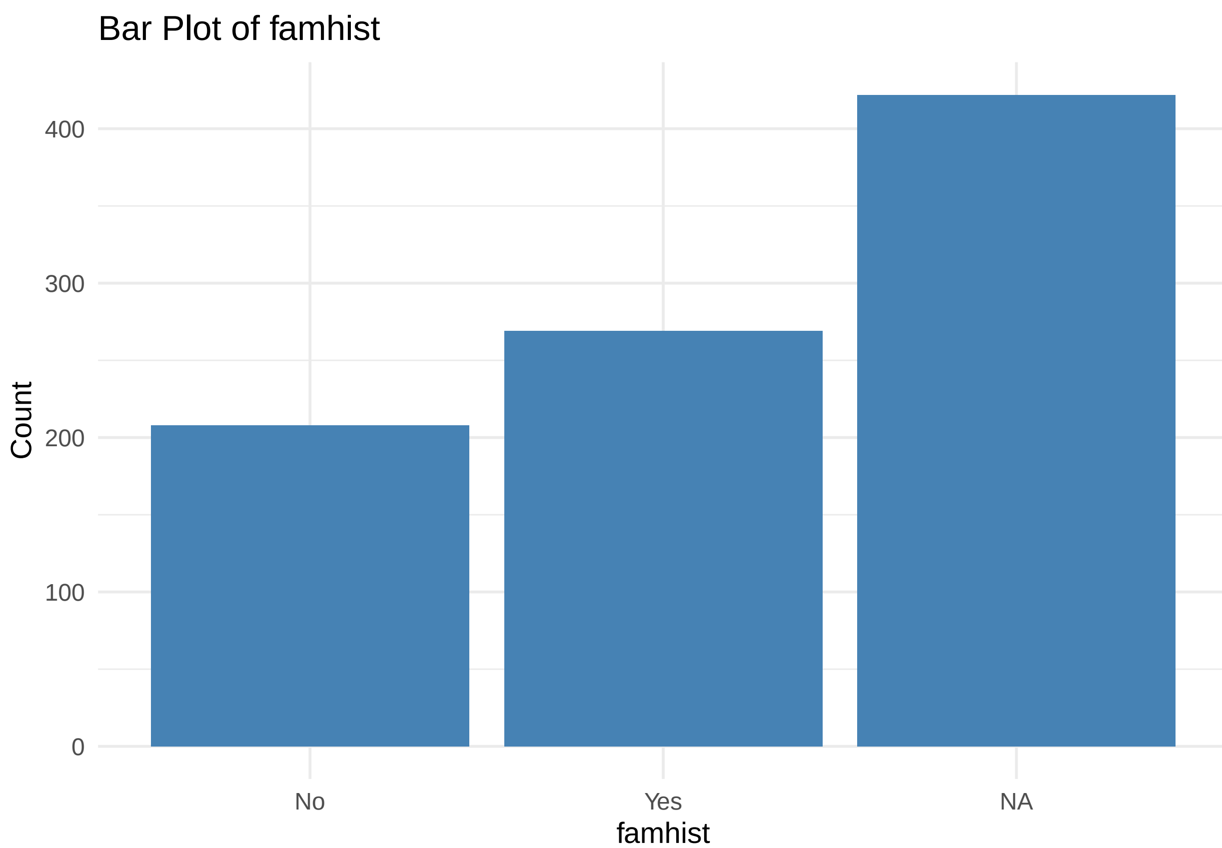



```r
df <- df %>%
  mutate(
    rldv5_missing = ifelse(is.na(rldv5), 1, 0),
    rldv5_imputed = ifelse(is.na(rldv5), median(rldv5, na.rm = TRUE), rldv5)
  )

# drop imputed column
df <- df %>%
  select(-rldv5)
```

Feature `rldv5` has a right skew therefore imputed with median.

Feature: `famhist`

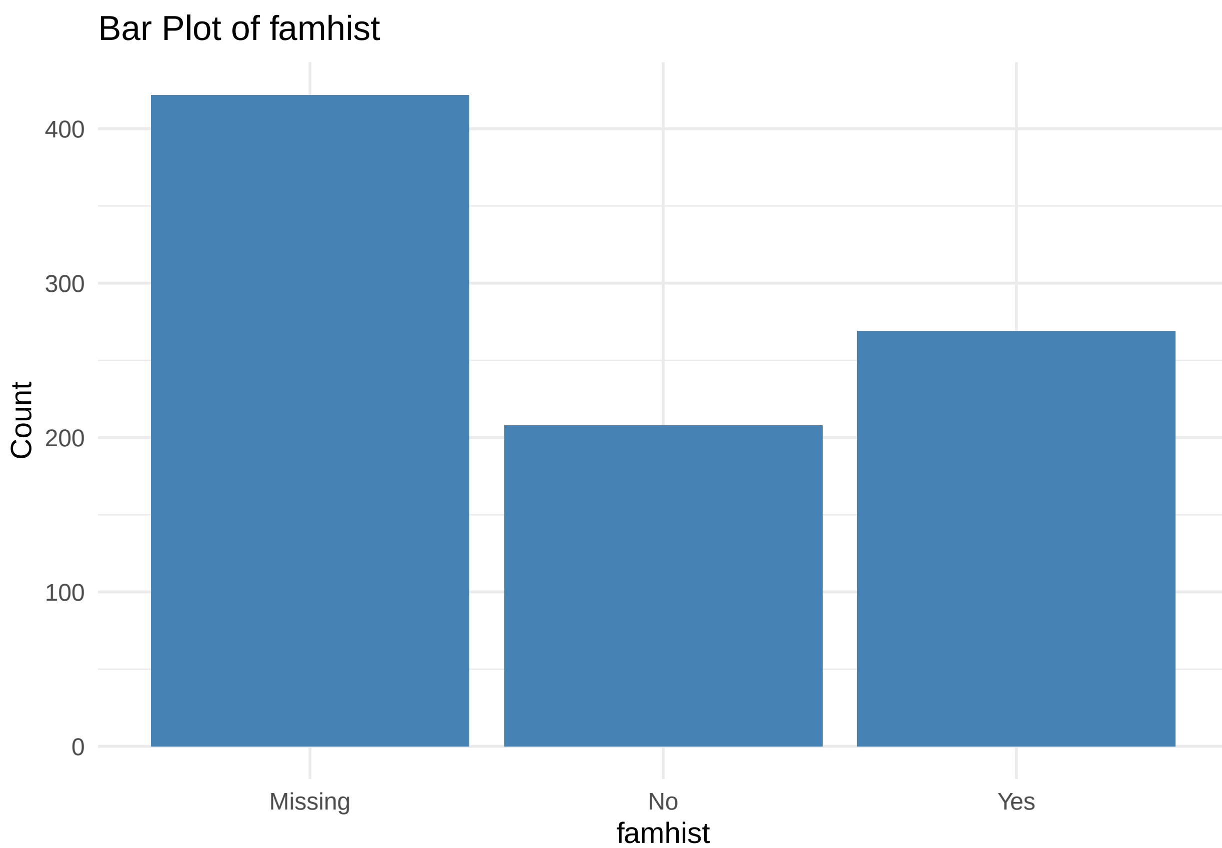
```r
show_barplot(df, famhist)
```




```r
unique(df$famhist)
```

```
## [1] Yes  No   <NA>
## Levels: No Yes
```

```r
# replace NA with "Missing"
df$famhist <- as.character(df$famhist)
df$famhist[is.na(df$famhist)] <- "Missing"
df$famhist <- factor(df$famhist)

show_barplot(df, famhist)
```



Missing values in `famhist` were imputed with 'missing'.

```r
# list of cols
# missing_gt40 <- c('ca', 'thal', 'thaltime', 'years', 'rldv5', 'famhist', 'cigs')
imputed_cols <-c('ca_imputed', 'ca_missing', 'thal_imputed', 'thal_missing',
                 'thaltime_missing', 'thaltime_imputed', 'smoke_profile',
                 'rldv5_missing', 'rldv5_imputed', 'famhist')
# drop date / time columns
drop_cols <- c('cmo', 'cday', 'cyr', 'ekgmo', 'ekgday', 'ekgyr')
df1 <- df %>%
  select(-all_of(drop_cols))

# compare dimensions
dim(df_cleaning)
```

Dropping original columns after imputation

```
## [1] 899  48
```

```r
dim(df)
```

```
## [1] 899  51
```

```r
dim(df1)
```

```
## [1] 899  45
```

```r
colnames(df1)
```

```
##  [1] "age"              "sex"              "painloc"          "painexer"
##  [5] "relrest"          "cp"               "trestbps"         "htn"
##  [9] "chol"             "fbs"              "famhist"          "restecg"
## [13] "prop"             "nitr"             "pro"              "diuretic"
## [17] "thaldur"          "met"              "thalach"          "thalrest"
## [21] "tpeakbps"         "tpeakbpd"         "trestbpd"         "exang"
## [25] "oldpeak"          "slope"            "rldv5e"           "num"
## [29] "lmt"              "ladprox"          "laddist"          "cxmain"
## [33] "om1"              "rcaprox"          "rcadist"          "location"
## [37] "ca_missing"       "ca_imputed"       "thal_missing"     "thal_imputed"
## [41] "thaltime_missing" "thaltime_imputed" "smoke_profile"    "rldv5_missing"
## [45] "rldv5_imputed"
```

```r
head(df1)
```

```
## # A tibble: 6 x 45
##     age sex    painloc painexer relrest cp    trestbps   htn  chol fbs   famhist
##   <dbl> <fct>  <fct>   <fct>    <fct>   <ord>    <dbl> <dbl> <dbl> <fct> <fct>
## 1    63 Male   <NA>    <NA>     <NA>    Typi~      145     1   233 Yes   Yes
## 2    67 Male   <NA>    <NA>     <NA>    Asym~      160     1   286 No    Yes
## 3    67 Male   <NA>    <NA>     <NA>    Asym~      120     1   229 No    Yes
## 4    37 Male   <NA>    <NA>     <NA>    Non-~      130     0   250 No    Yes
## 5    41 Female <NA>    <NA>     <NA>    Atyp~      130     1   204 No    Yes
## 6    56 Male   <NA>    <NA>     <NA>    Atyp~      120     1   236 No    Yes
## # i 34 more variables: restecg <ord>, prop <fct>, nitr <fct>, pro <fct>,
## #   diuretic <fct>, thaldur <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## #   tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, oldpeak <dbl>,
## #   slope <ord>, rldv5e <dbl>, num <dbl>, lmt <dbl>, ladprox <dbl>,
## #   laddist <dbl>, cxmain <dbl>, om1 <dbl>, rcaprox <dbl>, rcadist <dbl>,
## #   location <chr>, ca_missing <dbl>, ca_imputed <dbl>, thal_missing <dbl>,
## #   thal_imputed <dbl>, thaltime_missing <dbl>, thaltime_imputed <dbl>, ...
```

Missing $< 40\%$

Binary Variables

```r
# list binary var
df1 %>%
  summarise(across(everything(), ~ n_distinct(na.omit(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_unique") %>%
  filter(n_unique == 2)
```

```
## # A tibble: 21 x 2
##    variable n_unique
##    <chr>       <int>
##  1 sex             2
##  2 painloc         2
##  3 painexer        2
##  4 relrest         2
##  5 htn             2
##  6 fbs             2
##  7 prop            2
##  8 nitr            2
```

```
##   9 pro              2
## 10 diuretic         2
## # i 11 more rows
```

```r
# get binary var names
binary_vars <- df1 %>%
  select(where(~ n_distinct(na.omit(.)) == 2)) %>%
  names()

# convert binary vars to factors
df1[binary_vars] <- lapply(df1[binary_vars], as.factor)
# convert sex to male/female
#df$sex <- factor(df$sex, levels = c(0, 1), labels = c("female", "male"))

head(df1)
```

```
## # A tibble: 6 x 45
##     age sex    painloc painexer relrest cp    trestbps htn    chol fbs   famhist
##   <dbl> <fct>  <fct>   <fct>    <fct>   <ord>    <dbl> <fct> <dbl> <fct> <fct>
## 1    63 Male   <NA>    <NA>     <NA>    Typi~      145 1       233 Yes   Yes
## 2    67 Male   <NA>    <NA>     <NA>    Asym~      160 1       286 No    Yes
## 3    67 Male   <NA>    <NA>     <NA>    Asym~      120 1       229 No    Yes
## 4    37 Male   <NA>    <NA>     <NA>    Non--      130 0       250 No    Yes
## 5    41 Female <NA>    <NA>     <NA>    Atyp~      130 1       204 No    Yes
## 6    56 Male   <NA>    <NA>     <NA>    Atyp~      120 1       236 No    Yes
## # i 34 more variables: restecg <ord>, prop <fct>, nitr <fct>, pro <fct>,
## #   diuretic <fct>, thaldur <dbl>, met <dbl>, thalach <dbl>, thalrest <dbl>,
## #   tpeakbps <dbl>, tpeakbpd <dbl>, trestbpd <dbl>, exang <fct>, oldpeak <dbl>,
## #   slope <ord>, rldv5e <dbl>, num <dbl>, lmt <dbl>, ladprox <fct>,
## #   laddist <fct>, cxmain <fct>, om1 <fct>, rcaprox <fct>, rcadist <fct>,
## #   location <chr>, ca_missing <fct>, ca_imputed <dbl>, thal_missing <fct>,
## #   thal_imputed <dbl>, thaltime_missing <fct>, thaltime_imputed <dbl>, ...
```

Numerical Variables

```r
numeric_vars <- df1 %>%
  select(where(is.numeric)) %>%
  names()

numeric_na <- df1 %>%
  select(where(is.numeric)) %>%
  select(where(~ any(is.na(.)))) %>%
  names()

numeric_vars
```

```
##  [1] "age"              "trestbps"         "chol"             "thaldur"
##  [5] "met"              "thalach"          "thalrest"         "tpeakbps"
##  [9] "tpeakbpd"         "trestbpd"         "oldpeak"          "rldv5e"
## [13] "num"              "lmt"              "ca_imputed"       "thal_imputed"
## [17] "thaltime_imputed" "rldv5_imputed"
```

```r
df1[numeric_na]
```

```
## # A tibble: 899 x 12
##    trestbps  chol thaldur   met thalach thalrest tpeakbps tpeakbpd trestbpd
```

```
##        <dbl> <dbl>   <dbl> <dbl>   <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
##   1      145   233    10.5    13     150       60      190       90       85
##   2      160   286     9.5    13     108       64      160       90       90
##   3      120   229     8.5    10     129       78      140       80       80
##   4      130   250    13      17     187       84      195       68       78
##   5      130   204     7       9     172       71      160       74       86
##   6      120   236    11.3    16     178       73      165       70       75
##   7      140   268     6       7     160       83      180       84      100
##   8      120   354     9      10     163       84      165       80       80
##   9      130   254     8       9     147       75      120       70      105
## 10      140   203     5.5     7     155       86      185      120      100
## # i 889 more rows
## # i 3 more variables: oldpeak <dbl>, rldv5e <dbl>, lmt <dbl>
```

```r
# impute w median and create missingness columns
for (var in numeric_na) {
  median_val <- median(df1[[var]], na.rm = TRUE)
  df1[[paste0(var, "_missing")]] <- ifelse(is.na(df1[[var]]), 1, 0)
  df1[[paste0(var, "_imputed")]] <- ifelse(is.na(df1[[var]]), median_val, df1[[var]])
}

# drop imputed columns
df1 <- df1 %>%
  select(-all_of(numeric_na))
dim(df1)
```

```
## [1] 899  57
```

Numerical variables were imputed with the median value. Missing flags were also created for these variables.

Categorical Variables

```r
categorical_na <- df1 %>%
  select(where(~ is.factor(.) || is.character(.))) %>%
  select(where(~ any(is.na(.)))) %>%
  names()

# impute with Missing
for (var in categorical_na) {
  df1[[var]] <- as.character(df1[[var]])
  df1[[var]][is.na(df1[[var]])] <- "Missing"
  df1[[var]] <- factor(df1[[var]])
}
dim(df1)
```

```
## [1] 899  57
```

```r
anyNA(df1)
```

```
## [1] FALSE
```

For NAs in categorical variables, NAs were replaced with 'Missing'.

Create binary for target (num)

```r
# create binary target 0 = no disease, 1 - 3 = varying levels
df1 <- df1 %>%
```

```r
  mutate(
    has_disease = factor(ifelse(num == 0, "No", "Yes"), levels = c("No", "Yes"))
  )

head(df1)
```

```
## # A tibble: 6 x 58
##     age sex    painloc painexer relrest cp     htn   fbs   famhist restecg prop
##   <dbl> <fct>  <fct>   <fct>    <fct>   <ord>  <fct> <fct> <fct>   <fct>   <fct>
## 1    63 Male   Missing Missing  Missing Typic~ 1     Yes   Yes     Hypert~ No
## 2    67 Male   Missing Missing  Missing Asymp~ 1     No    Yes     Hypert~ Yes
## 3    67 Male   Missing Missing  Missing Asymp~ 1     No    Yes     Hypert~ Yes
## 4    37 Male   Missing Missing  Missing Non-a~ 0     No    Yes     Normal  Yes
## 5    41 Female Missing Missing  Missing Atypi~ 1     No    Yes     Hypert~ No
## 6    56 Male   Missing Missing  Missing Atypi~ 1     No    Yes     Normal  No
## # i 47 more variables: nitr <fct>, pro <fct>, diuretic <fct>, exang <fct>,
## #   slope <fct>, num <dbl>, ladprox <fct>, laddist <fct>, cxmain <fct>,
## #   om1 <fct>, rcaprox <fct>, rcadist <fct>, location <chr>, ca_missing <fct>,
## #   ca_imputed <dbl>, thal_missing <fct>, thal_imputed <dbl>,
## #   thaltime_missing <fct>, thaltime_imputed <dbl>, smoke_profile <fct>,
## #   rldv5_missing <fct>, rldv5_imputed <dbl>, trestbps_missing <dbl>,
## #   trestbps_imputed <dbl>, chol_missing <dbl>, chol_imputed <dbl>, ...
```

Binary target column for the target variable `num` was created. The column `num` was not removed from the dataset.

Export finalized dataset

```r
# export final dataset
saveRDS(df1, here("Data/Clean", "heart_disease_final.rds"))
```

Univariate Analysis

```r
knitr::opts_chunk$set(echo = TRUE)
```

```r
df <- readRDS(here("Data/Clean", "heart_disease_final.rds"))
head(df)
```

```
## # A tibble: 6 x 58
##     age sex    painloc painexer relrest cp     htn   fbs   famhist restecg prop
##   <dbl> <fct>  <fct>   <fct>    <fct>   <ord>  <fct> <fct> <fct>   <fct>   <fct>
## 1    63 Male   Missing Missing  Missing Typic~ 1     Yes   Yes     Hypert~ No
## 2    67 Male   Missing Missing  Missing Asymp~ 1     No    Yes     Hypert~ Yes
## 3    67 Male   Missing Missing  Missing Asymp~ 1     No    Yes     Hypert~ Yes
## 4    37 Male   Missing Missing  Missing Non-a~ 0     No    Yes     Normal  Yes
## 5    41 Female Missing Missing  Missing Atypi~ 1     No    Yes     Hypert~ No
## 6    56 Male   Missing Missing  Missing Atypi~ 1     No    Yes     Normal  No
## # i 47 more variables: nitr <fct>, pro <fct>, diuretic <fct>, exang <fct>,
## #   slope <fct>, num <dbl>, ladprox <fct>, laddist <fct>, cxmain <fct>,
## #   om1 <fct>, rcaprox <fct>, rcadist <fct>, location <chr>, ca_missing <fct>,
## #   ca_imputed <dbl>, thal_missing <fct>, thal_imputed <dbl>,
## #   thaltime_missing <fct>, thaltime_imputed <dbl>, smoke_profile <fct>,
## #   rldv5_missing <fct>, rldv5_imputed <dbl>, trestbps_missing <dbl>,
## #   trestbps_imputed <dbl>, chol_missing <dbl>, chol_imputed <dbl>, ...
```

```r
# separate target from variables
target_vars <- c("num", "has_disease")
y_df <- df %>%
  select(all_of(target_vars))

X_df <- df %>%
  select(-all_of(target_vars))
```

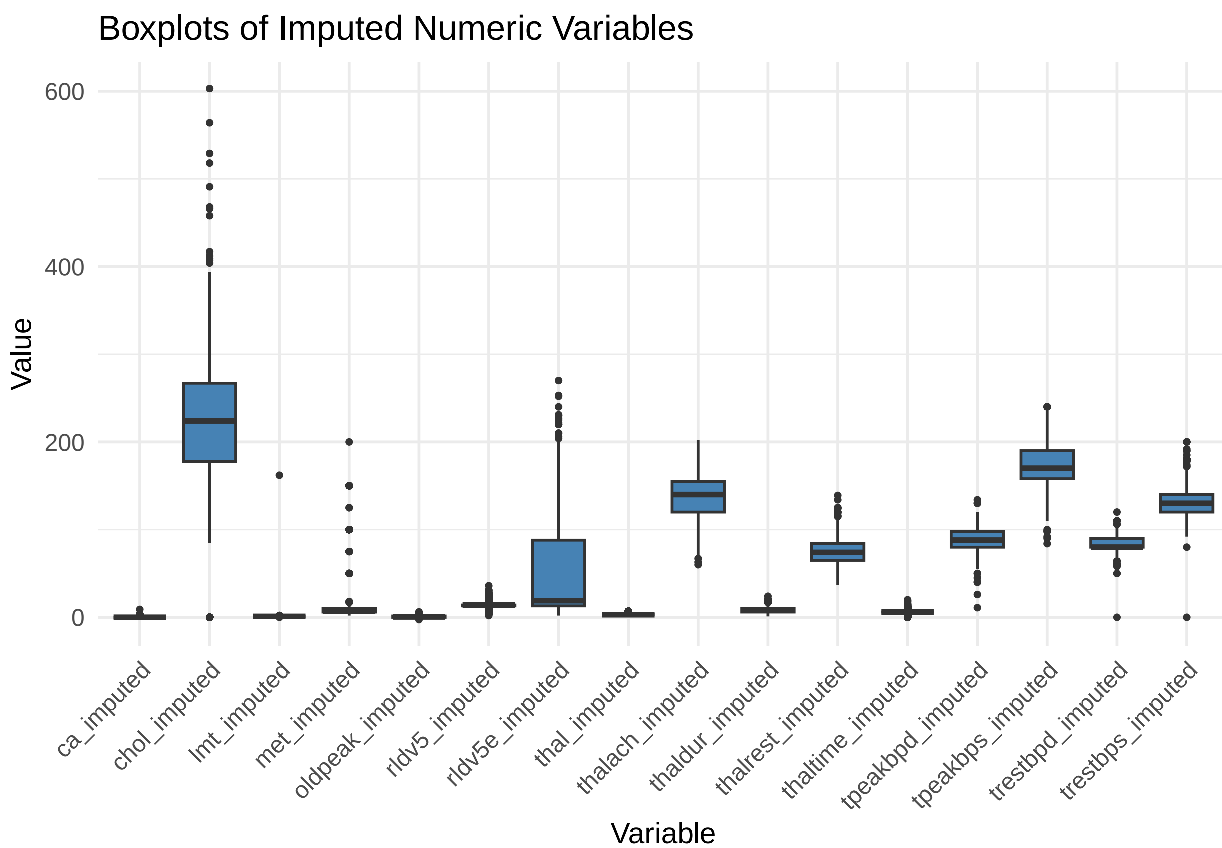
Numerical vars

```r
num_vars <-X_df %>%
  select(where(is.numeric)) %>%
  names()

# summary
X_df %>%
  select(all_of(num_vars)) %>%
  summary()
```

```
##       age          ca_imputed      thal_imputed  thaltime_imputed
##  Min.   :28.00   Min.   :0.0000   Min.   :3.00   Min.   : 0.000
##  1st Qu.:47.00   1st Qu.:0.0000   1st Qu.:3.00   1st Qu.: 6.000
##  Median :54.00   Median :0.0000   Median :3.00   Median : 6.000
##  Mean   :53.48   Mean   :0.2258   Mean   :3.95   Mean   : 5.846
##  3rd Qu.:60.00   3rd Qu.:0.0000   3rd Qu.:3.00   3rd Qu.: 6.000
##  Max.   :77.00   Max.   :9.0000   Max.   :7.00   Max.   :20.000
##  rldv5_imputed   trestbps_missing  trestbps_imputed  chol_missing
##  Min.   : 2.00   Min.   :0.00000   Min.   :  0      Min.   :0.00000
##  1st Qu.:13.00   1st Qu.:0.00000   1st Qu.:120      1st Qu.:0.00000
##  Median :14.00   Median :0.00000   Median :130      Median :0.00000
##  Mean   :14.21   Mean   :0.06563   Mean   :132      Mean   :0.03337
##  3rd Qu.:14.00   3rd Qu.:0.00000   3rd Qu.:140      3rd Qu.:0.00000
##  Max.   :36.00   Max.   :1.00000   Max.   :200      Max.   :1.00000
##   chol_imputed   thaldur_missing   thaldur_imputed   met_missing
##  Min.   :  0.0   Min.   :0.00000   Min.   : 1.000   Min.   :0.0000
##  1st Qu.:177.5   1st Qu.:0.00000   1st Qu.: 6.000   1st Qu.:0.0000
##  Median :224.0   Median :0.00000   Median : 8.100   Median :0.0000
##  Mean   :199.6   Mean   :0.06229   Mean   : 8.621   Mean   :0.1168
##  3rd Qu.:267.0   3rd Qu.:0.00000   3rd Qu.:10.300   3rd Qu.:0.0000
##  Max.   :603.0   Max.   :1.00000   Max.   :24.000   Max.   :1.0000
##   met_imputed     thalach_missing   thalach_imputed thalrest_missing
##  Min.   :  2.00   Min.   :0.00000   Min.   : 60.0   Min.   :0.00000
##  1st Qu.:  6.00   1st Qu.:0.00000   1st Qu.:120.0   1st Qu.:0.00000
##  Median :  7.00   Median :0.00000   Median :140.0   Median :0.00000
##  Mean   : 15.38   Mean   :0.06118   Mean   :137.5   Mean   :0.06229
##  3rd Qu.: 10.00   3rd Qu.:0.00000   3rd Qu.:155.0   3rd Qu.:0.00000
##  Max.   :200.00   Max.   :1.00000   Max.   :202.0   Max.   :1.00000
##  thalrest_imputed tpeakbps_missing  tpeakbps_imputed tpeakbpd_missing
##  Min.   : 37.00   Min.   :0.00000   Min.   : 84.0    Min.   :0.00000
##  1st Qu.: 65.00   1st Qu.:0.00000   1st Qu.:158.0    1st Qu.:0.00000
##  Median : 74.00   Median :0.00000   Median :170.0    Median :0.00000
##  Mean   : 75.39   Mean   :0.07008   Mean   :171.5    Mean   :0.07008
##  3rd Qu.: 84.00   3rd Qu.:0.00000   3rd Qu.:190.0    3rd Qu.:0.00000
```

```
## Max.   :139.00   Max.   :1.00000   Max.   :240.0    Max.   :1.00000
## tpeakbpd_imputed trestbpd_missing  trestbpd_imputed oldpeak_missing
## Min.   : 11.00   Min.   :0.00000   Min.   :  0.00   Min.   :0.00000
## 1st Qu.: 80.00   1st Qu.:0.00000   1st Qu.: 80.00   1st Qu.:0.00000
## Median : 88.00   Median :0.00000   Median : 80.00   Median :0.00000
## Mean   : 87.34   Mean   :0.06563   Mean   : 83.29   Mean   :0.06897
## 3rd Qu.: 98.00   3rd Qu.:0.00000   3rd Qu.: 90.00   3rd Qu.:0.00000
## Max.   :134.00   Max.   :1.00000   Max.   :120.00   Max.   :1.00000
## oldpeak_imputed   rldv5e_missing  rldv5e_imputed    lmt_missing
## Min.   :-2.6000   Min.   :0.000   Min.   :  2.00   Min.   :0.0000
## 1st Qu.: 0.0000   1st Qu.:0.000   1st Qu.: 13.00   1st Qu.:0.0000
## Median : 0.5000   Median :0.000   Median : 19.00   Median :0.0000
## Mean   : 0.8449   Mean   :0.158   Mean   : 49.24   Mean   :0.3059
## 3rd Qu.: 1.5000   3rd Qu.:0.000   3rd Qu.: 88.00   3rd Qu.:1.0000
## Max.   : 6.2000   Max.   :1.000   Max.   :270.00   Max.   :1.0000
##  lmt_imputed
## Min.   :  0.000
## 1st Qu.:  1.000
## Median :  1.000
## Mean   :  1.225
## 3rd Qu.:  1.000
## Max.   :162.000
```

```r
# Example: just the *_imputed columns (exclude *_missing)
imputed_vars <- names(X_df)[str_detect(names(X_df), "_imputed") & !str_detect(names(X_df), "_missing")]

X_df %>%
  select(all_of(imputed_vars)) %>%
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = variable, y = value)) +
  geom_boxplot(fill = "steelblue", outlier.shape = 16, outlier.size = 1) +
  theme_minimal() +
  labs(title = "Boxplots of Imputed Numeric Variables", x = "Variable", y = "Value") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# summary statistics
X_df %>%
  select(where(is.numeric)) %>%
  skimr::skim()
```

Table 1: Data summary

Name	Piped data
Number of rows	899
Number of columns	29
Column type frequency:	
numeric	29
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	53.48	9.44	28.0	47.0	54.0	60.0	77.0	
ca_imputed	0	1	0.23	0.68	0.0	0.0	0.0	0.0	9.0	
thal_imputed	0	1	3.95	1.66	3.0	3.0	3.0	3.0	7.0	
thaltime_imputed	0	1	5.85	2.82	0.0	6.0	6.0	6.0	20.0	
rldv5_imputed	0	1	14.21	4.14	2.0	13.0	14.0	14.0	36.0	
trestbps_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
trestbps_imputed	0	1	131.96	18.52	0.0	120.0	130.0	140.0	200.0	
chol_missing	0	1	0.03	0.18	0.0	0.0	0.0	0.0	1.0	
chol_imputed	0	1	199.60	110.04	0.0	177.5	224.0	267.0	603.0	
thaldur_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thaldur_imputed	0	1	8.62	3.63	1.0	6.0	8.1	10.3	24.0	
met_missing	0	1	0.12	0.32	0.0	0.0	0.0	0.0	1.0	
met_imputed	0	1	15.38	29.08	2.0	6.0	7.0	10.0	200.0	
thalach_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thalach_imputed	0	1	137.46	25.17	60.0	120.0	140.0	155.0	202.0	
thalrest_missing	0	1	0.06	0.24	0.0	0.0	0.0	0.0	1.0	
thalrest_imputed	0	1	75.39	14.27	37.0	65.0	74.0	84.0	139.0	
tpeakbps_missing	0	1	0.07	0.26	0.0	0.0	0.0	0.0	1.0	
tpeakbps_imputed	0	1	171.53	24.82	84.0	158.0	170.0	190.0	240.0	
tpeakbpd_missing	0	1	0.07	0.26	0.0	0.0	0.0	0.0	1.0	
tpeakbpd_imputed	0	1	87.34	14.21	11.0	80.0	88.0	98.0	134.0	
trestbpd_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	
trestbpd_imputed	0	1	83.29	9.95	0.0	80.0	80.0	90.0	120.0	
oldpeak_missing	0	1	0.07	0.25	0.0	0.0	0.0	0.0	1.0	
oldpeak_imputed	0	1	0.84	1.05	-2.6	0.0	0.5	1.5	6.2	
rldv5e_missing	0	1	0.16	0.36	0.0	0.0	0.0	0.0	1.0	
rldv5e_imputed	0	1	49.24	56.87	2.0	13.0	19.0	88.0	270.0	
lmt_missing	0	1	0.31	0.46	0.0	0.0	0.0	1.0	1.0	
lmt_imputed	0	1	1.22	5.37	0.0	1.0	1.0	1.0	162.0	

Categorical vars

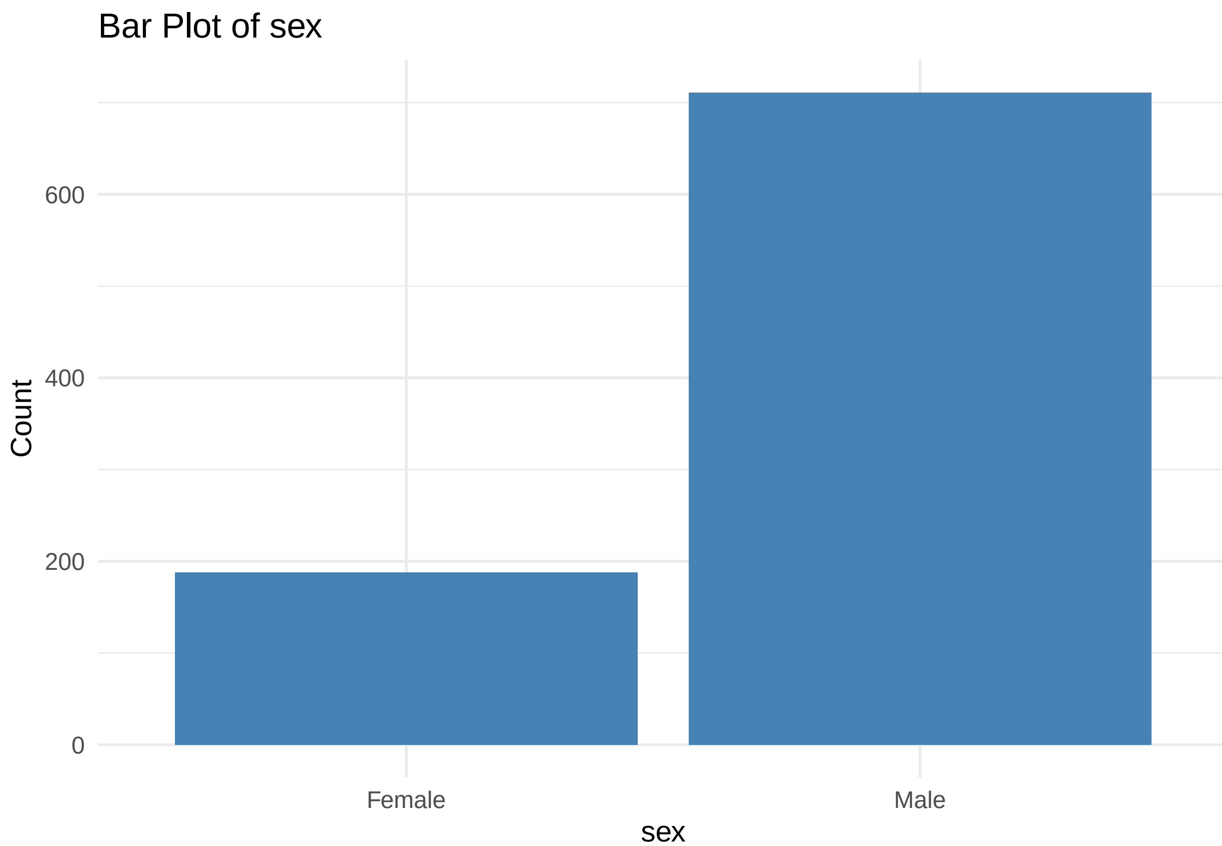
```
cat_vars <- X_df %>%
  select(where(~ is.factor(.) || is.character(.))) %>%
  names()

# relative frequency tables
for (var in cat_vars) {
  cat("----", var, "----\n")
  print(prop.table(table(X_df[[var]])))
  cat("\n")
}
```

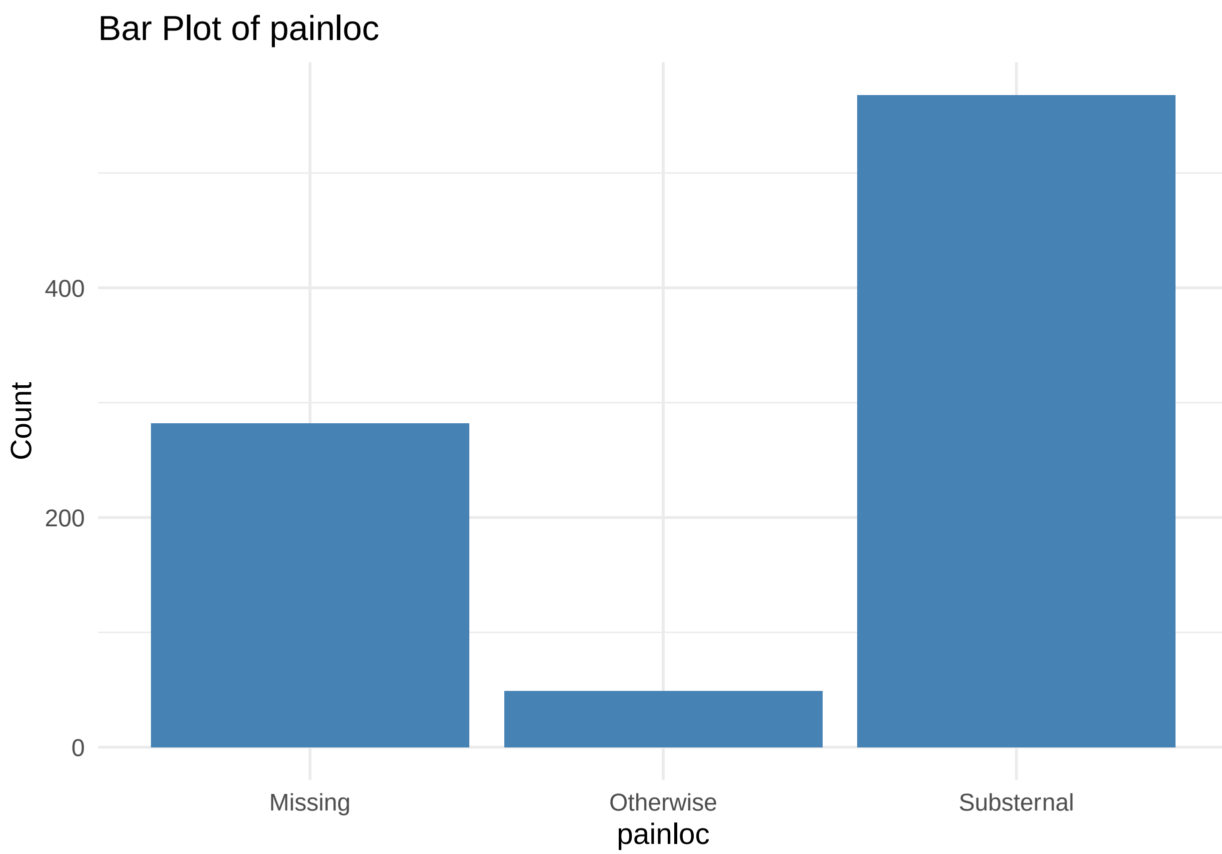
```
## ---- sex ----
##
##    Female      Male
## 0.2091212 0.7908788
##
## ---- painloc ----
##
##    Missing  Otherwise Substernal
## 0.31368187 0.05450501 0.63181313
##
## ---- painexer ----
##
##              Missing            Otherwise Provoked by Exertion
##            0.3136819            0.2791991            0.4071190
##
```

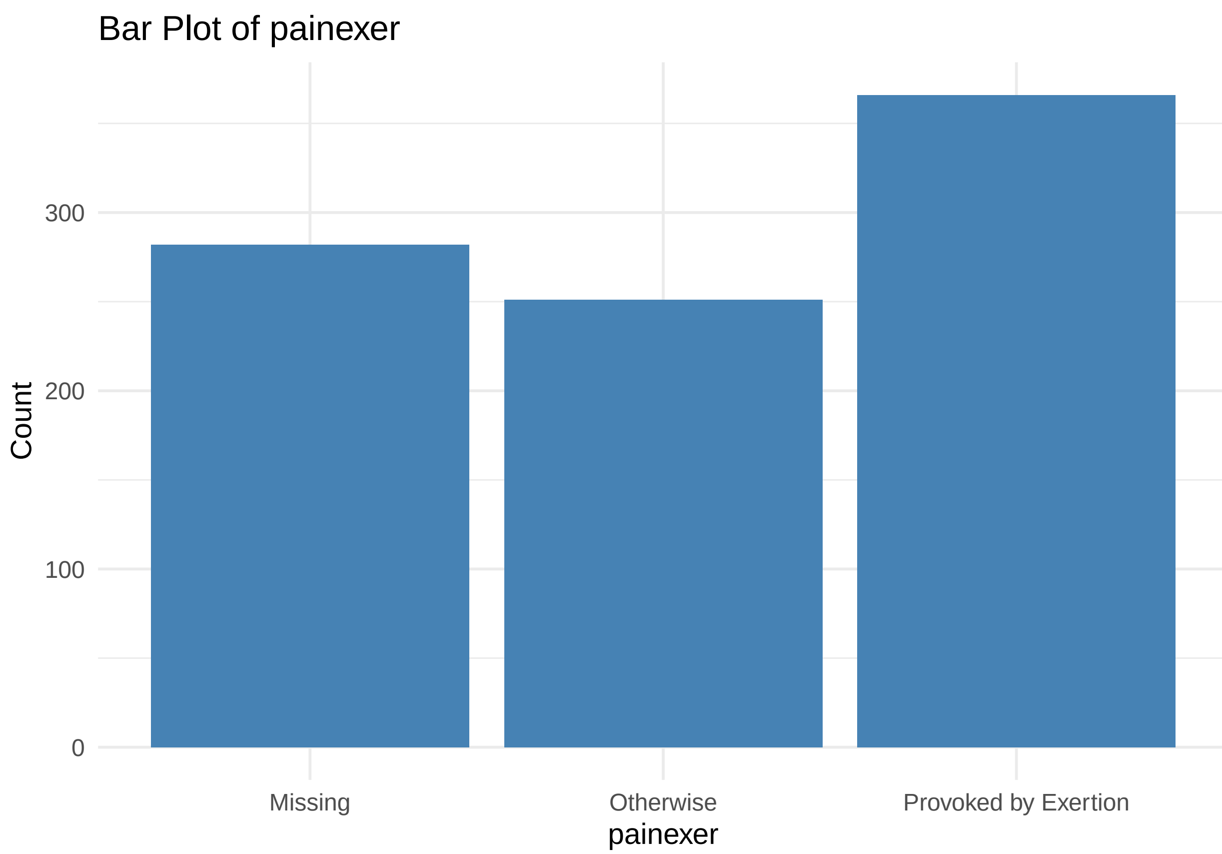
```
## ---- relrest ----
##
##             Missing           Otherwise Relieved after rest
##           0.3181313           0.2235818           0.4582870
##
## ---- cp ----
##
##   Typical Angina  Atypical Angina Non-anginal Pain     Asymptomatic
##       0.05005562       0.18576196       0.22469410       0.53948832
##
## ---- htn ----
##
##         0         1   Missing
## 0.5038932 0.4582870 0.0378198
##
## ---- fbs ----
##
##   Missing        No       Yes
## 0.1001112 0.7497219 0.1501669
##
## ---- famhist ----
##
##   Missing        No       Yes
## 0.4694105 0.2313682 0.2992214
##
## ---- restecg ----
##
##    Abnormal Hypertrophy     Missing      Normal
## 0.196885428 0.202447164 0.002224694 0.598442714
##
## ---- prop ----
##
##    Missing         No        Yes
## 0.07452725 0.68743048 0.23804227
##
## ---- nitr ----
##
##    Missing         No        Yes
## 0.07230256 0.68075640 0.24694105
##
## ---- pro ----
##
##    Missing         No        Yes
## 0.07007786 0.76974416 0.16017798
##
## ---- diuretic ----
##
##    Missing         No        Yes
## 0.09121246 0.80645161 0.10233593
##
## ---- exang ----
##
##    Missing         No        Yes
## 0.06117909 0.57174638 0.36707453
```

```
##
## ---- slope ----
##
## Downsloping        Flat     Missing   Upsloping
##  0.06674082  0.37152392  0.34371524  0.21802002
##
## ---- ladprox ----
##
##         1         2   Missing
## 0.4961068 0.2413793 0.2625139
##
## ---- laddist ----
##
##         1         2   Missing
## 0.5461624 0.1802002 0.2736374
##
## ---- cxmain ----
##
##         1         2   Missing
## 0.5194661 0.2191324 0.2614016
##
## ---- om1 ----
##
##         1         2   Missing
## 0.5750834 0.1234705 0.3014461
##
## ---- rcaprox ----
##
##         1         2   Missing
## 0.4783092 0.2491657 0.2725250
##
## ---- rcadist ----
##
##         1         2   Missing
## 0.5795328 0.1201335 0.3003337
##
## ---- location ----
##
##     Cleveland       Hungary   switzerland VA Long Beach
##     0.3136819     0.3270300     0.1368187     0.2224694
##
## ---- ca_missing ----
##
##        0        1
## 0.323693 0.676307
##
## ---- thal_missing ----
##
##         0         1
## 0.4593993 0.5406007
##
## ---- thaltime_missing ----
##
##         0         1
```

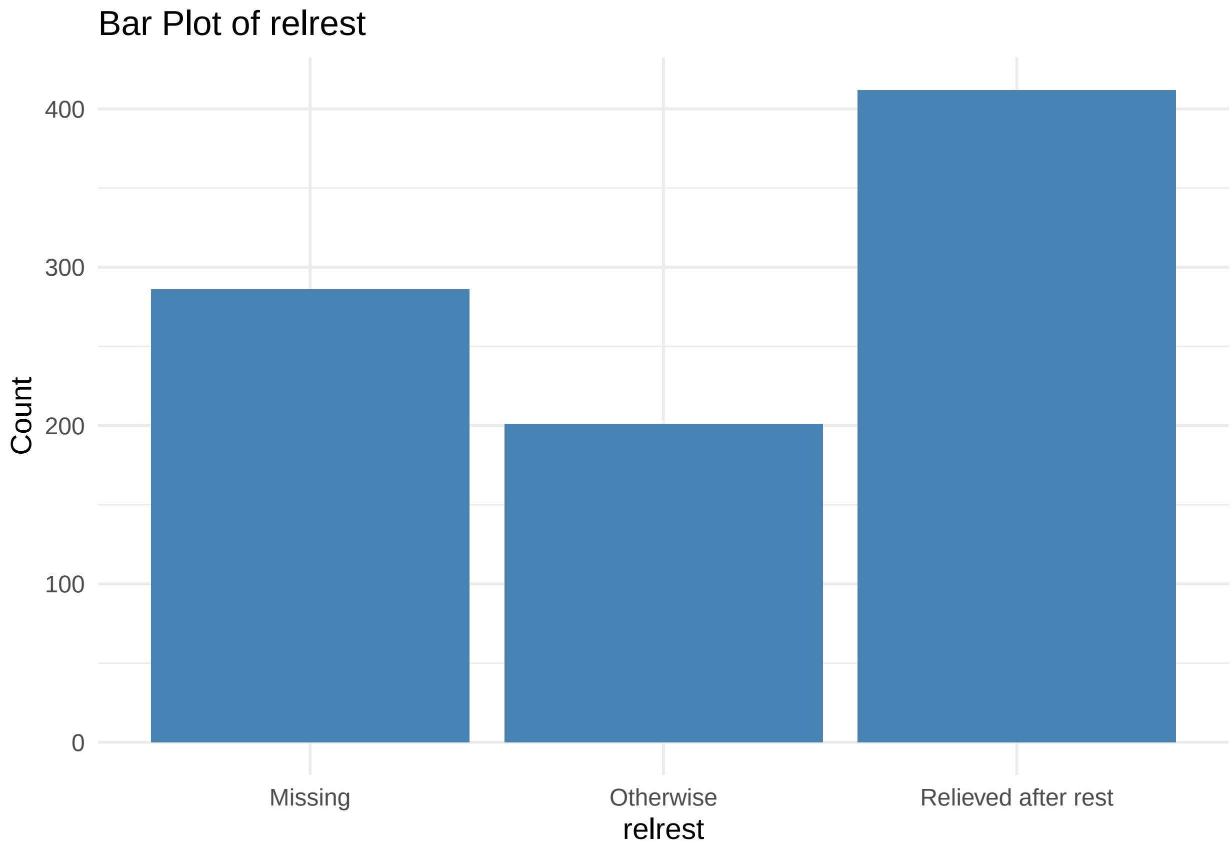
```
## 0.4961068 0.5038932
##
## ---- smoke_profile ----
##
##           heavy_long         heavy_medium          heavy_short
##          0.073414905          0.027808676          0.003337041
##      heavy_very_long           light_long         light_medium
##          0.012235818          0.013348165          0.015572859
##          light_short      light_very_long              missing
##          0.007786429          0.003337041          0.483870968
##        moderate_long      moderate_medium       moderate_short
##          0.087875417          0.032258065          0.016685206
##   moderate_very_long                  non      very_heavy_long
##          0.013348165          0.170189099          0.021134594
##    very_heavy_medium     very_heavy_short very_heavy_very_long
##          0.008898776          0.006674082          0.002224694
##
## ---- rldv5_missing ----
##
##          0         1
## 0.5272525 0.4727475
```

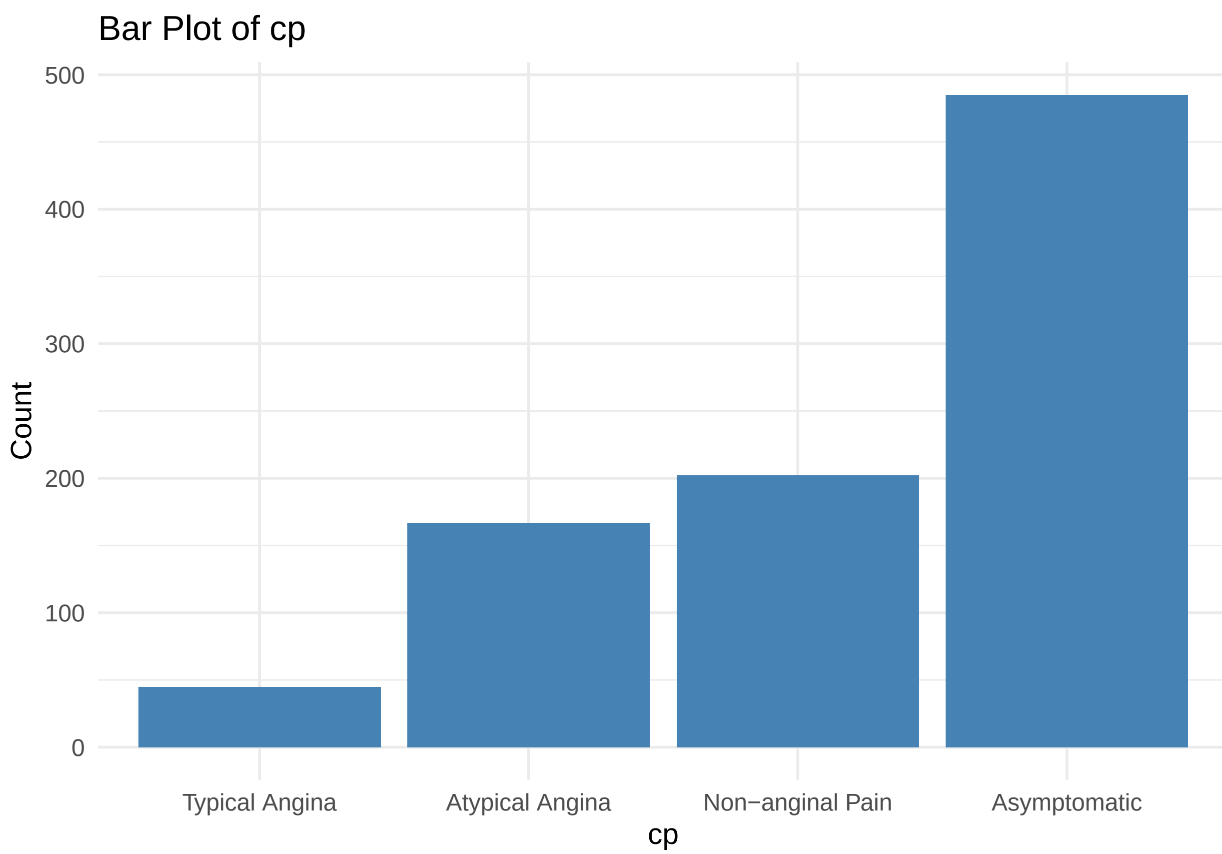


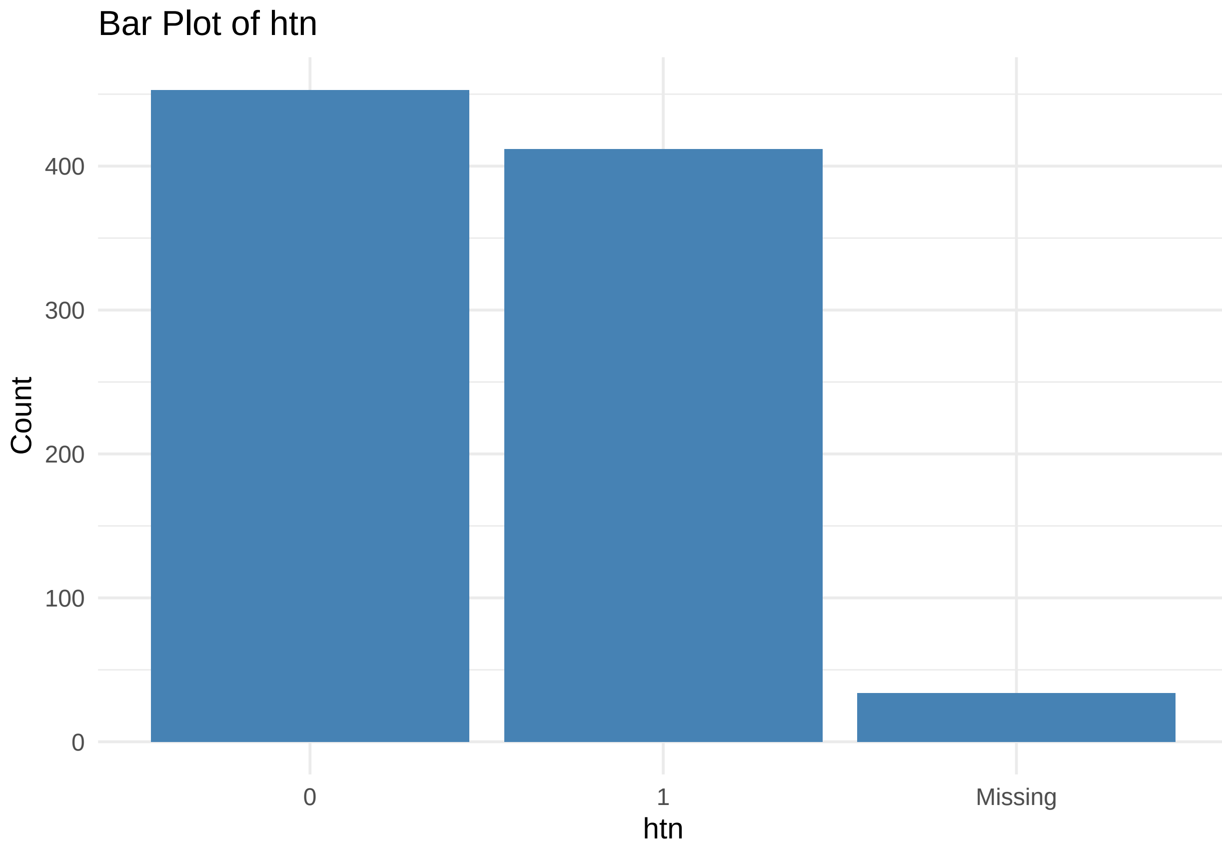

Bar Plot of painloc
Count
400
200
0
Missing
Otherwise
Substernal
painloc

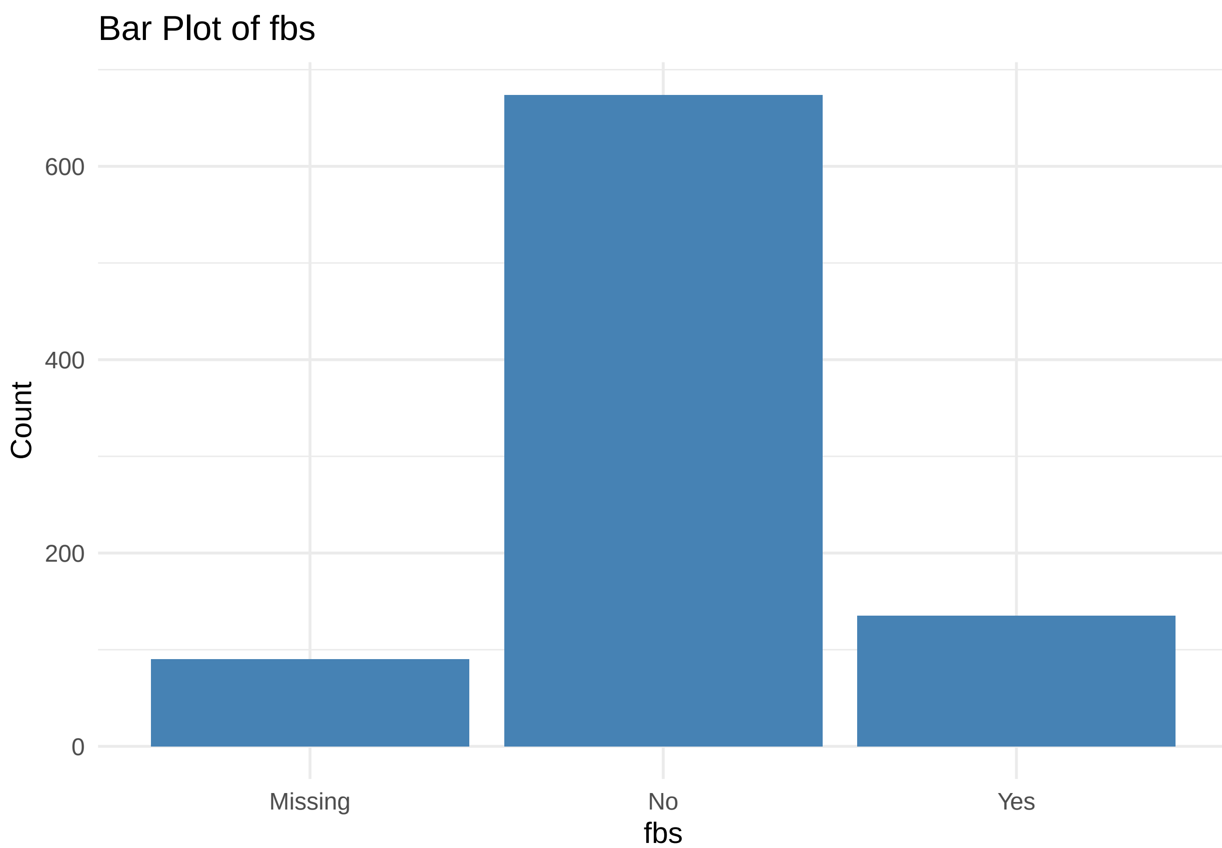


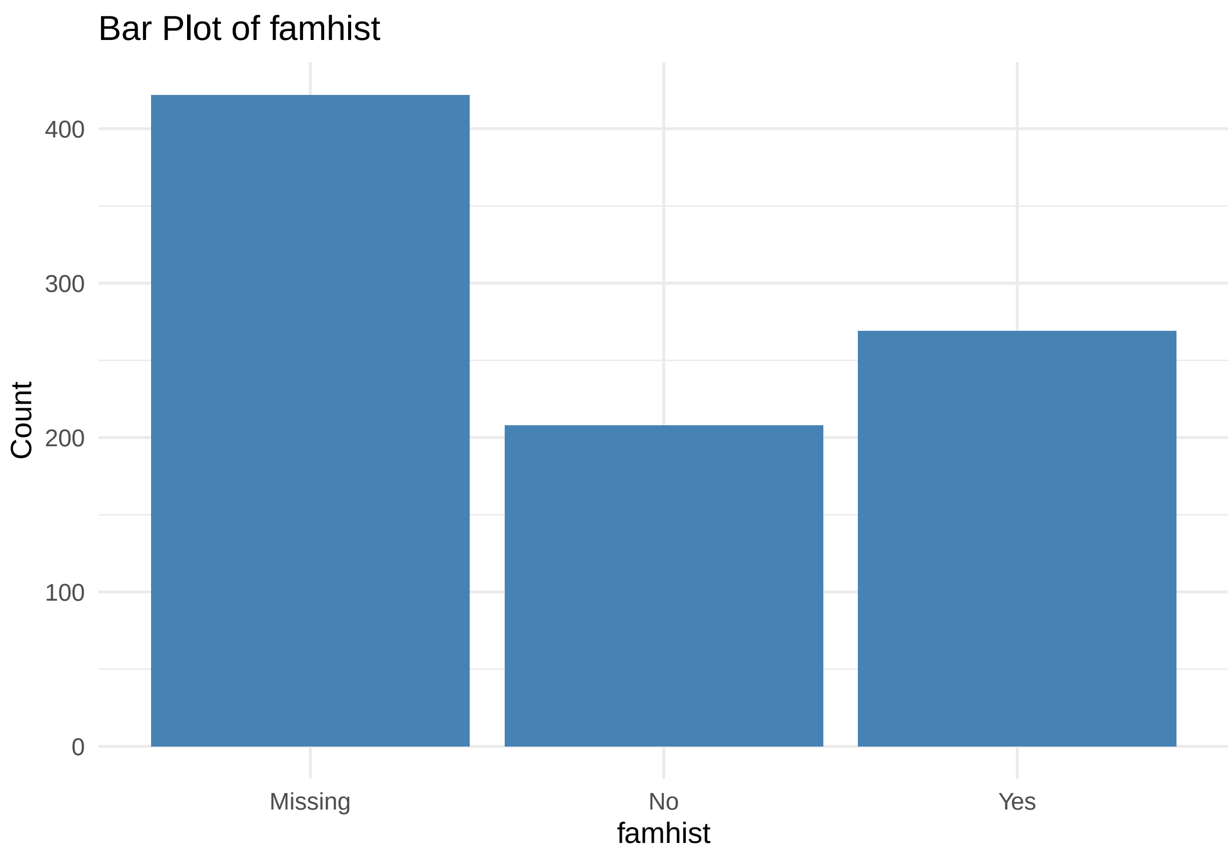
Bar Plot of painexer
Count
0
100
200
300
Missing
Otherwise
Provoked by Exertion
painexer

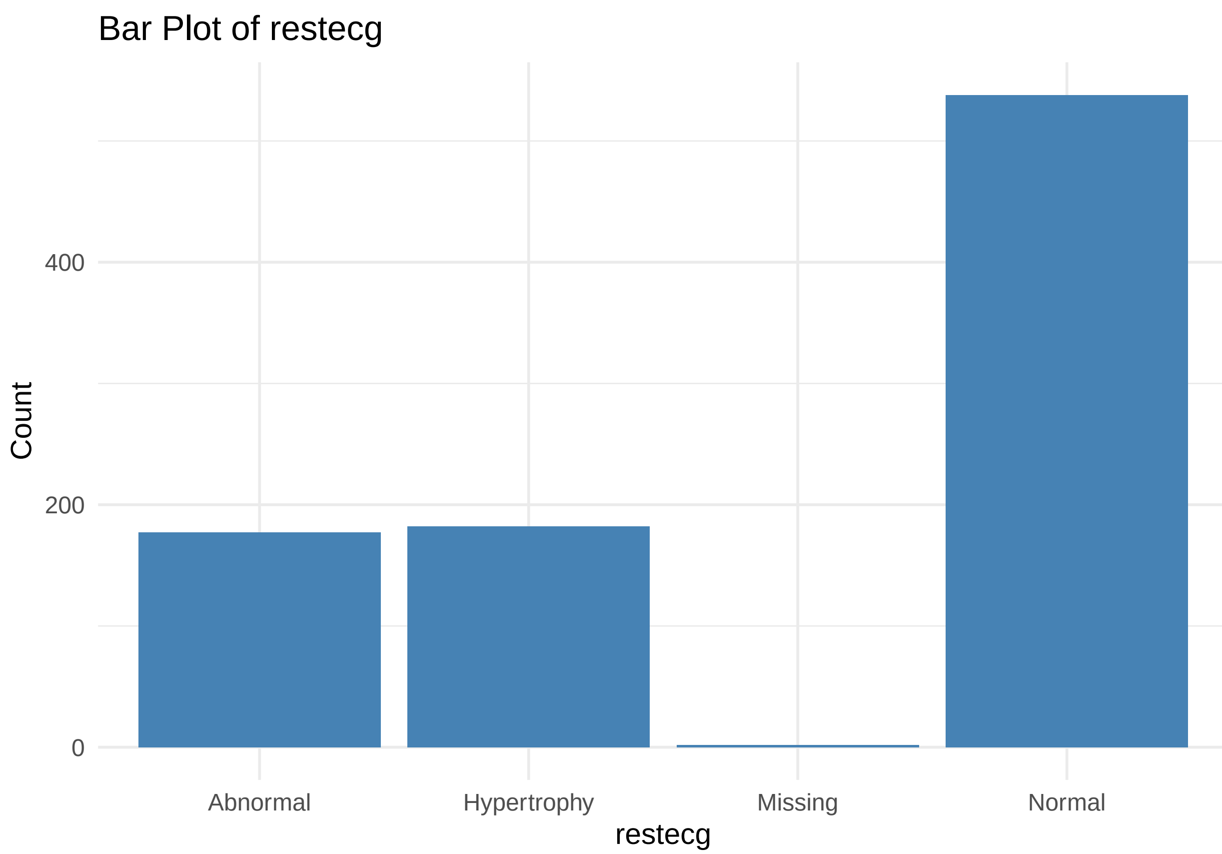

Bar Plot of relrest
400
300
200
100
0
Count
Missing
Otherwise
Relieved after rest
relrest

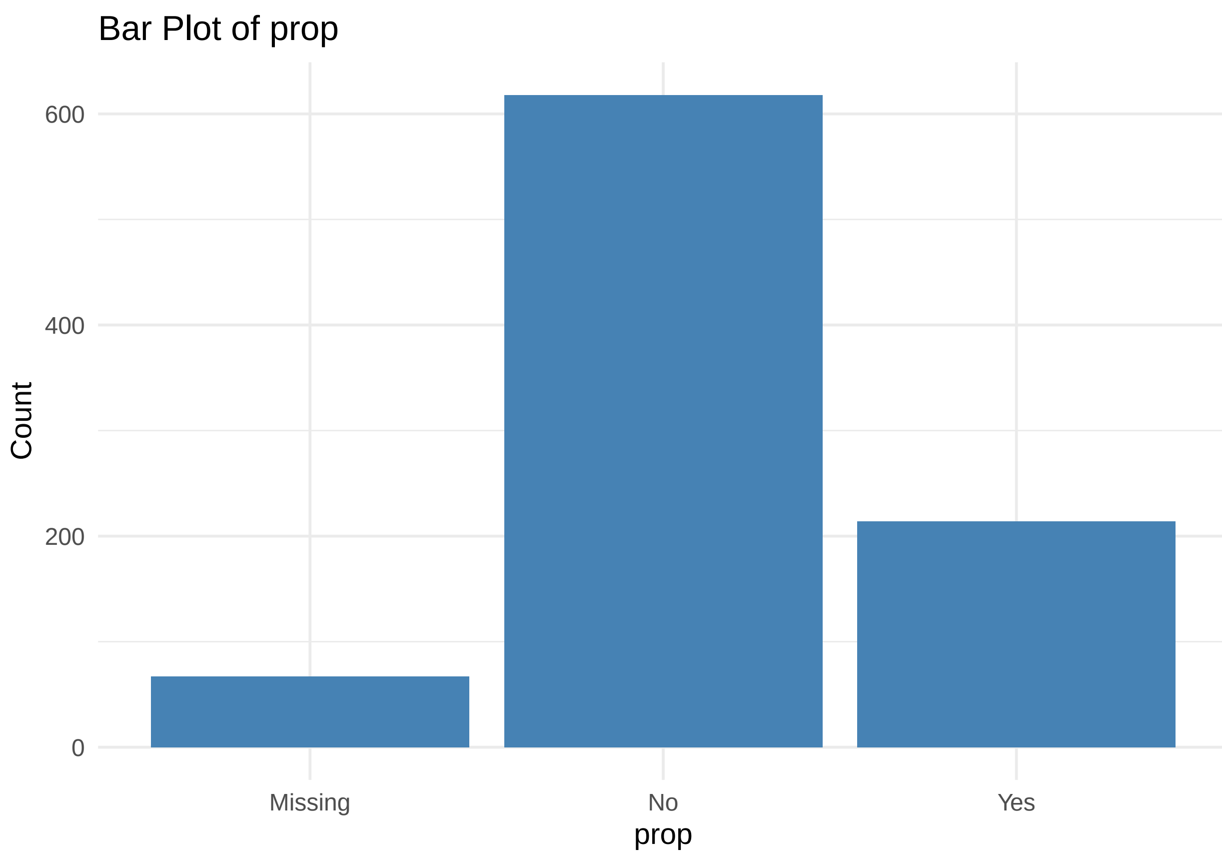

Bar Plot of cp
500
400
300
200
100
0
Count
Typical Angina
Atypical Angina
Non−anginal Pain
Asymptomatic
cp

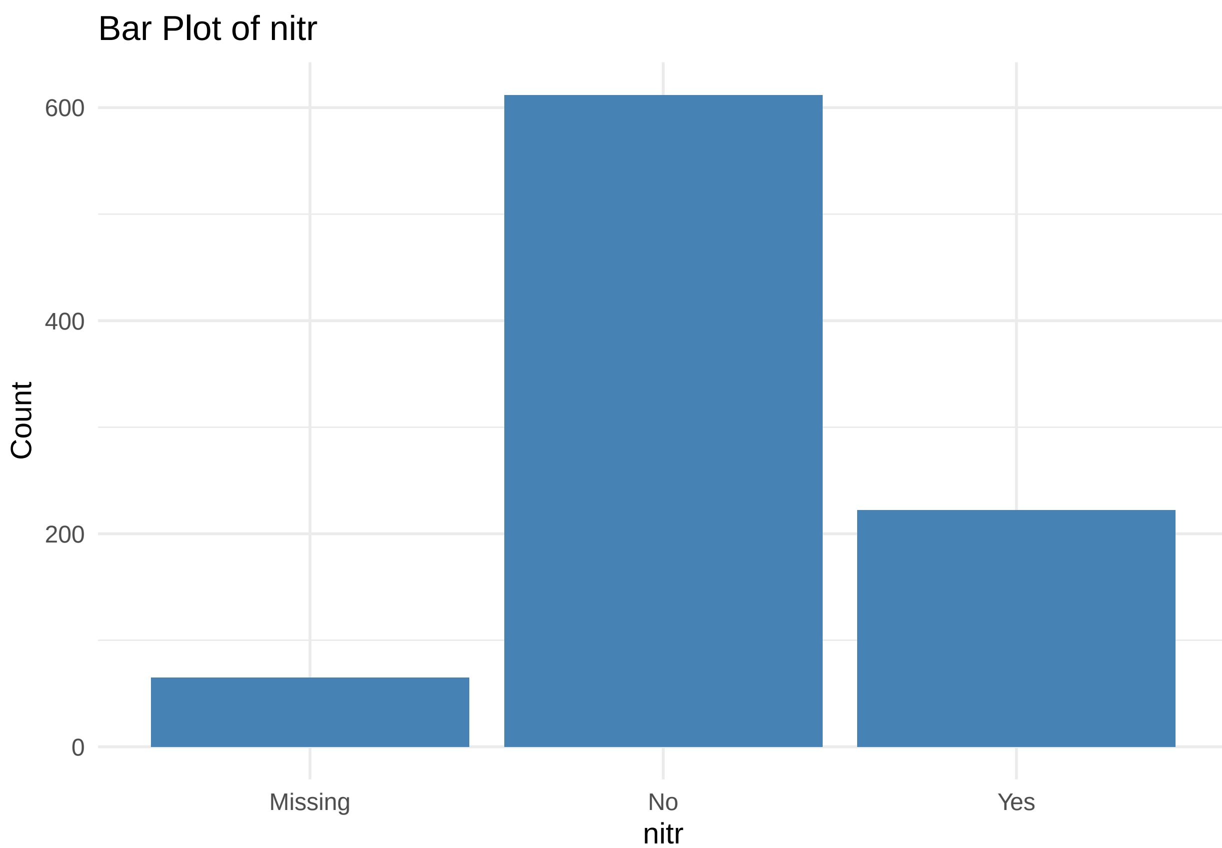

Bar Plot of htn
Count
400
300
200
100
0
0
1
Missing
htn

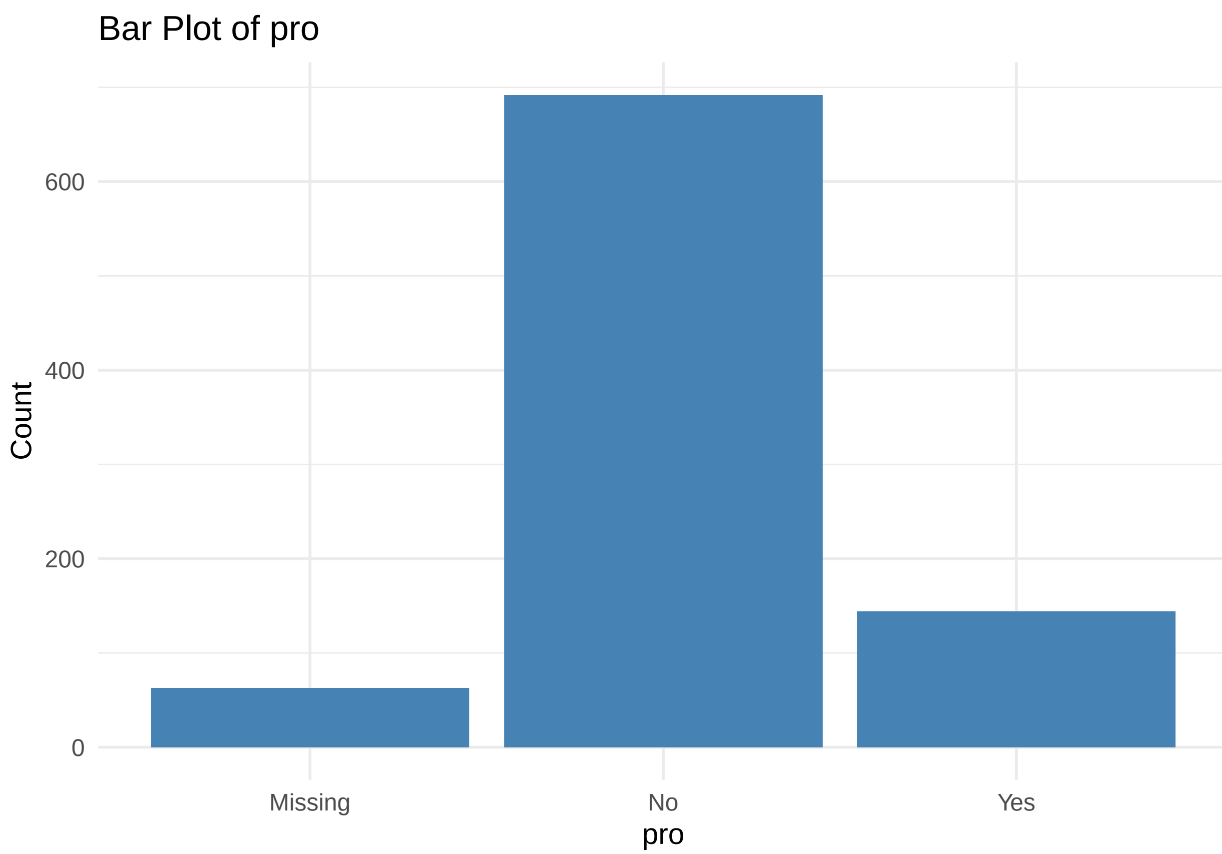

Bar Plot of fbs
Count
600
400
200
0
Missing
No
Yes
fbs

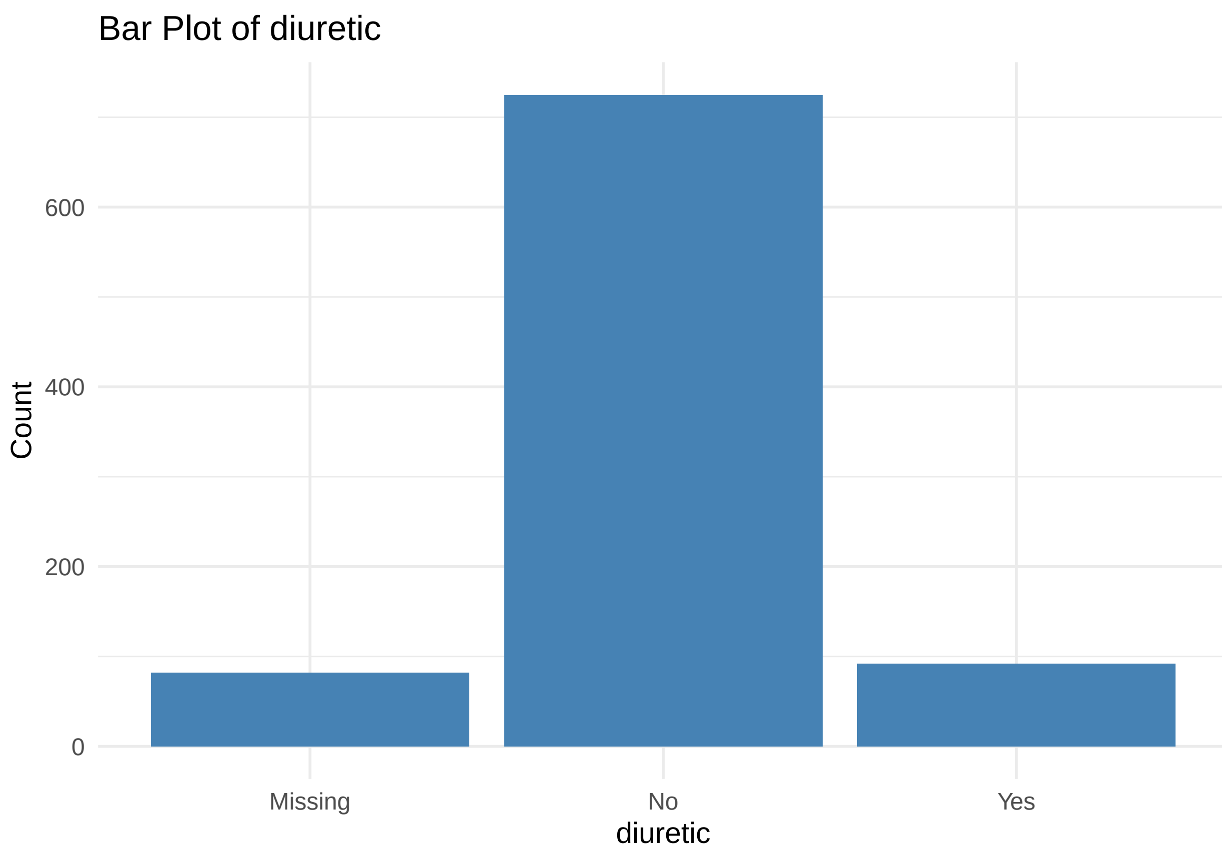

Bar Plot of famhist
Count
400
300
200
100
0
Missing
No
Yes
famhist

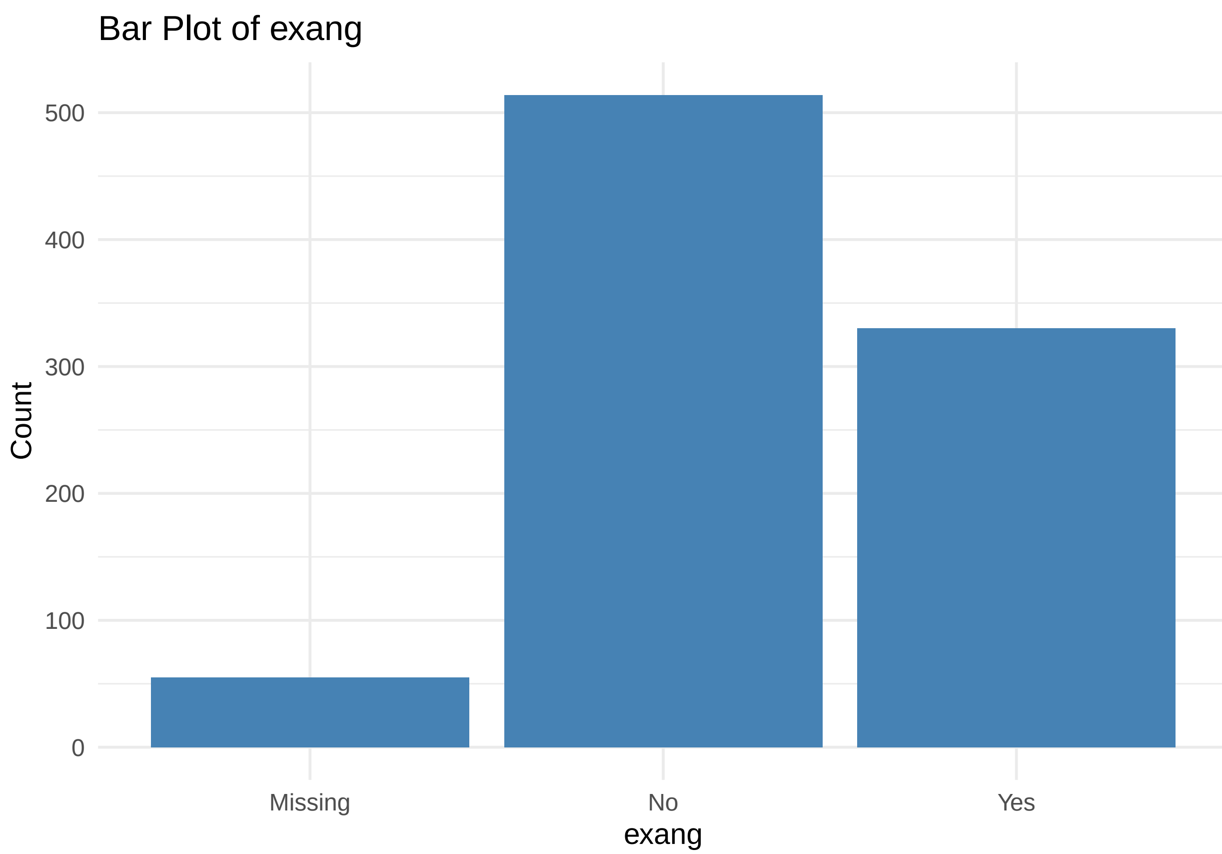

Bar Plot of restecg
Count
400
200
0
Abnormal
Hypertrophy
Missing
Normal
restecg

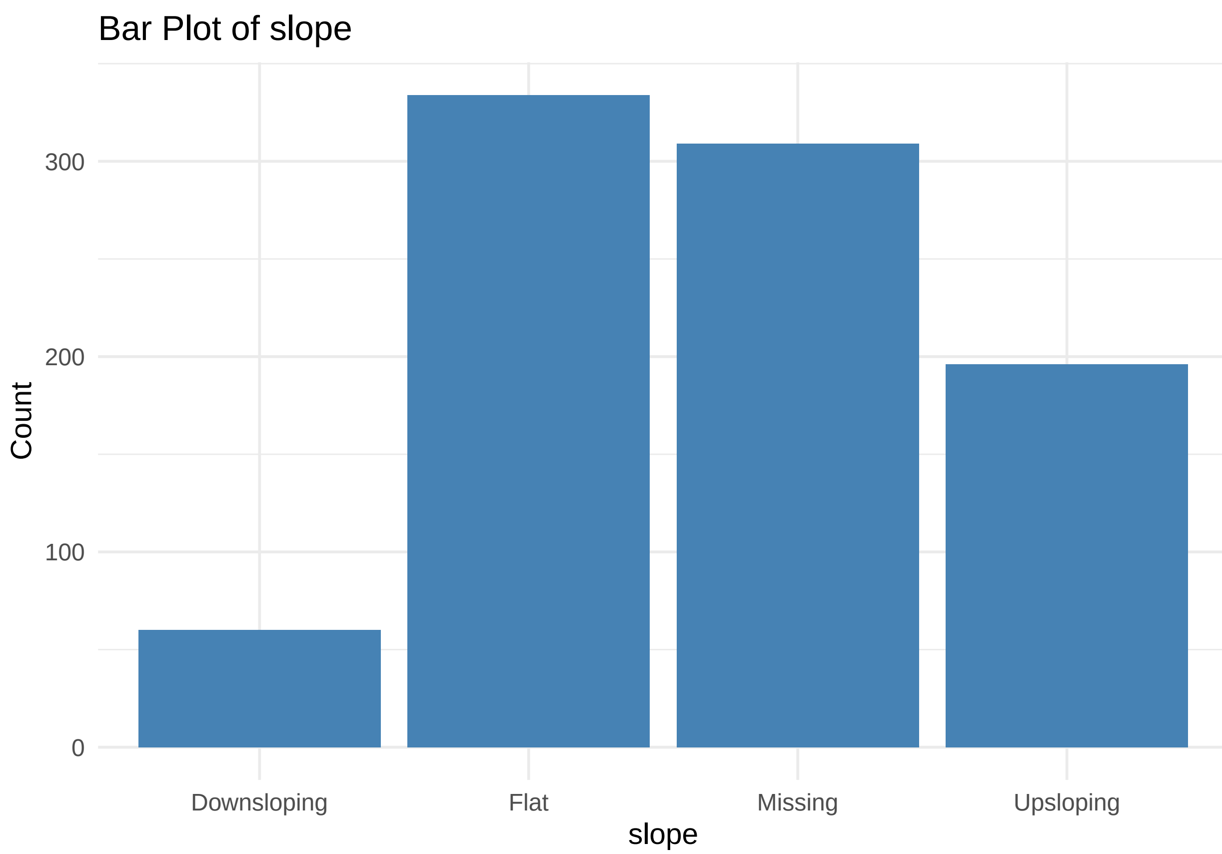

Bar Plot of prop
600
400
200
0
Count
Missing
No
Yes
prop

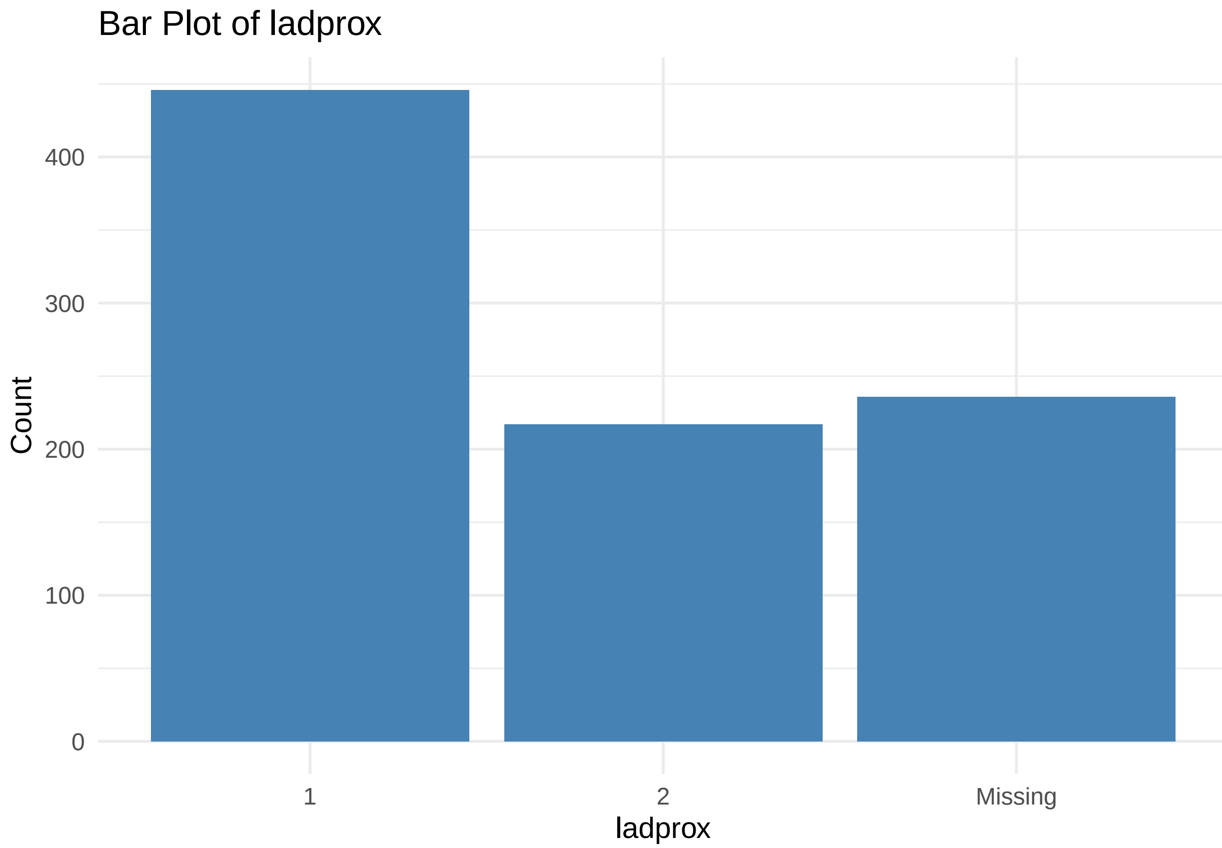

Bar Plot of nitr
Count
600
400
200
0
Missing
No
Yes
nitr

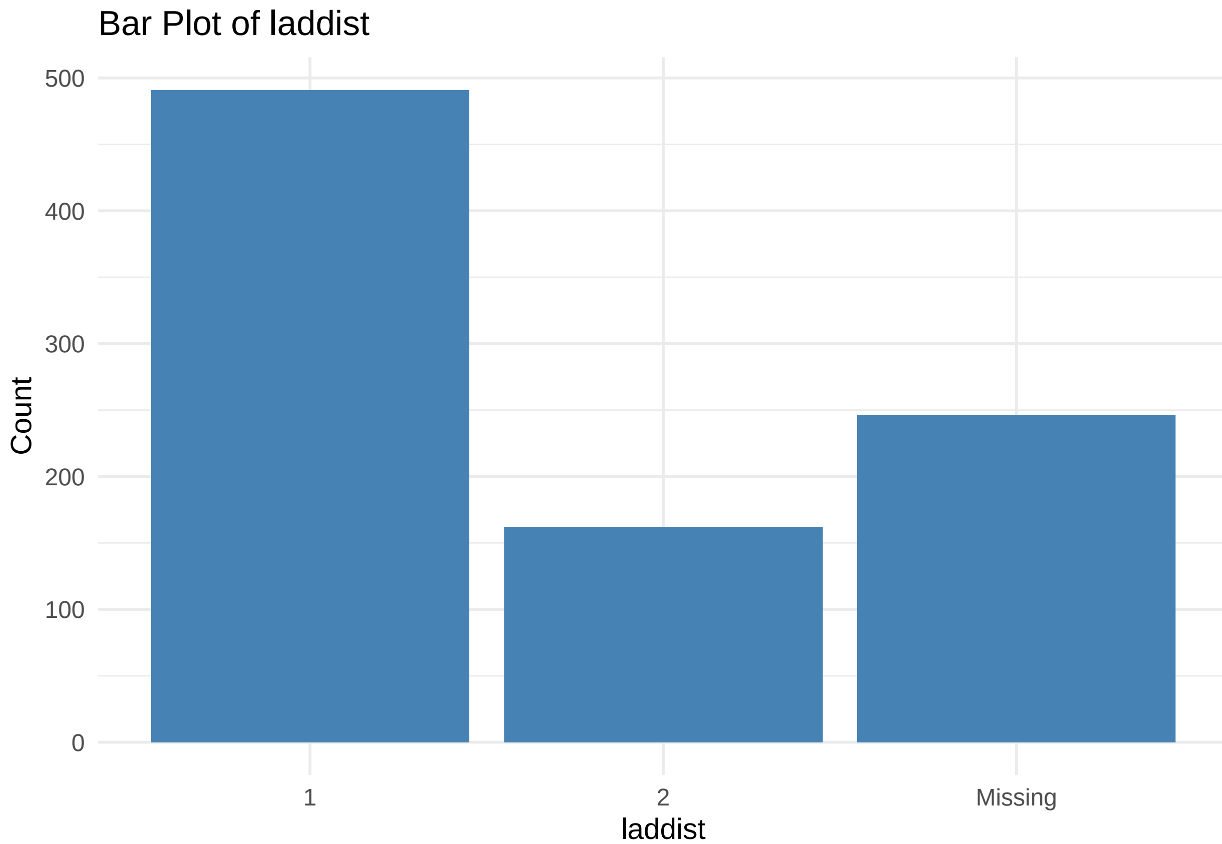

Bar Plot of pro
Count
600
400
200
0
Missing
No
Yes
pro

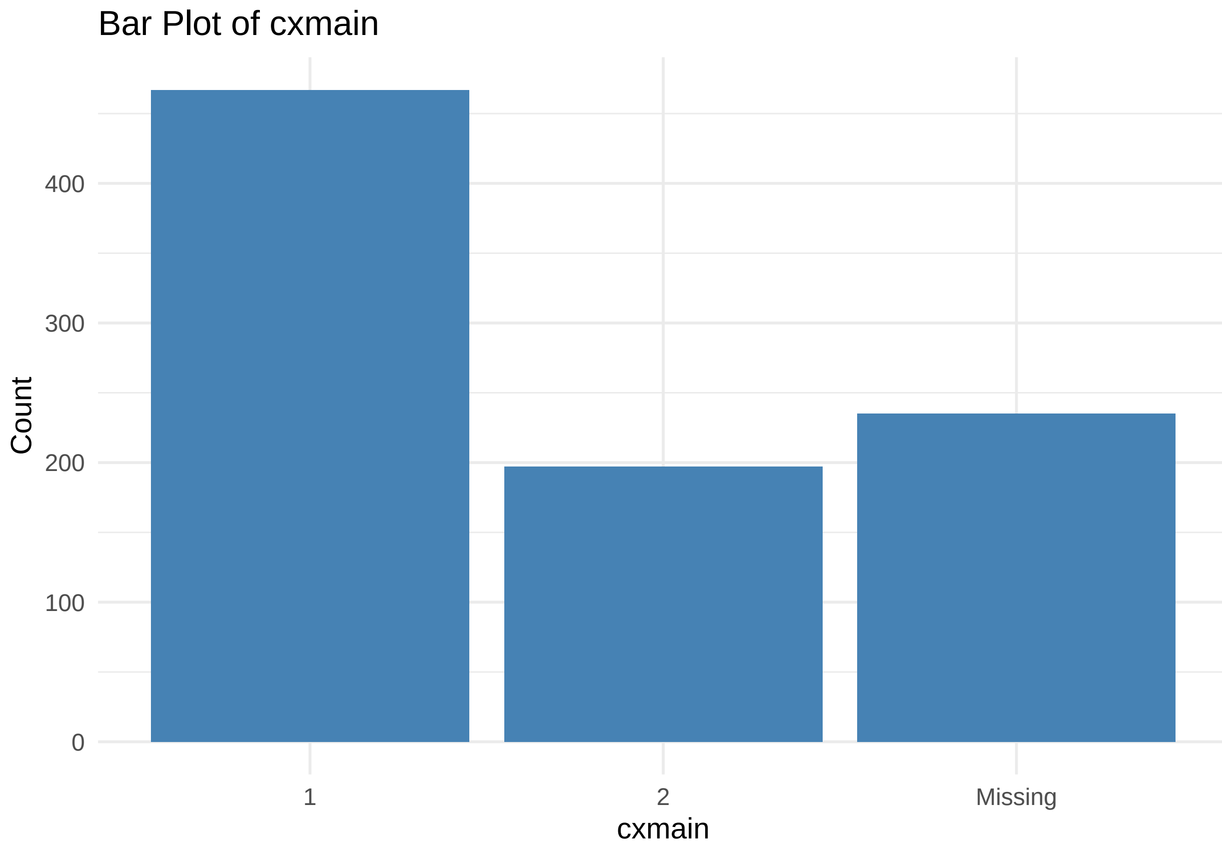

Bar Plot of diuretic
Count
600
400
200
0
Missing
No
Yes
diuretic

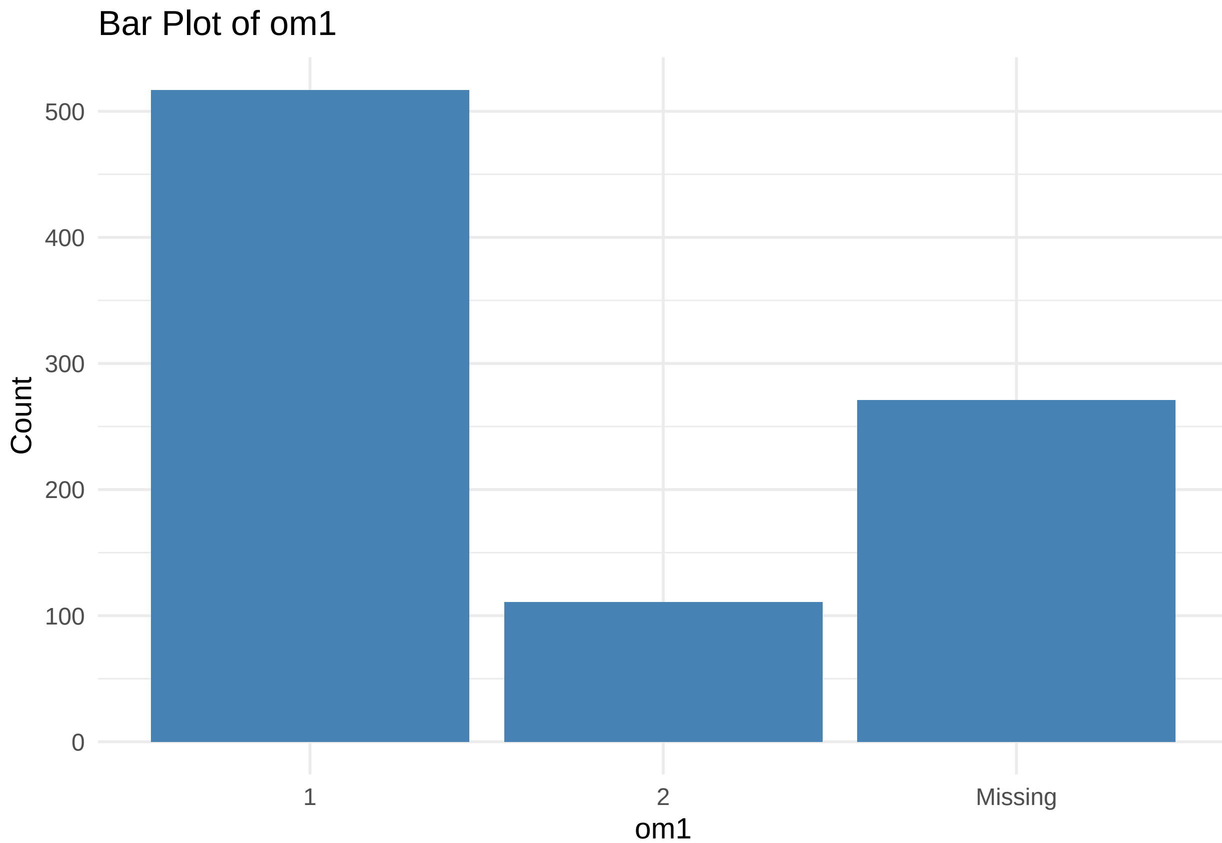

Bar Plot of exang
Count
500
400
300
200
100
0
Missing
No
Yes
exang

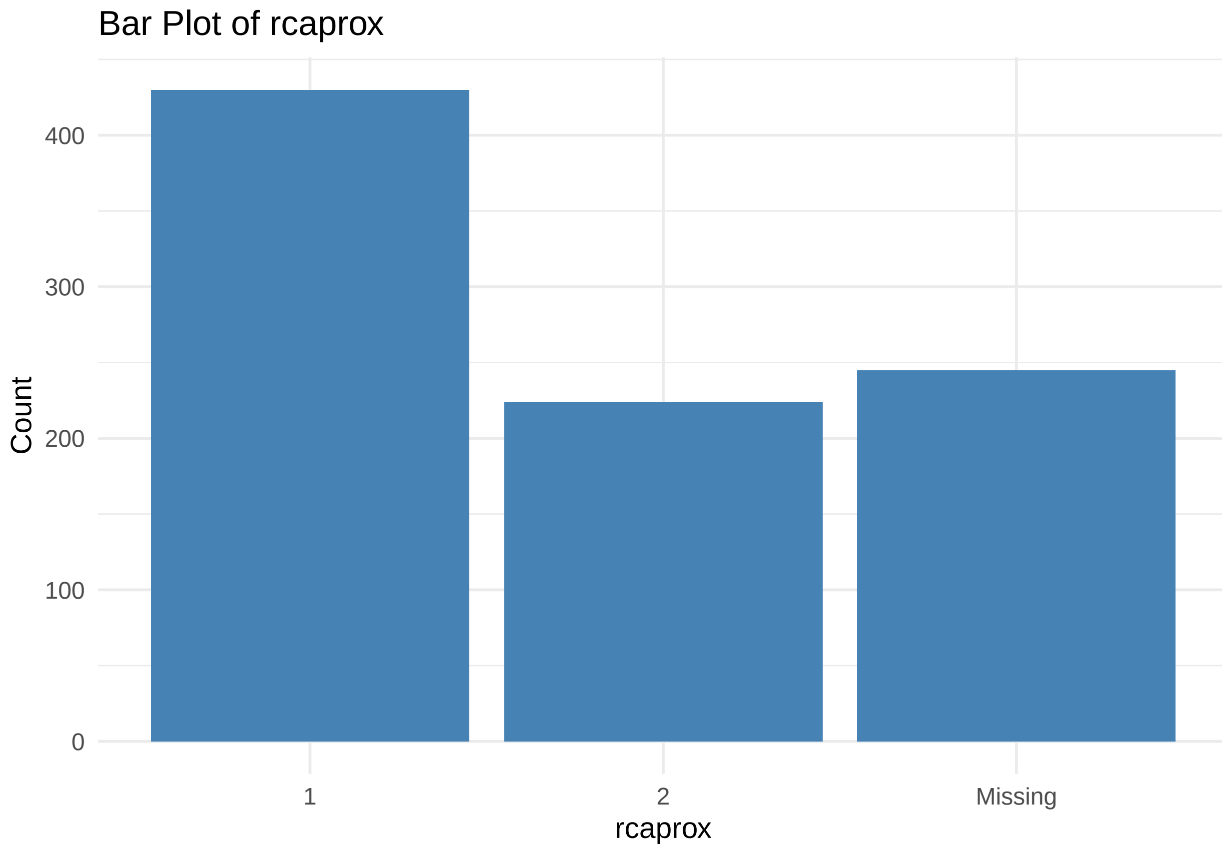

Bar Plot of slope
300
200
100
0
Count
Downsloping
Flat
Missing
Upsloping
slope

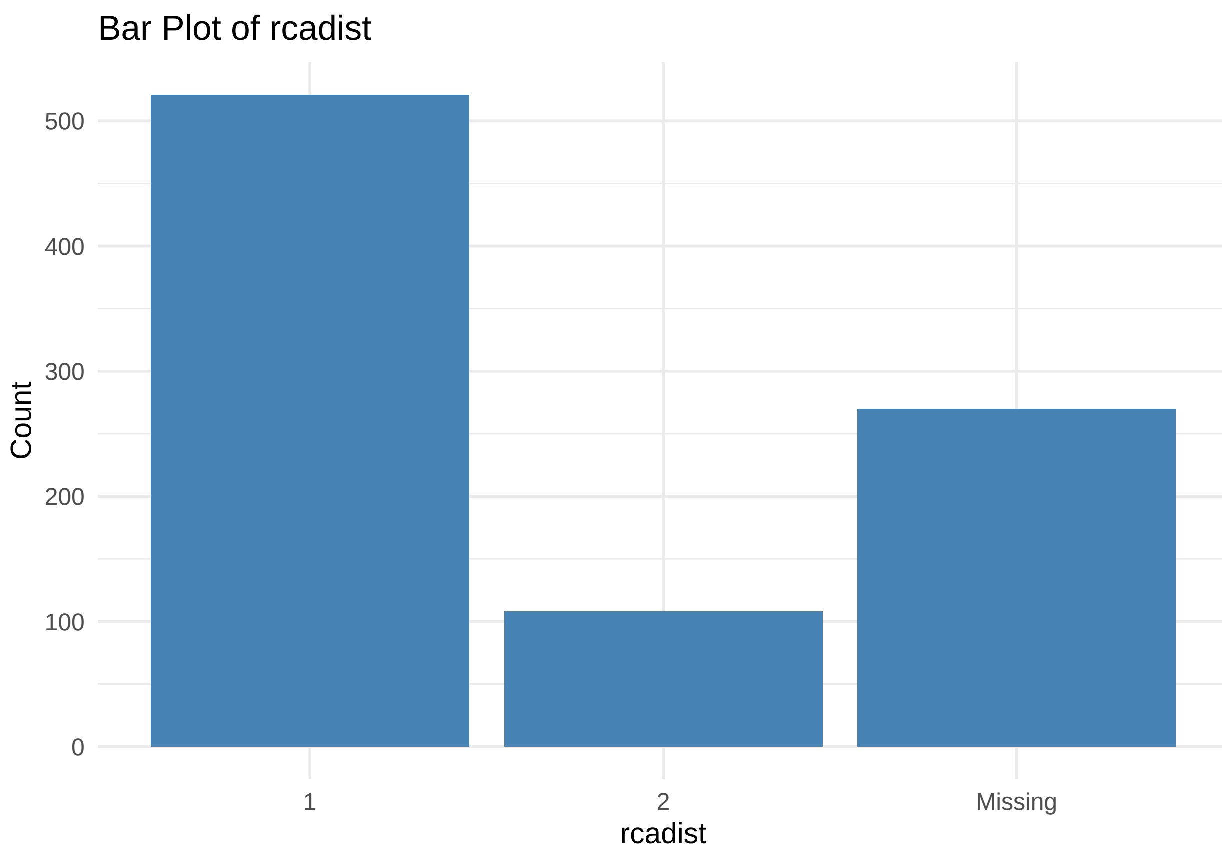

Bar Plot of ladprox
Count
400
300
200
100
0
1
2
Missing
ladprox

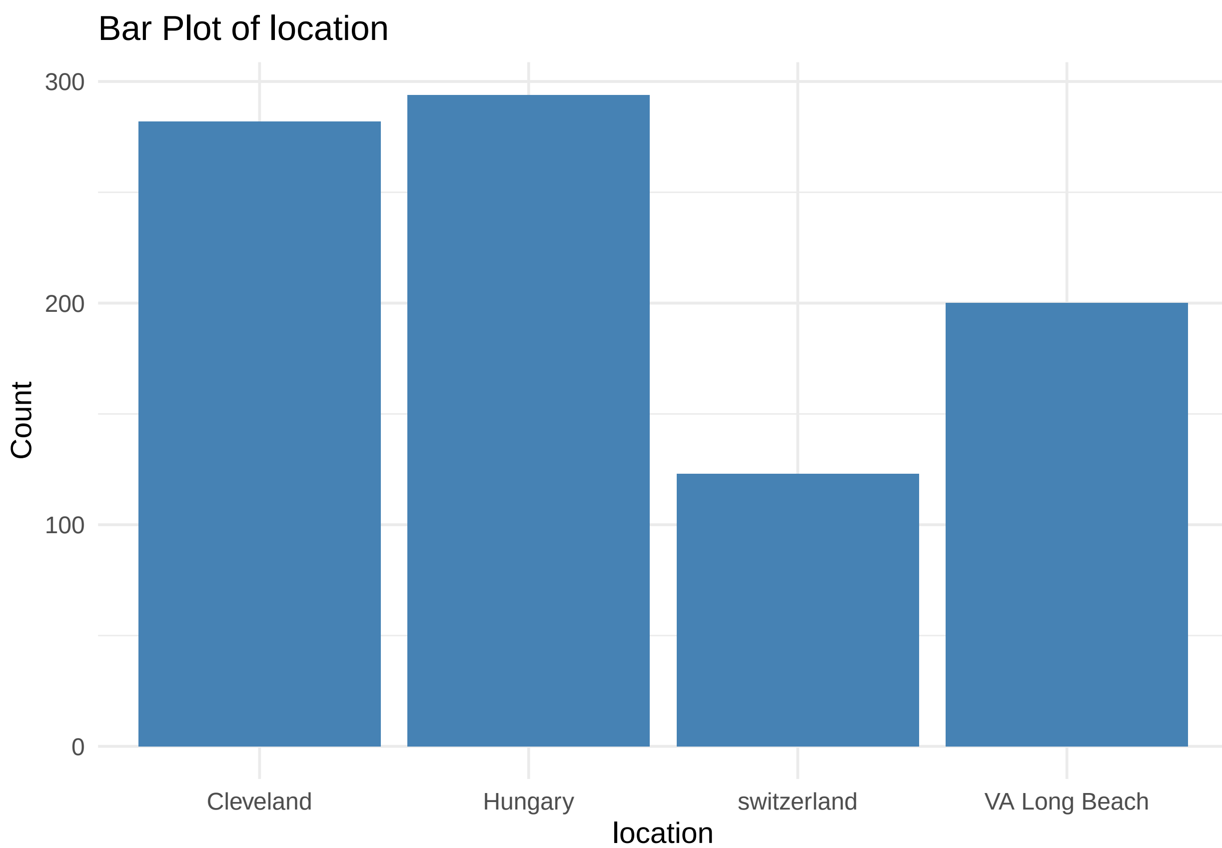

Bar Plot of laddist
Count
500
400
300
200
100
0
1
2
Missing
laddist

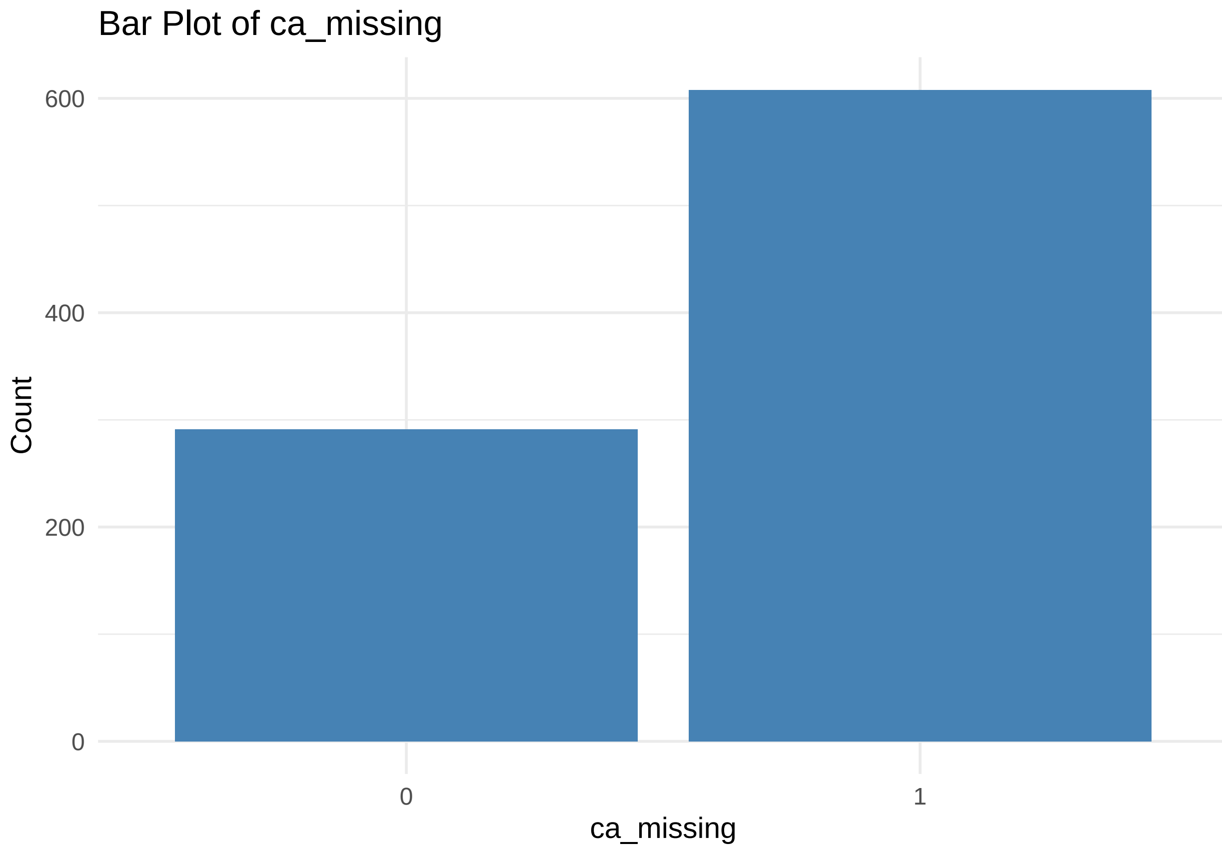

Bar Plot of cxmain
Count
400
300
200
100
0
1
2
Missing
cxmain

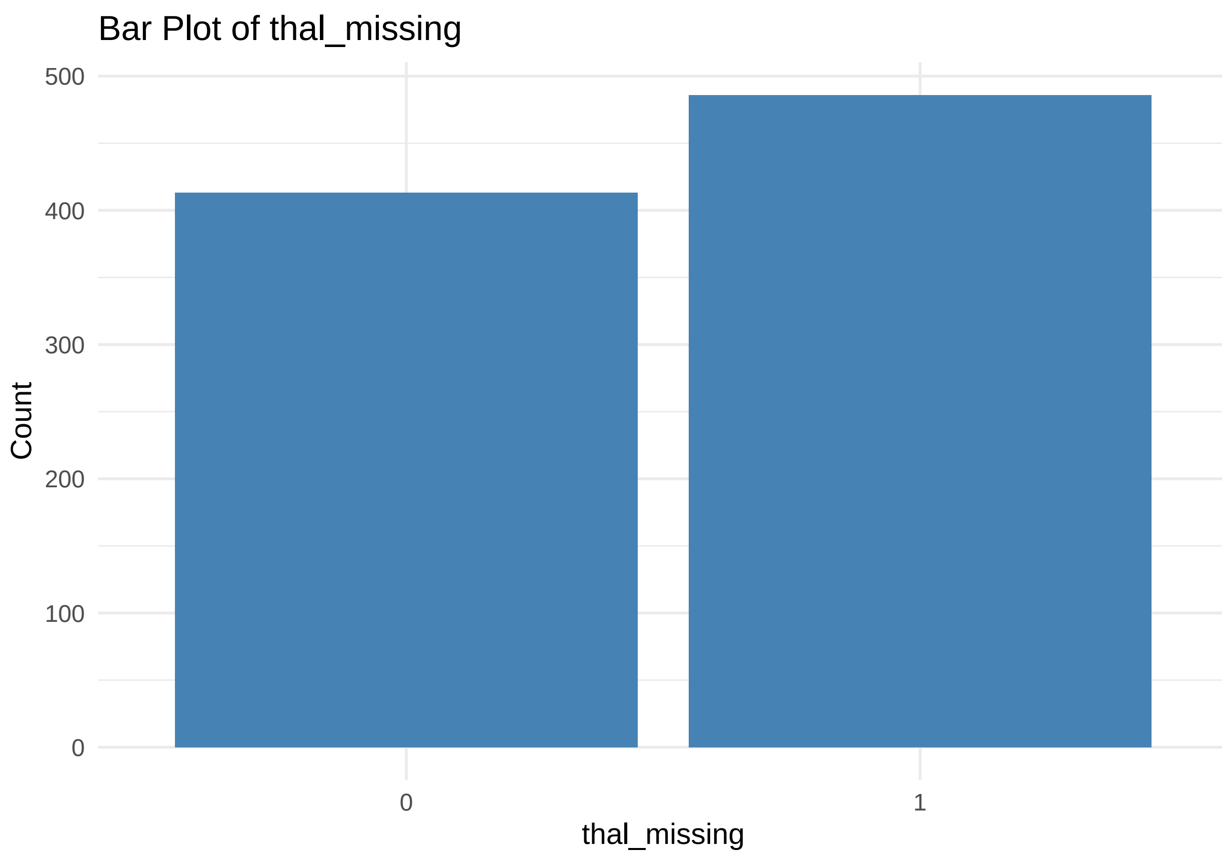

Bar Plot of om1
Count
500
400
300
200
100
0
1
2
Missing
om1


Bar Plot of rcaprox
Count
400
300
200
100
0
1
2
Missing
rcaprox

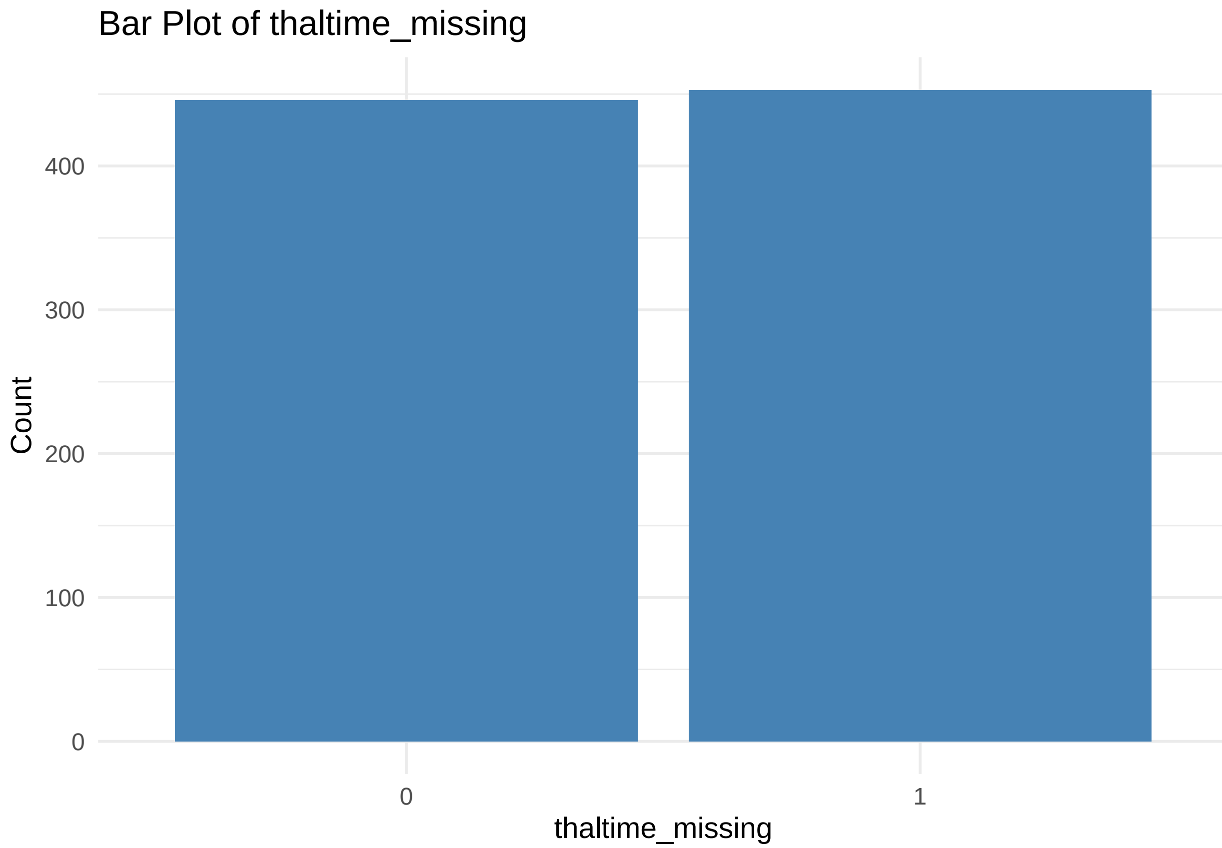

Bar Plot of rcadist
Count
500
400
300
200
100
0
1
2
Missing
rcadist

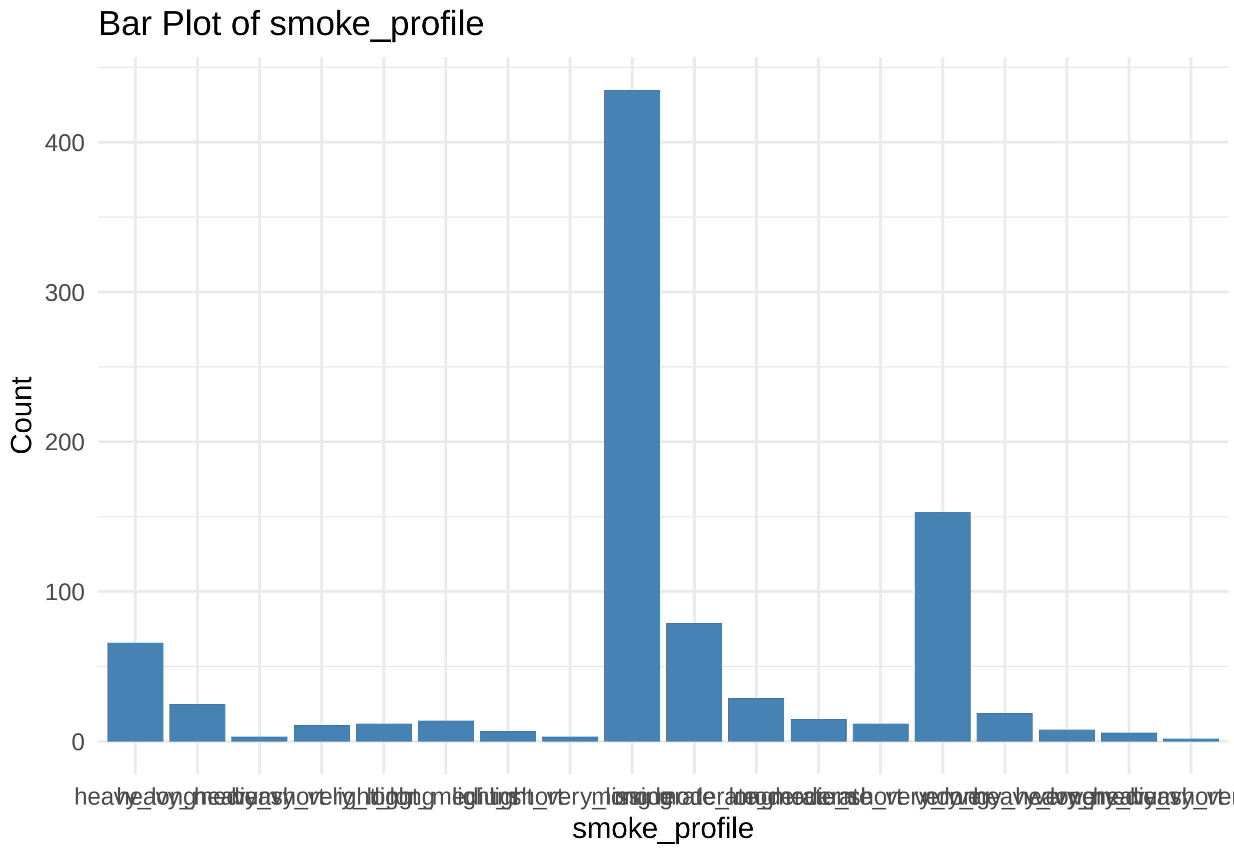

Bar Plot of location
300
200
100
0
Count
Cleveland
Hungary
switzerland
VA Long Beach
location

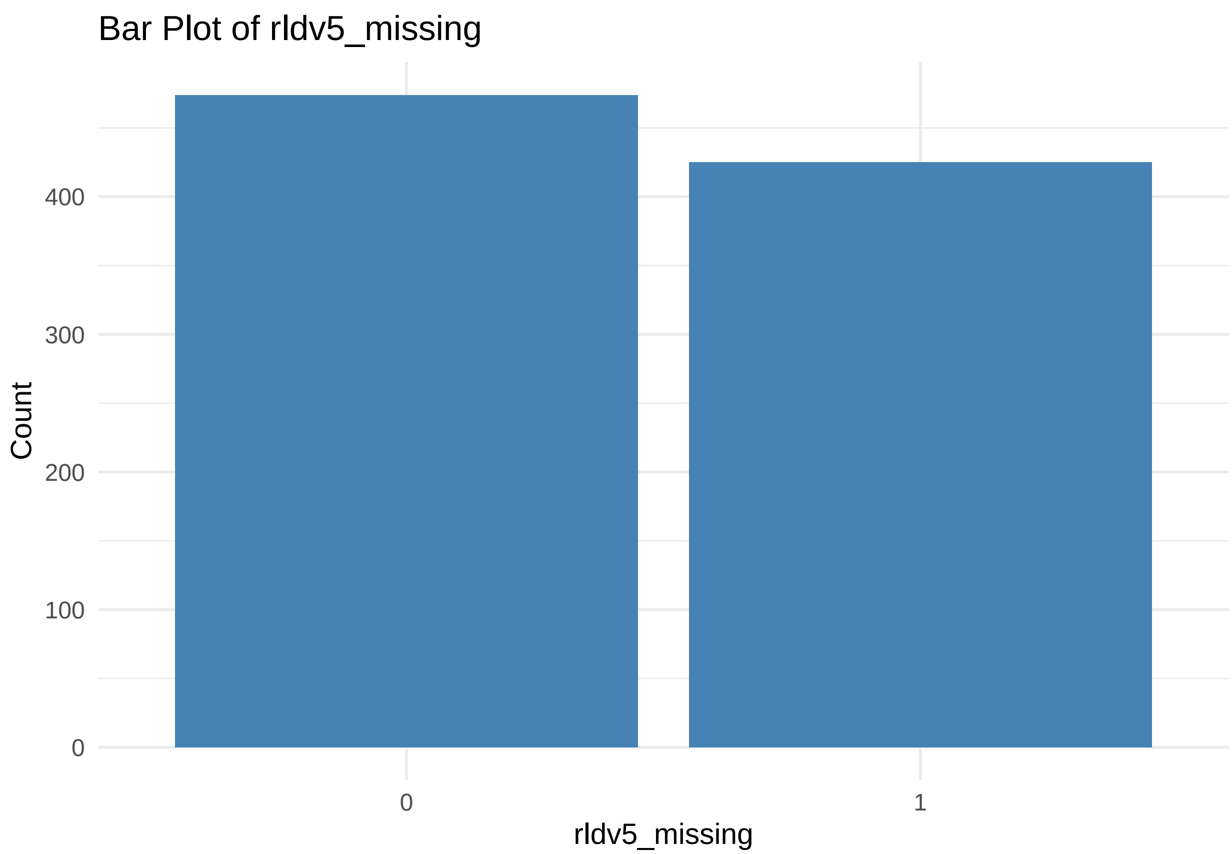

Bar Plot of ca_missing
Count
600
400
200
0
0
1
ca_missing



Bar Plot of thal_missing
500
400
300
200
100
0
Count
0
1
thal_missing


Bar Plot of thaltime_missing
Count
400
300
200
100
0
0
1
thaltime_missing


Bar Plot of smoke_profile
Count
smoke_profile
0
100
200
300
400



Feature Types

```
X_df %>%
  summarise(
    numeric = sum(sapply(., is.numeric)),
    categorical = sum(sapply(., is.factor))
  )
```

```
## # A tibble: 1 x 2
##   numeric categorical
##     <int>       <int>
## 1      29          26
```

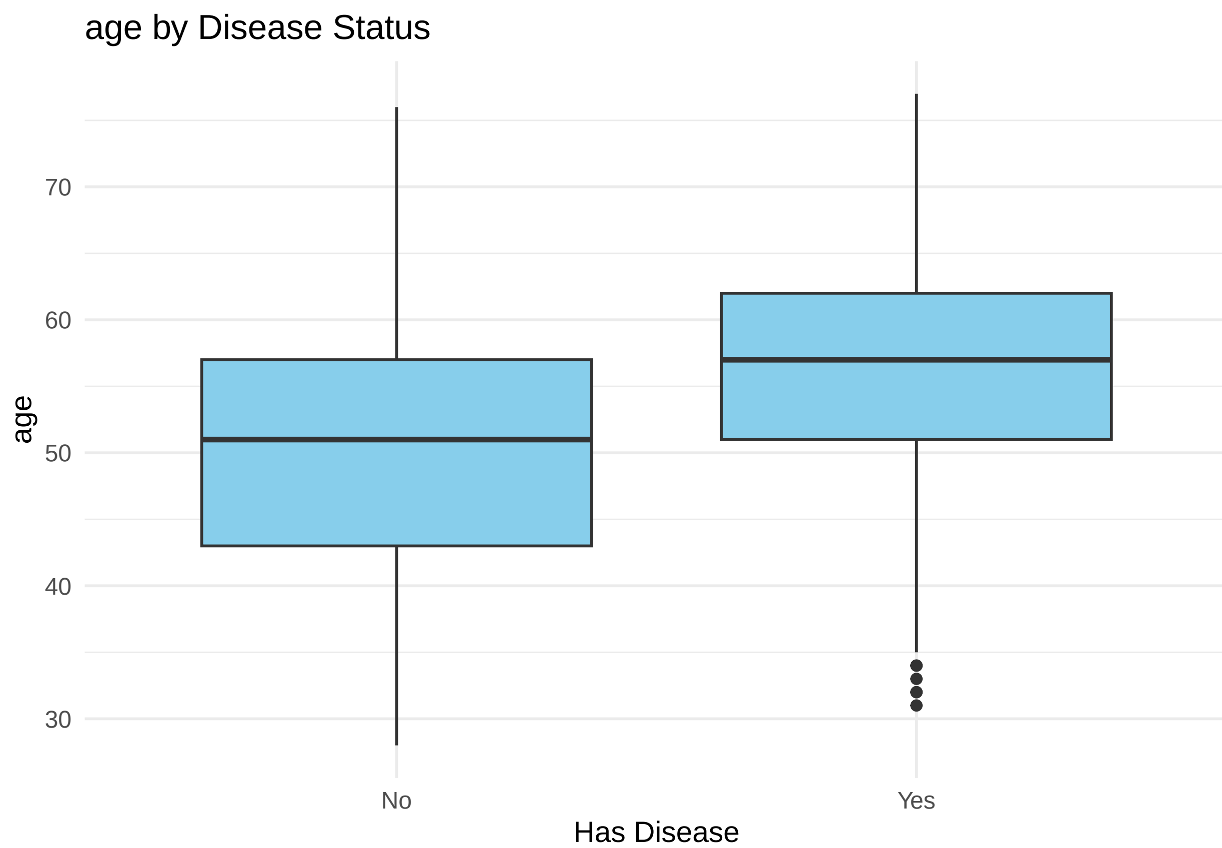
Bivariate Analysis

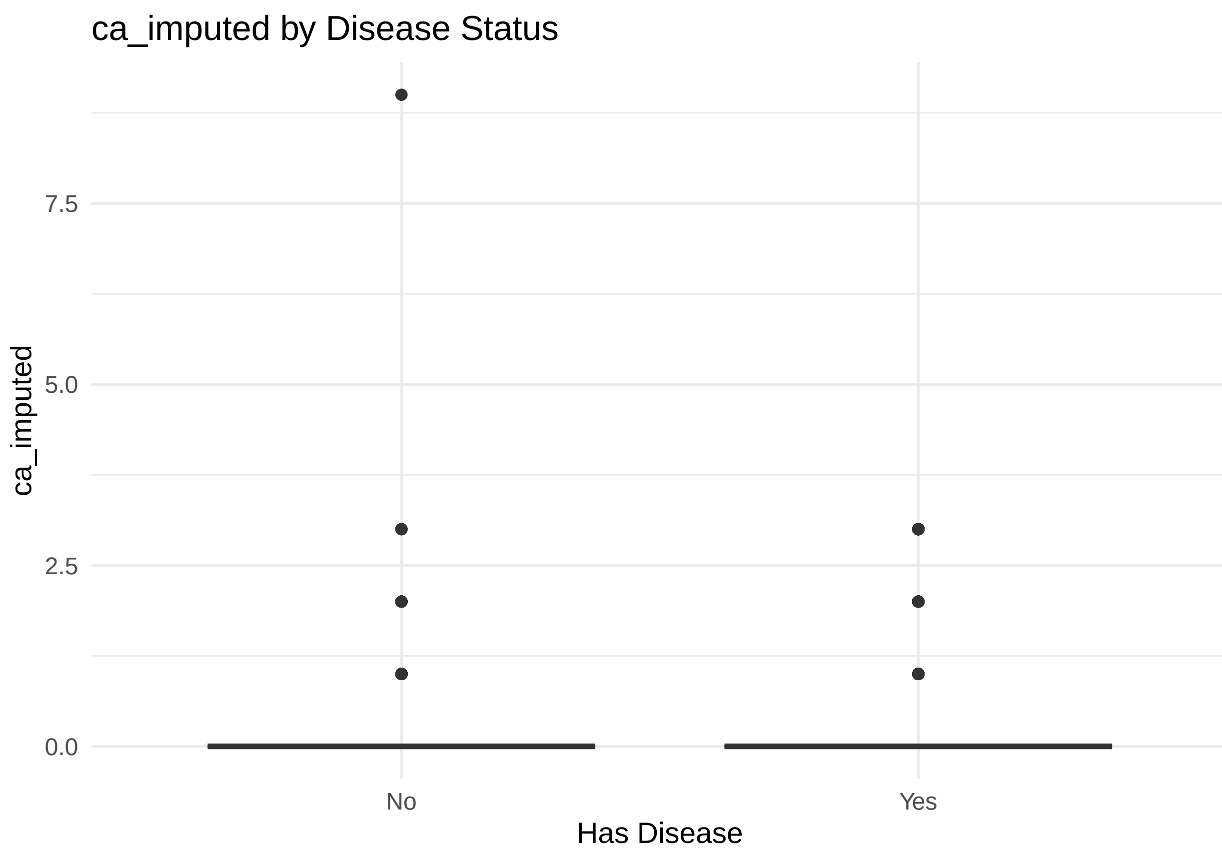
Numerical Vars

Box Plots

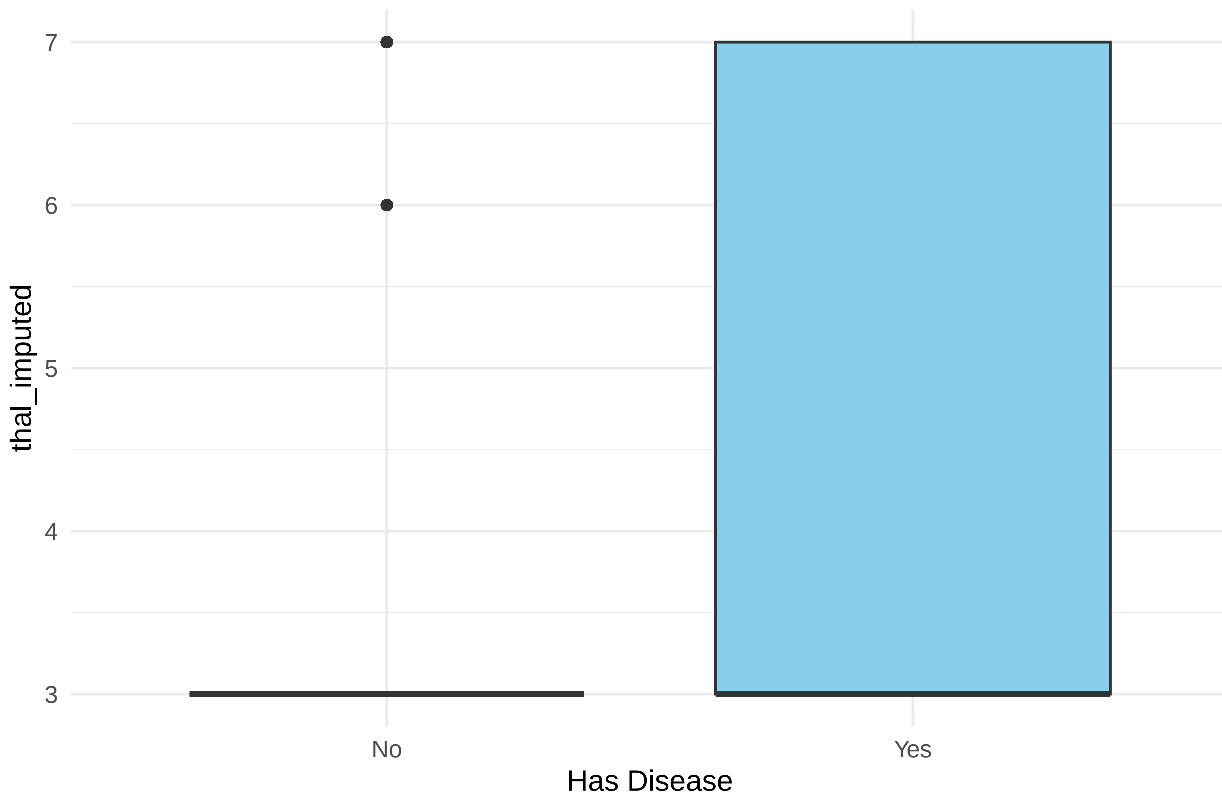
```
plot_bivariate_num <- function(var) {
  ggplot(df, aes(x = factor(has_disease), y = .data[[var]])) +
    geom_boxplot(fill = "skyblue") +
    labs(title = paste(var, "by Disease Status"), x = "Has Disease", y = var) +
    theme_minimal()
}
```

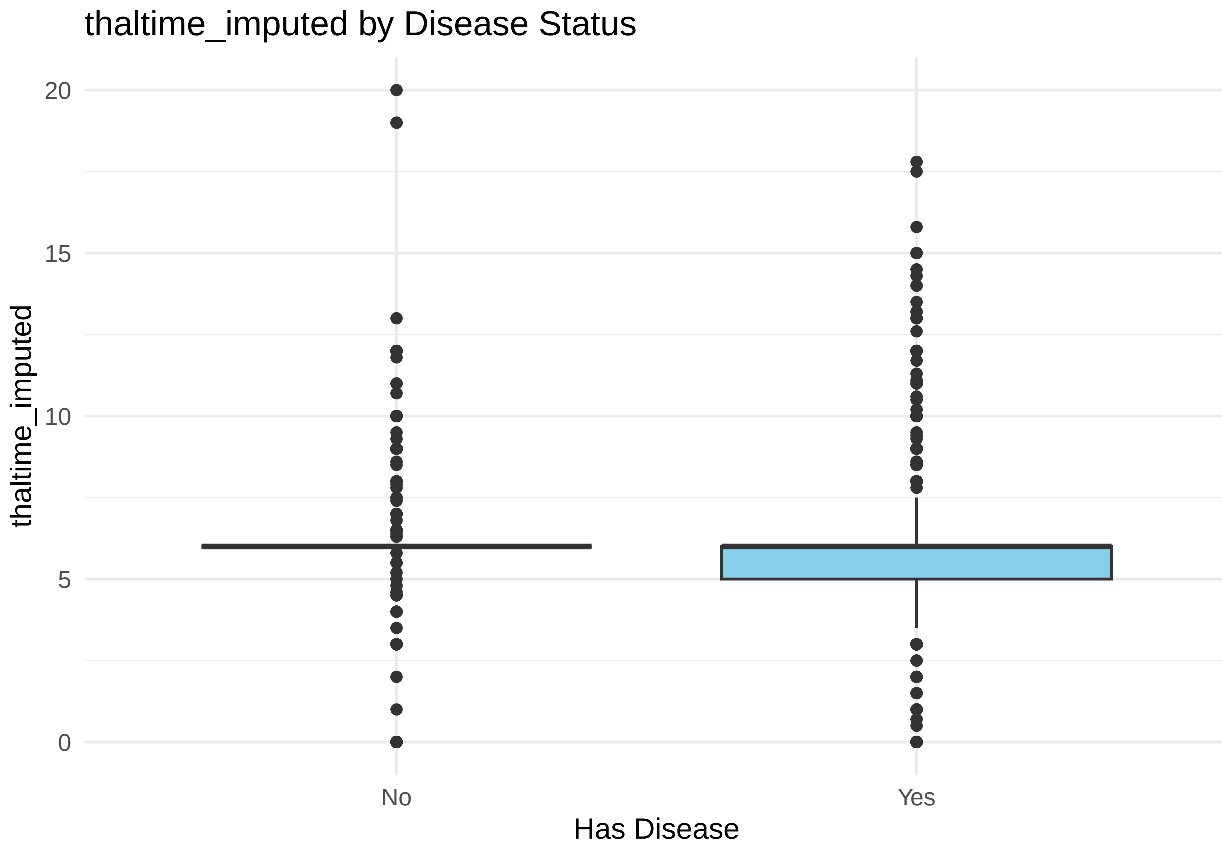
```r
for (var in num_vars){
print(plot_bivariate_num(var))
}
```

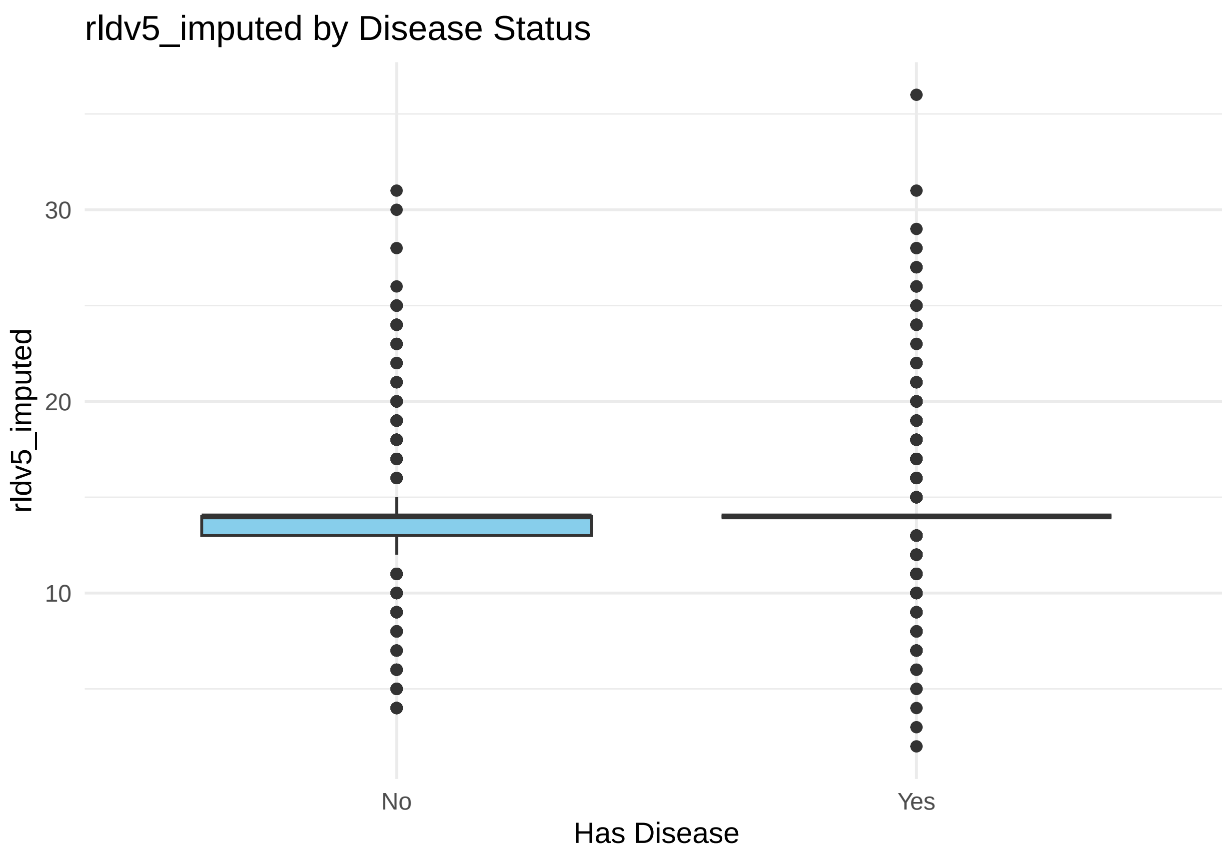


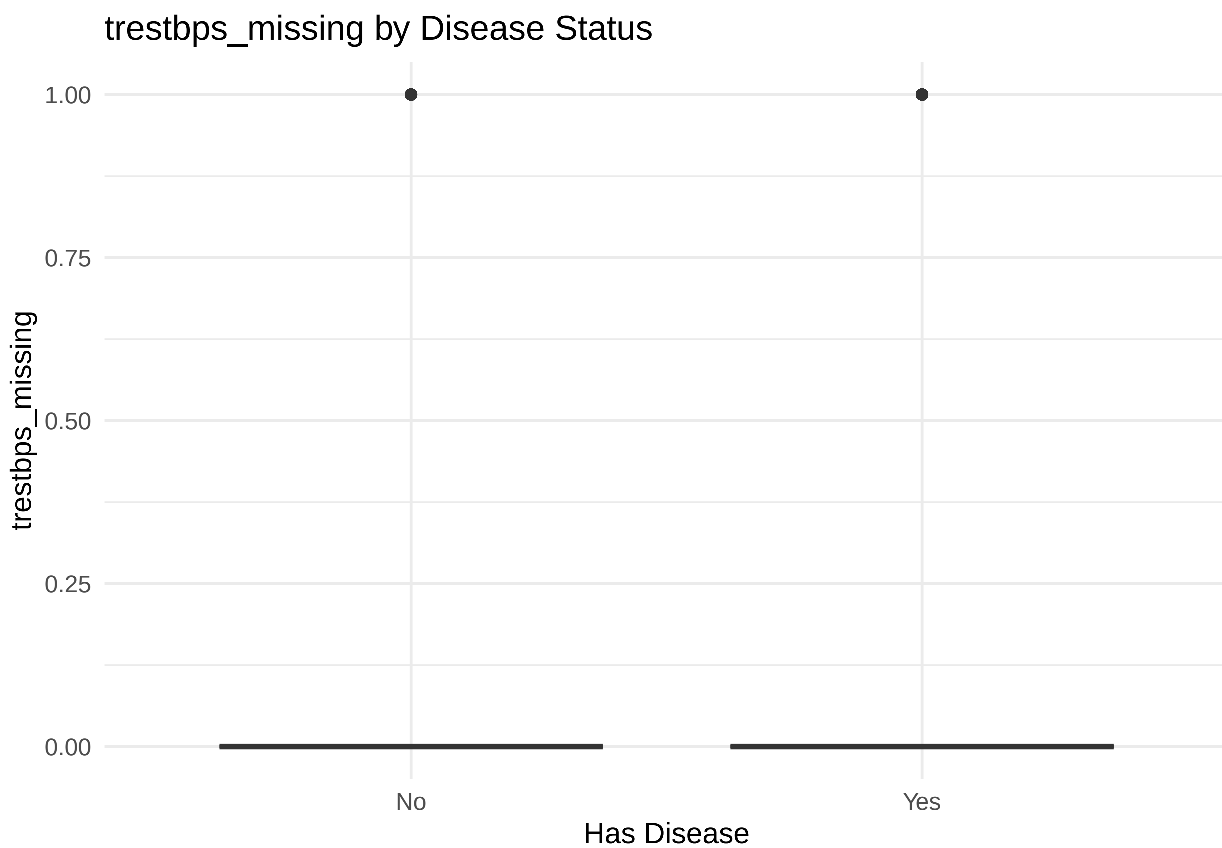

ca_imputed by Disease Status
7.5
5.0
2.5
0.0
ca_imputed
No
Yes
Has Disease

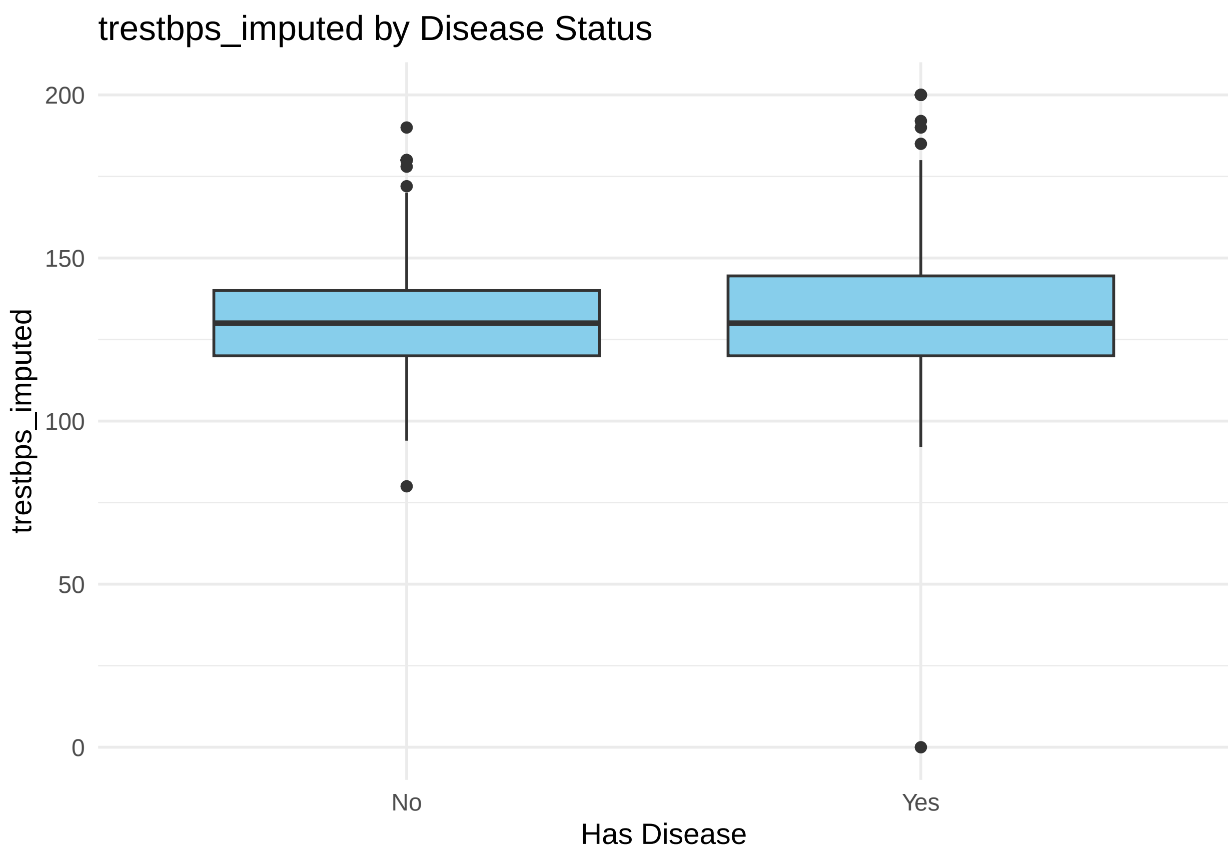
thal_imputed by Disease Status

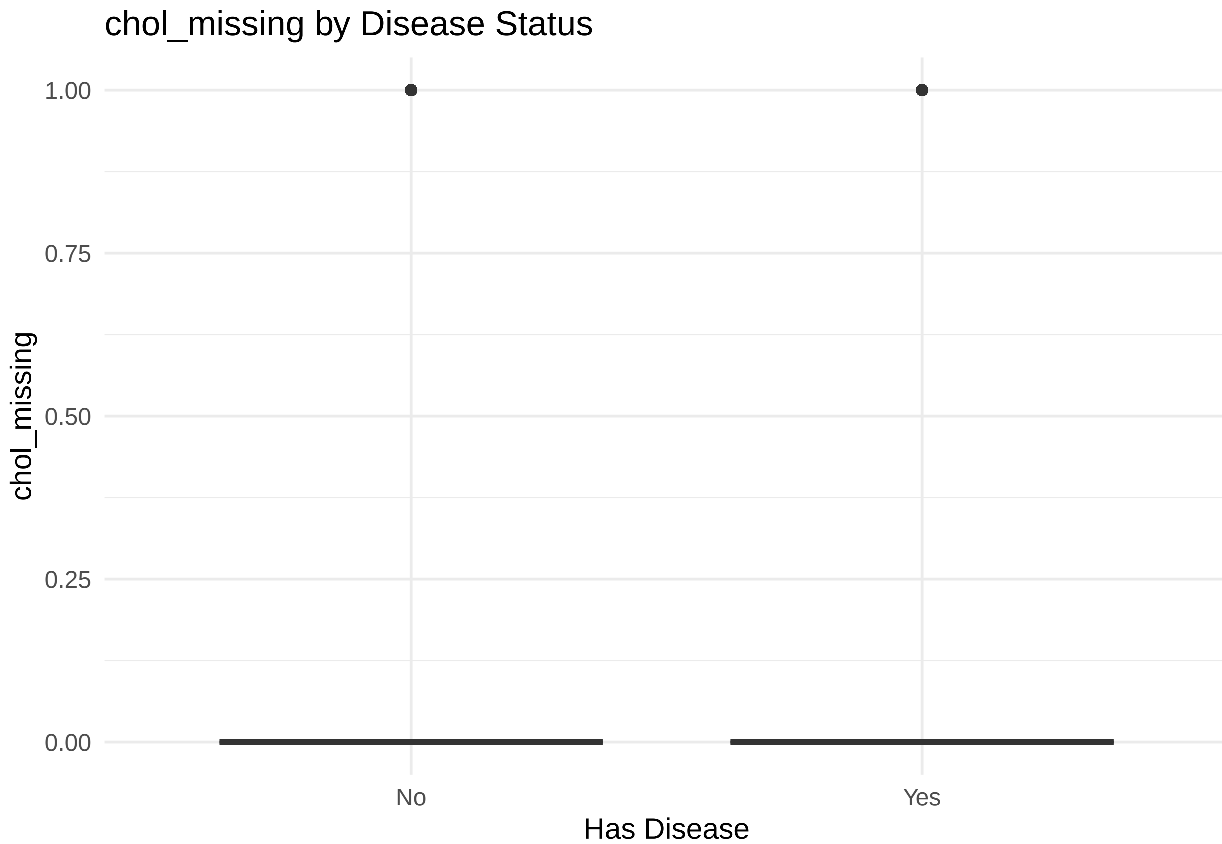


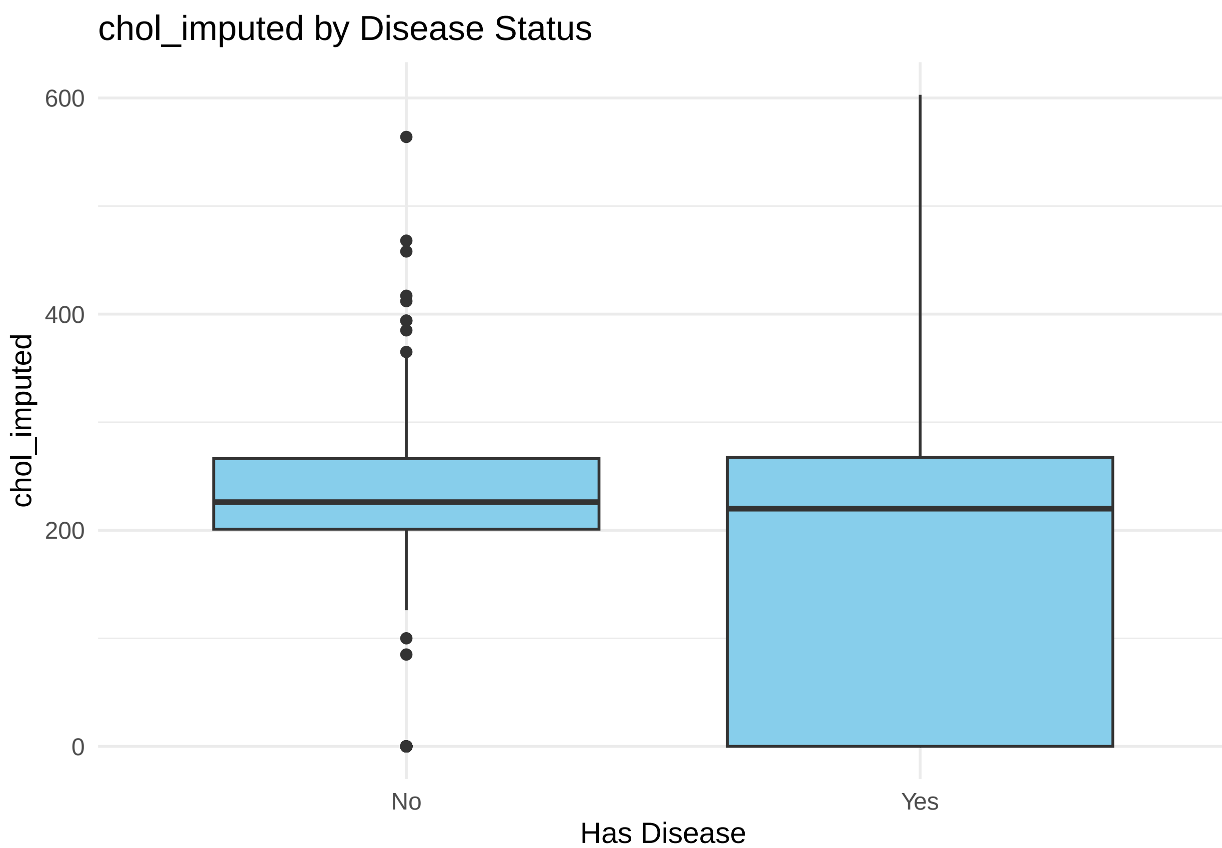

thaltime_imputed by Disease Status
20
15
10
5
0
thaltime_imputed
No
Yes
Has Disease

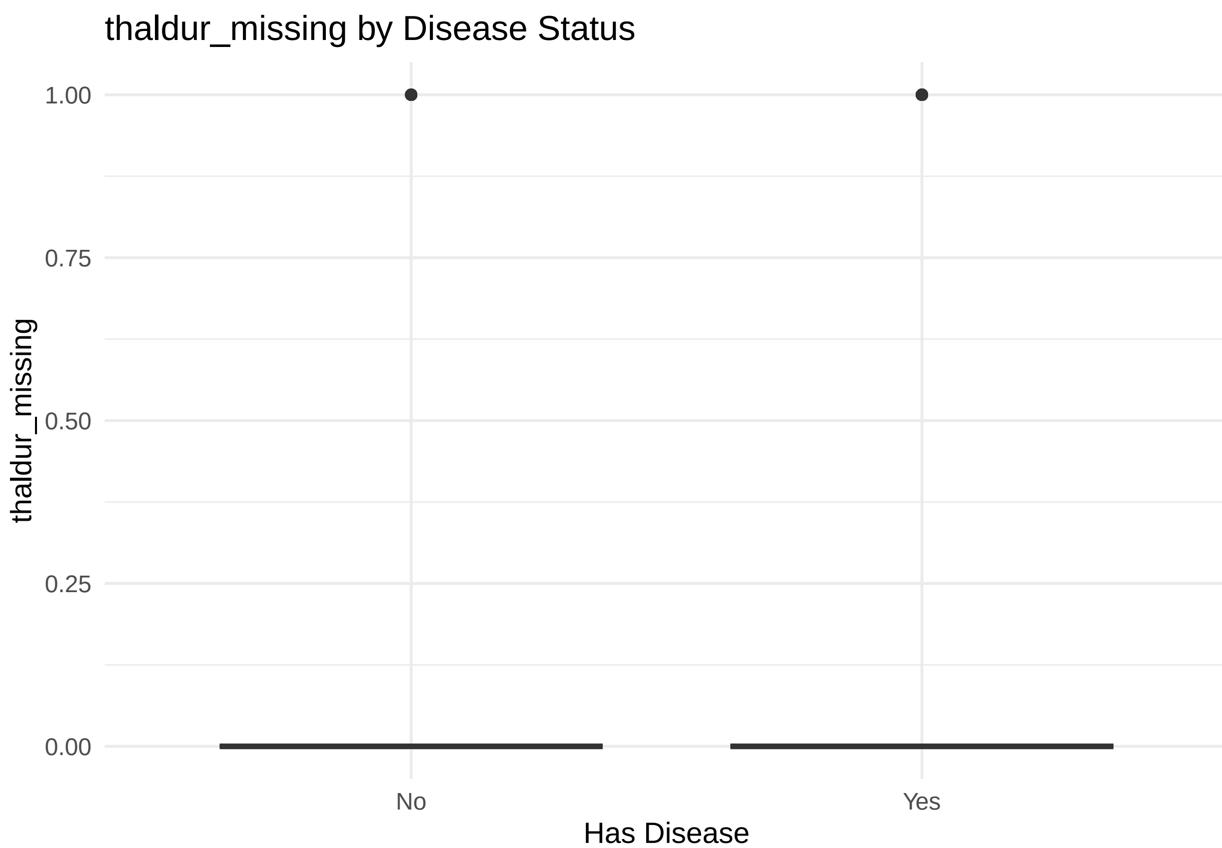

rldv5_imputed by Disease Status
30
20
10
rldv5_imputed
No
Yes
Has Disease

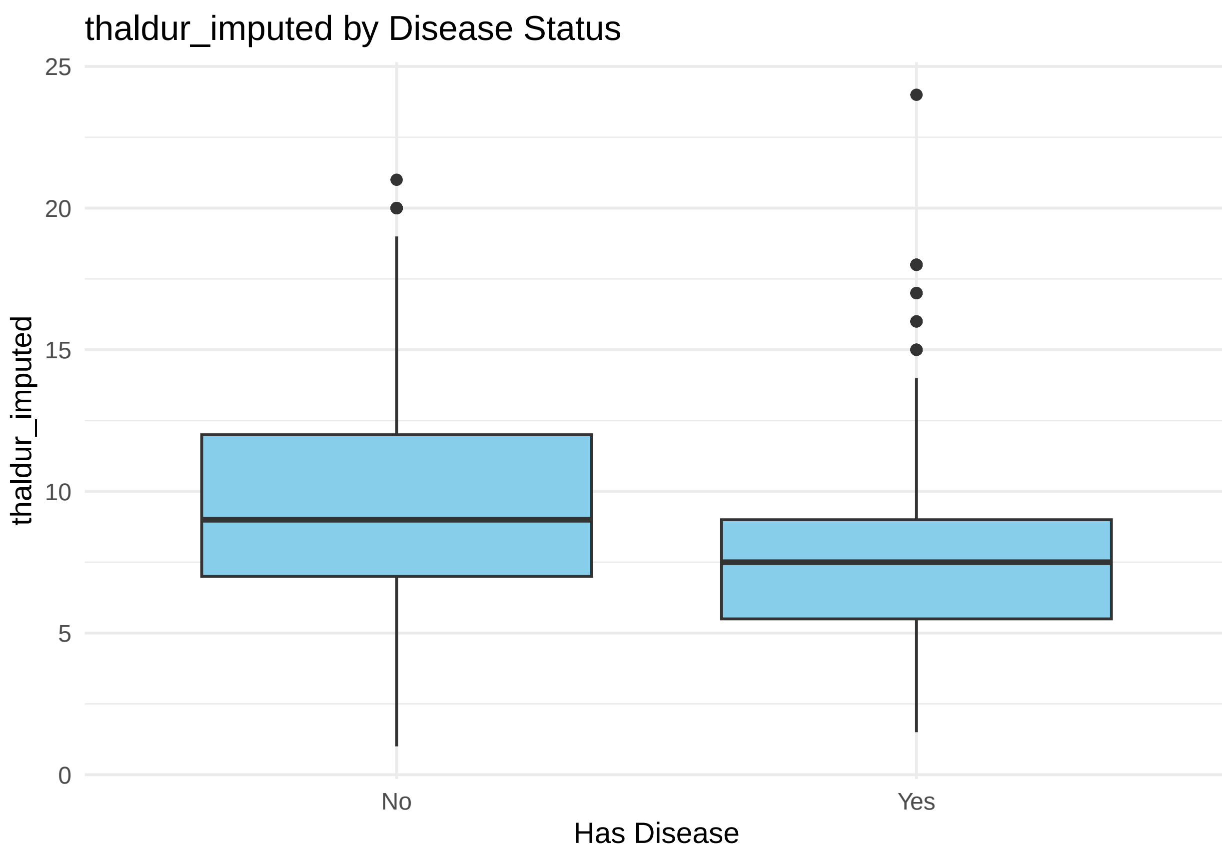

trestbps_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
trestbps_missing
No
Yes
Has Disease

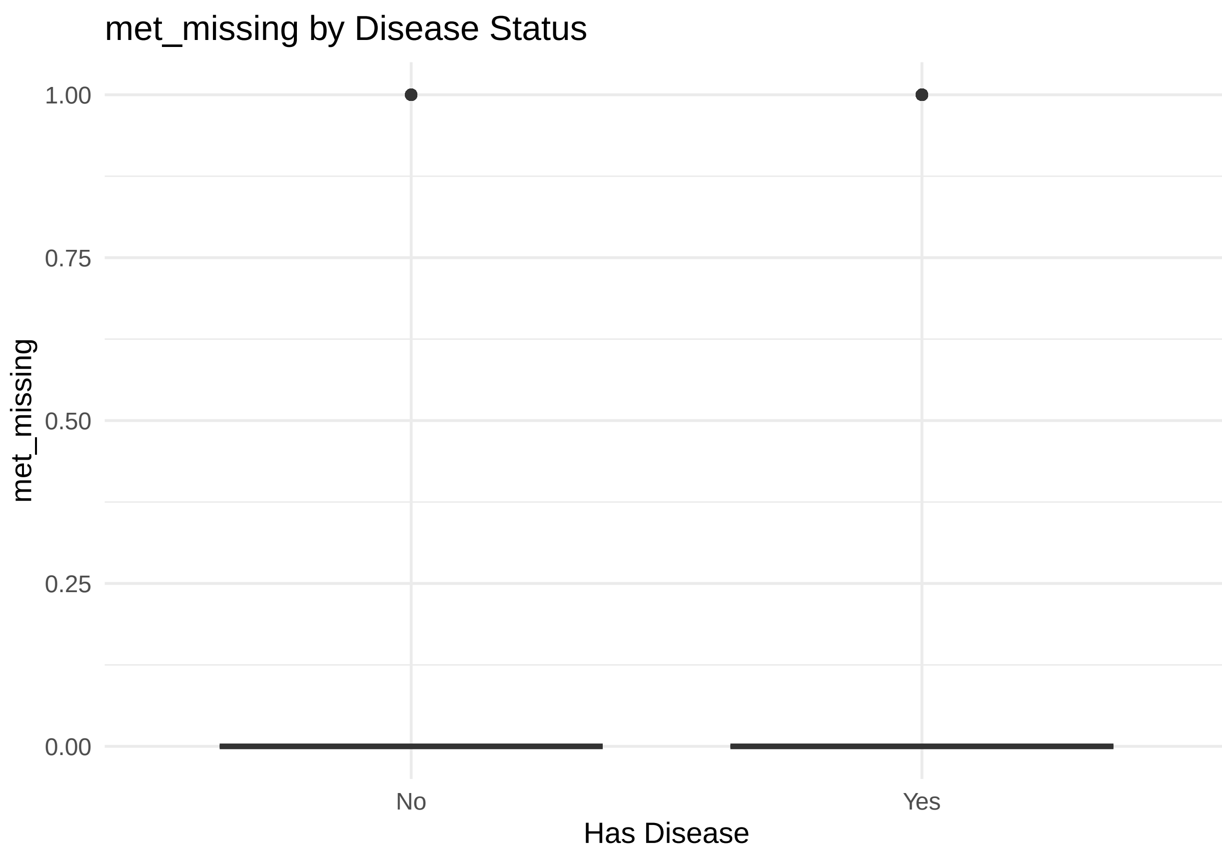

trestbps_imputed by Disease Status
200
150
100
50
0
trestbps_imputed
No
Yes
Has Disease

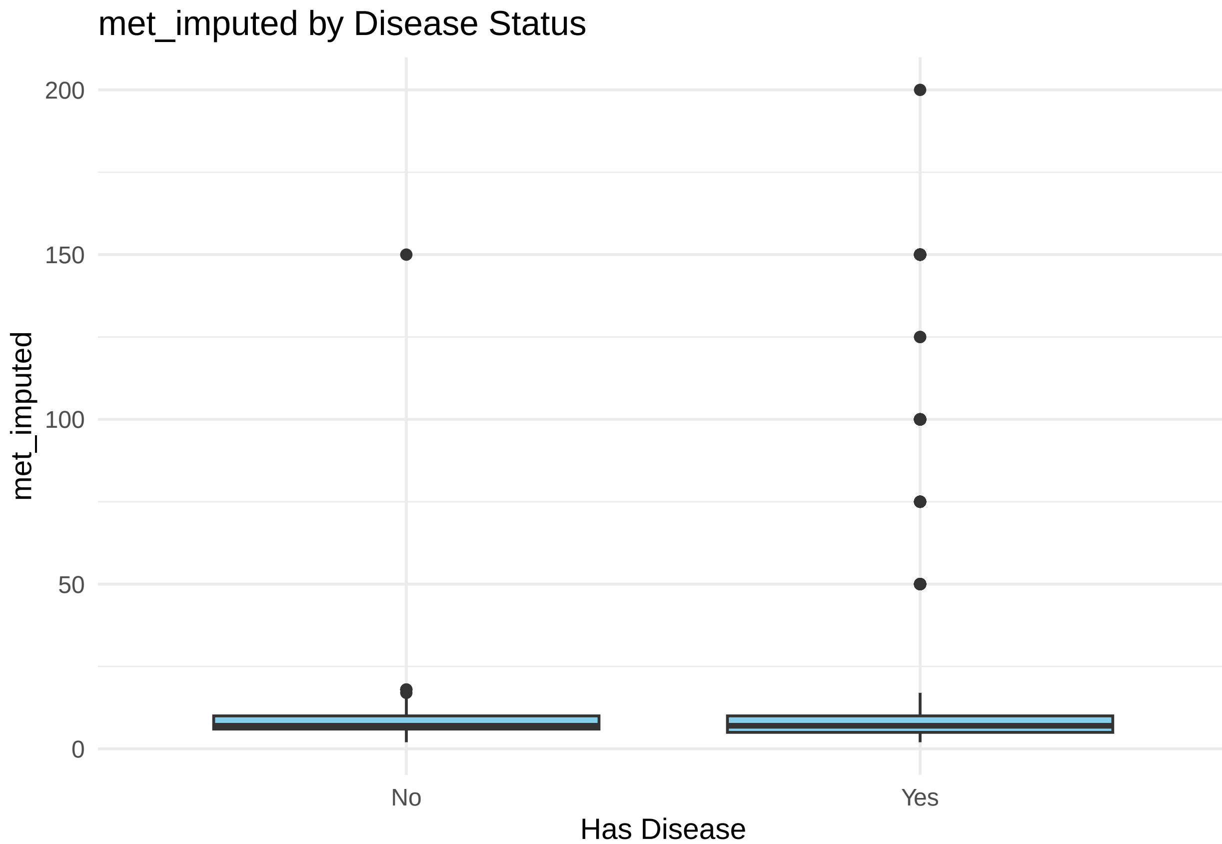

chol_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
chol_missing
No
Yes
Has Disease

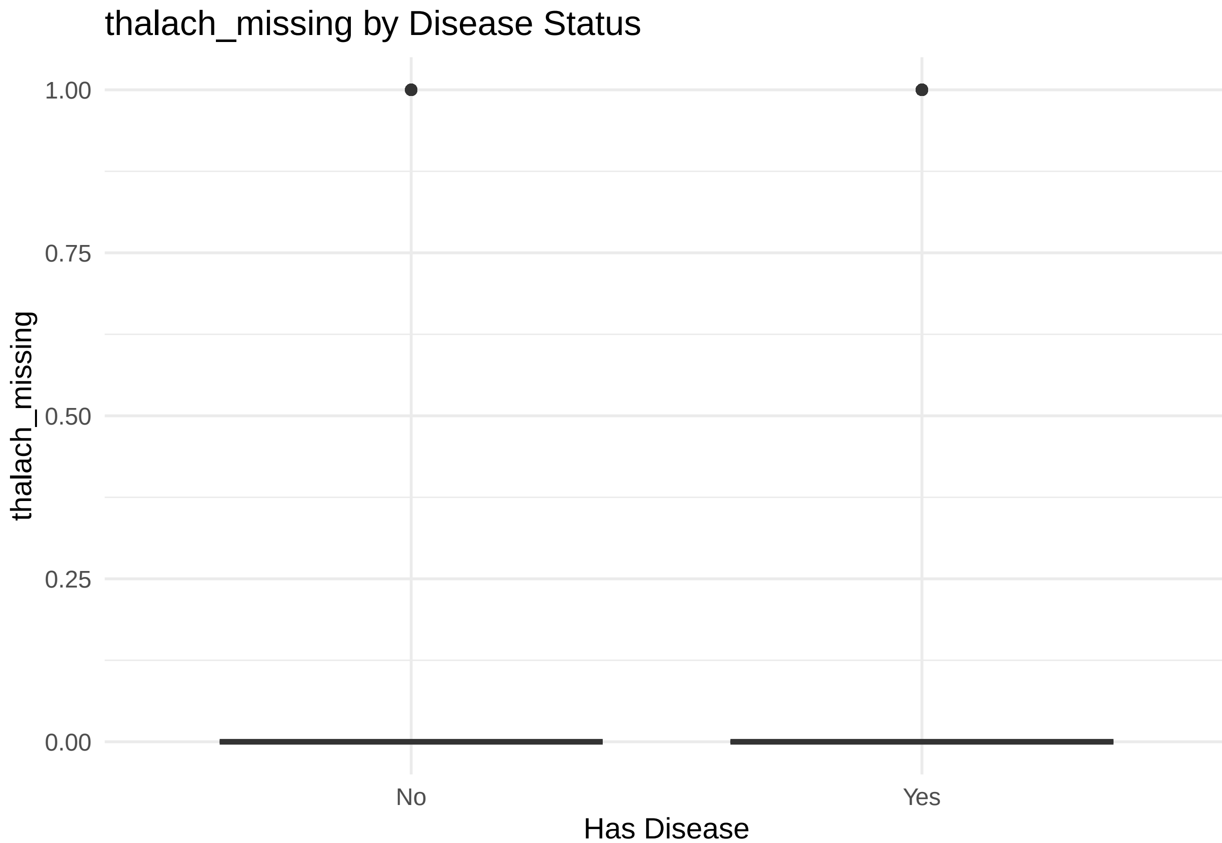

chol_imputed by Disease Status
600
400
200
0
chol_imputed
No
Yes
Has Disease


thaldur_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
thaldur_missing
No
Yes
Has Disease

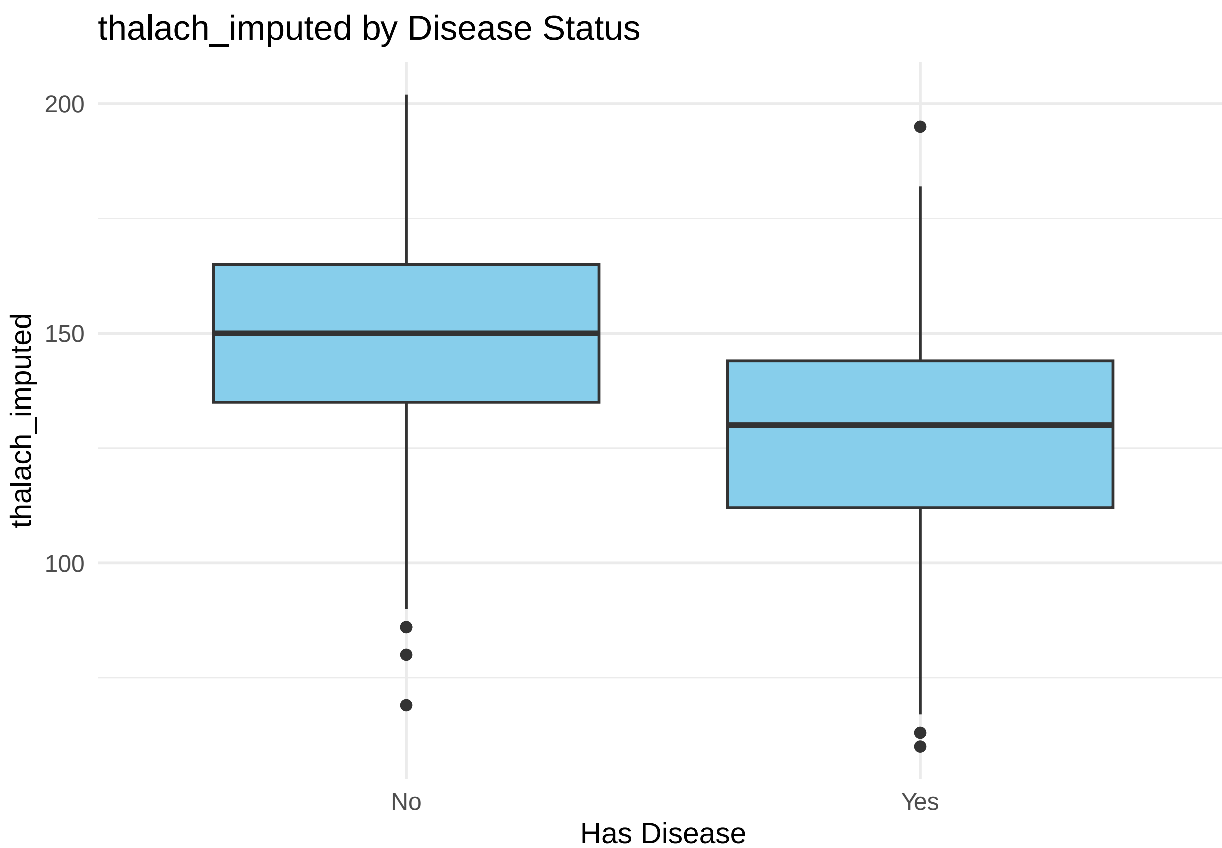

thaldur_imputed by Disease Status
25
20
15
10
5
0
thaldur_imputed
No
Yes
Has Disease

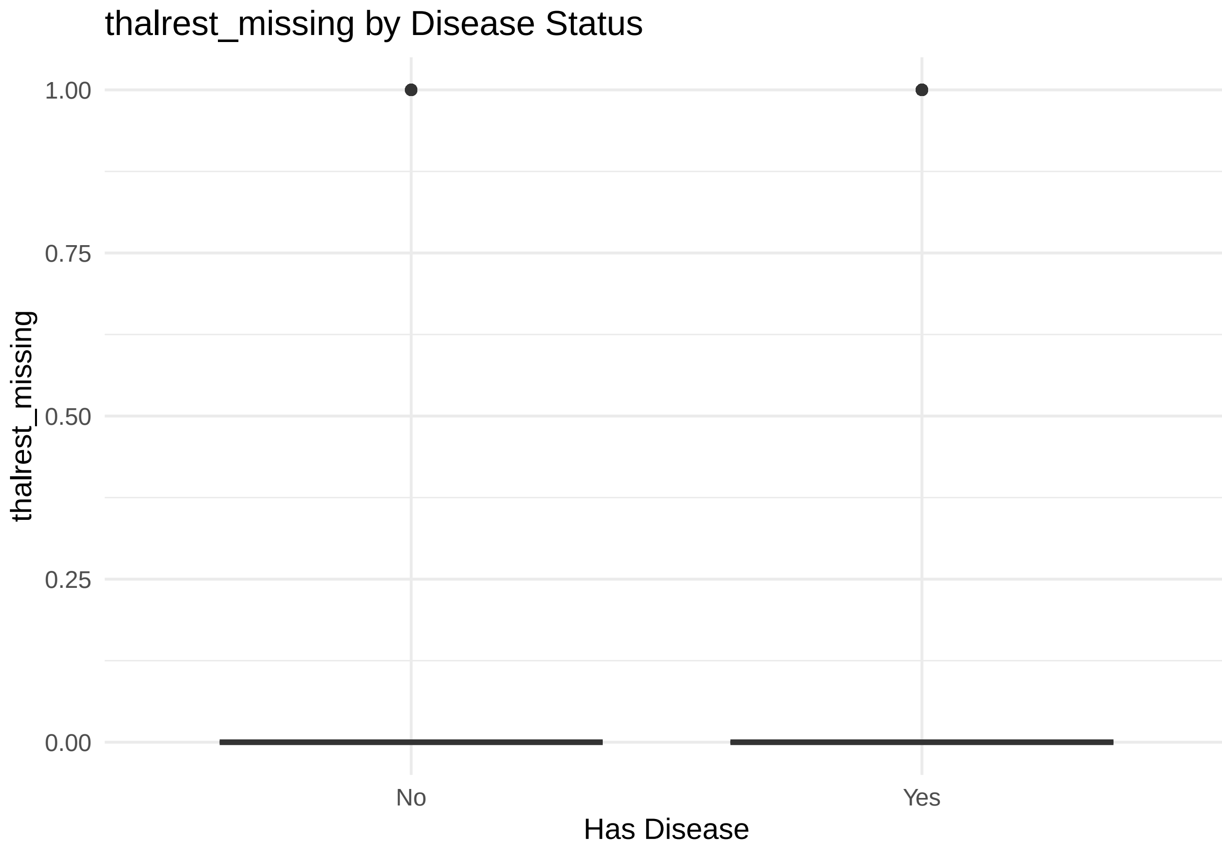

met_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
met_missing
No
Yes
Has Disease

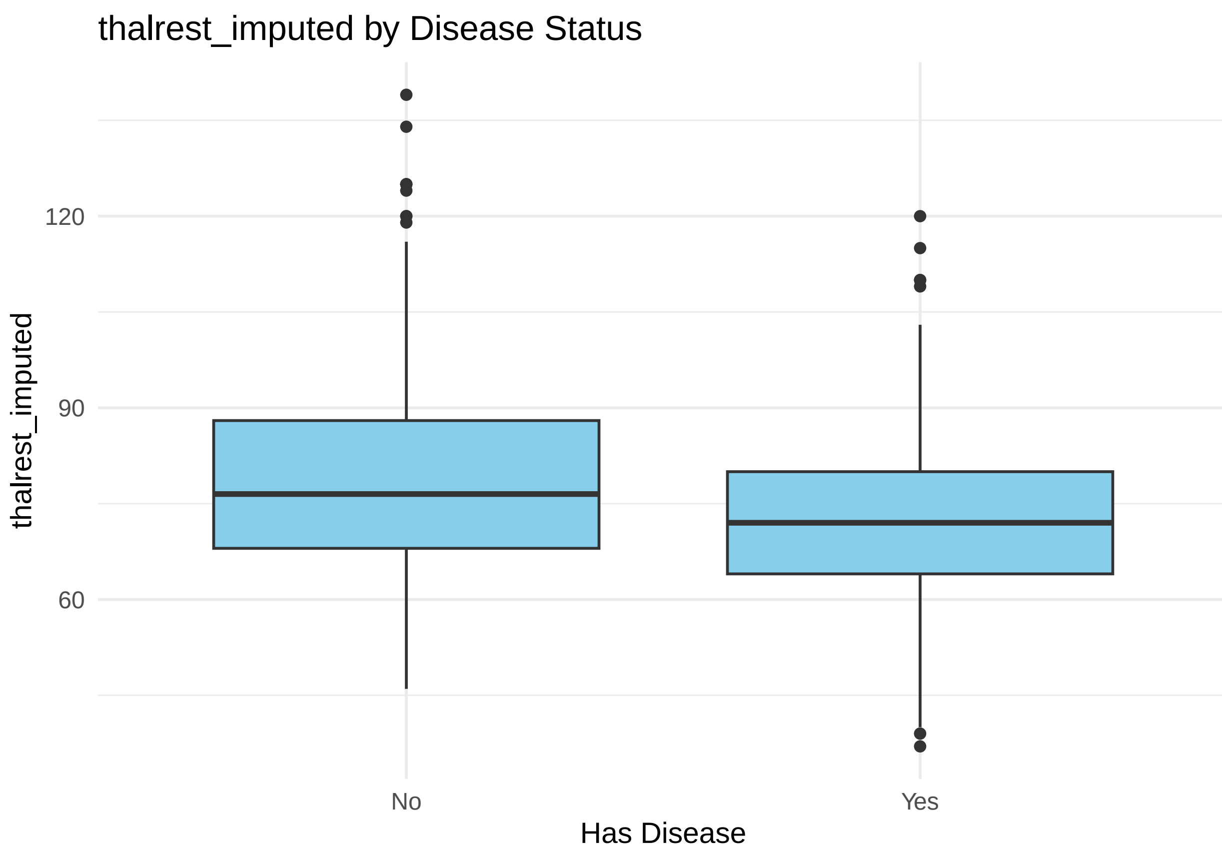

met_imputed by Disease Status
200
150
100
50
0
met_imputed
No
Yes
Has Disease

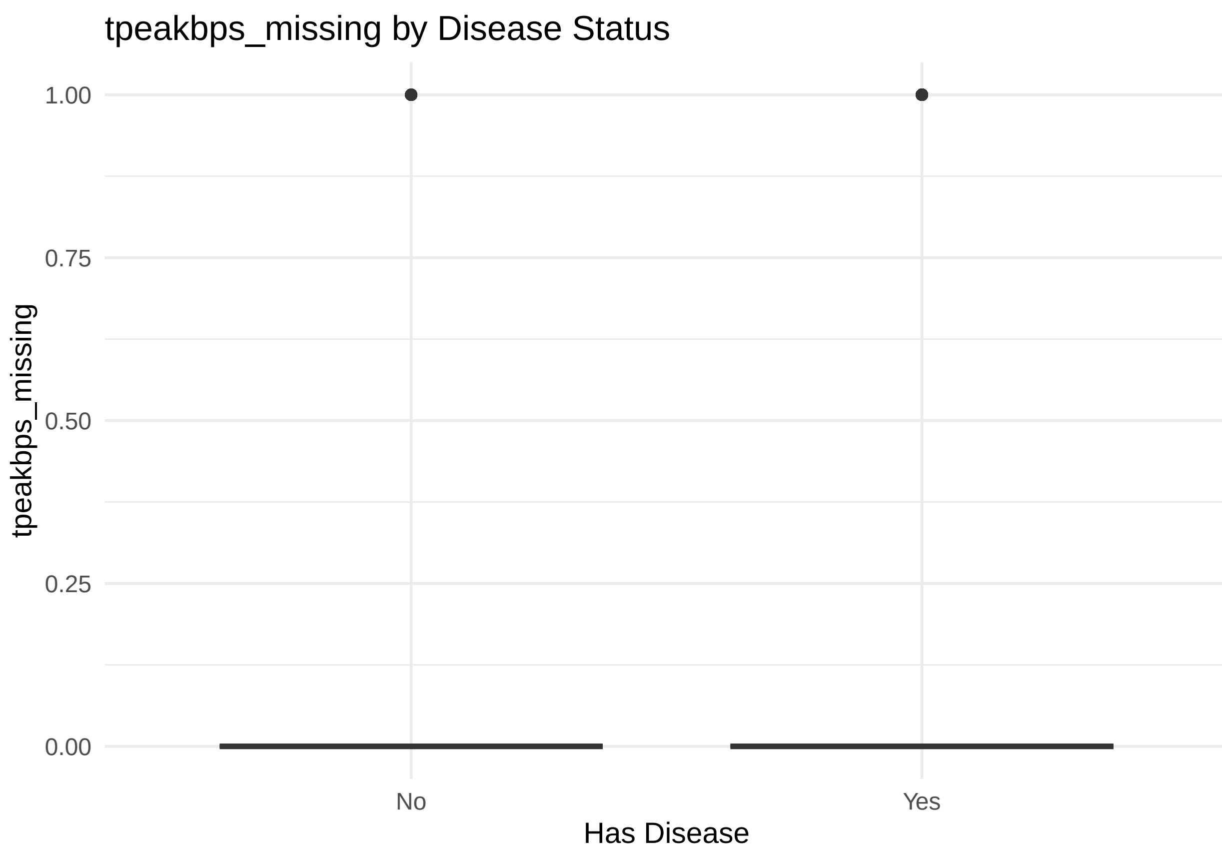


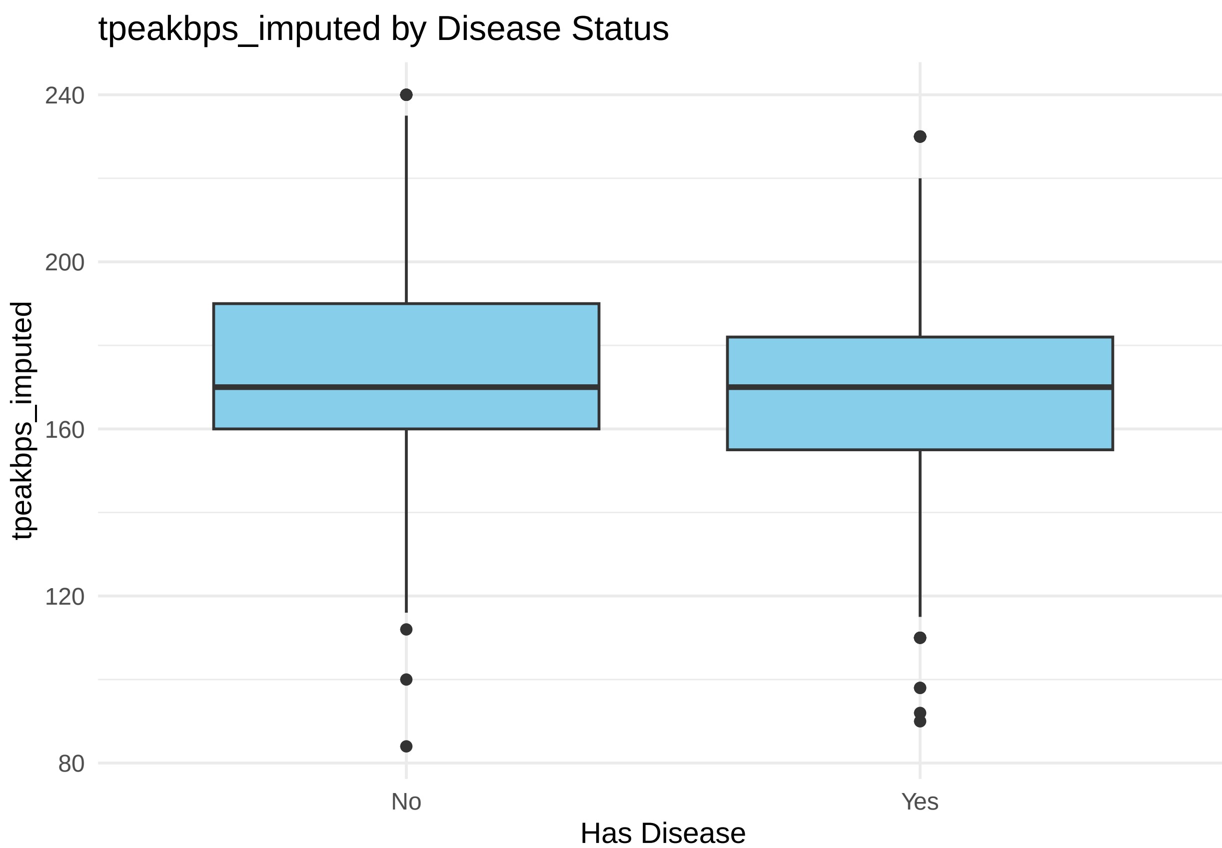
thalach_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
thalach_missing
No
Yes
Has Disease

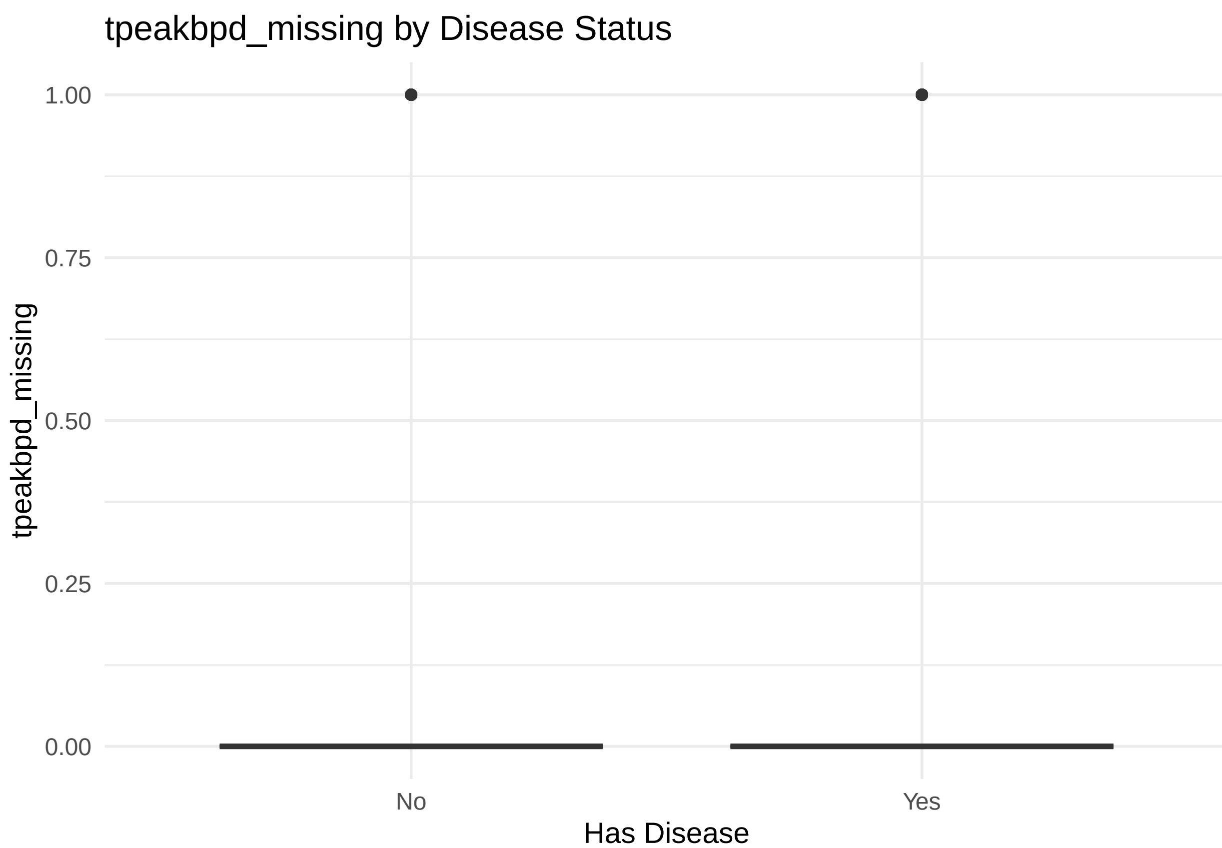

thalach_imputed by Disease Status
200
150
100
thalach_imputed
No
Yes
Has Disease

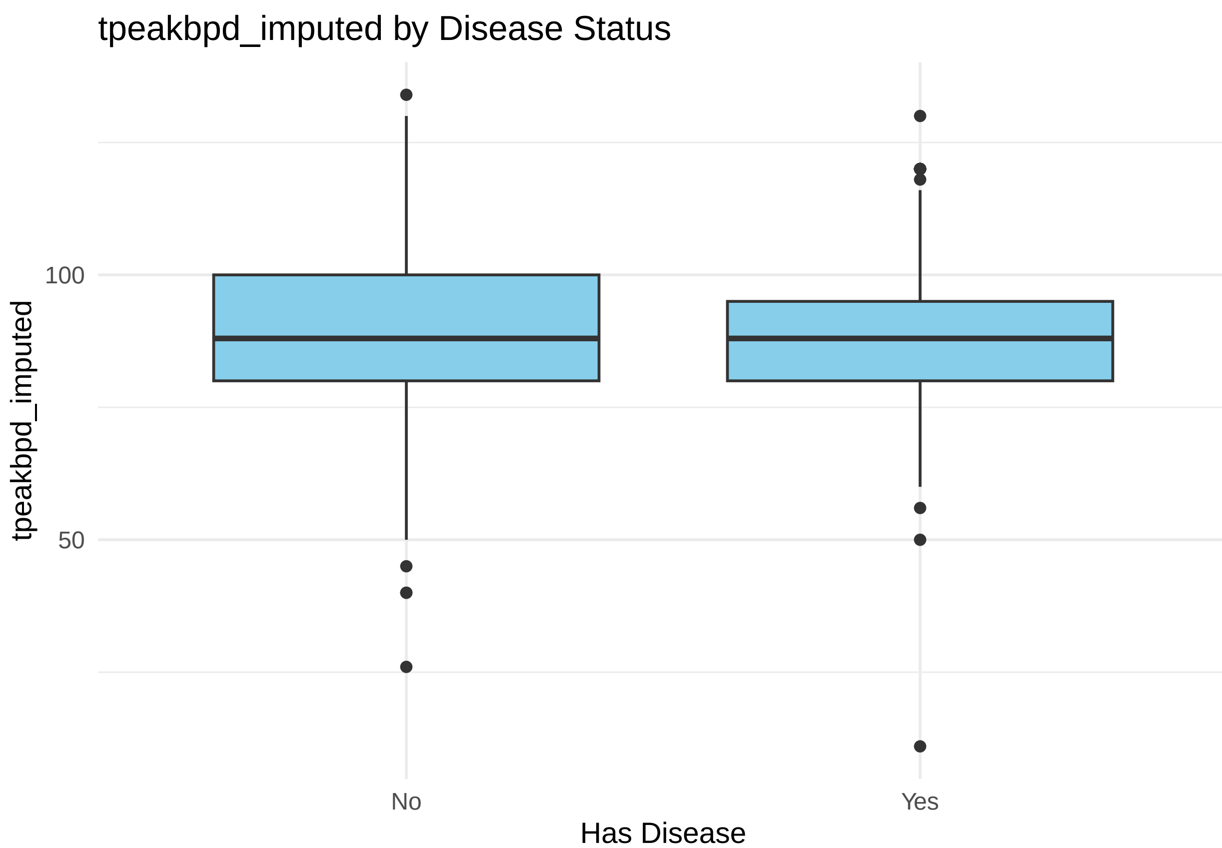

thalrest_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
thalrest_missing
No
Yes
Has Disease


thalrest_imputed by Disease Status
120
90
60
thalrest_imputed
No
Yes
Has Disease

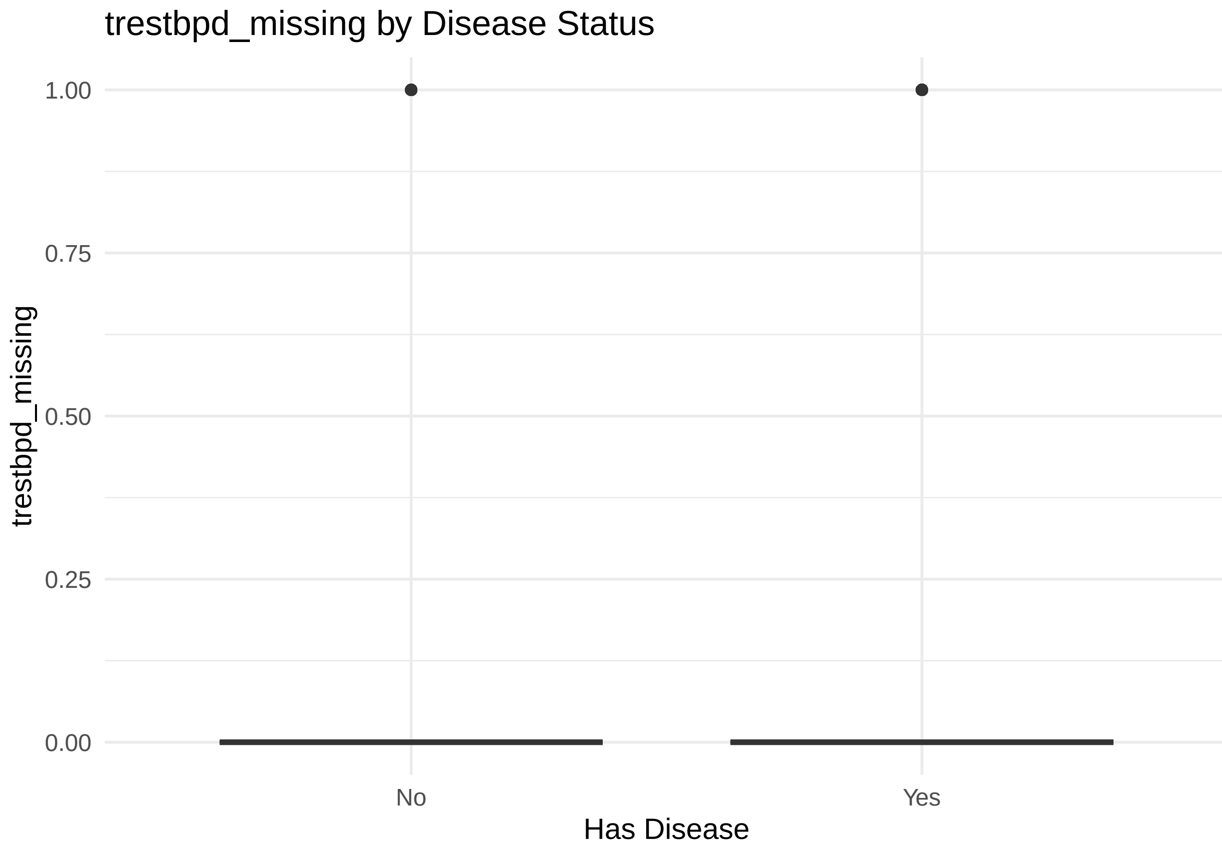

tpeakbps_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
tpeakbps_missing
No
Yes
Has Disease


tpeakbps_imputed by Disease Status
240
200
160
120
80
tpeakbps_imputed
No
Yes
Has Disease

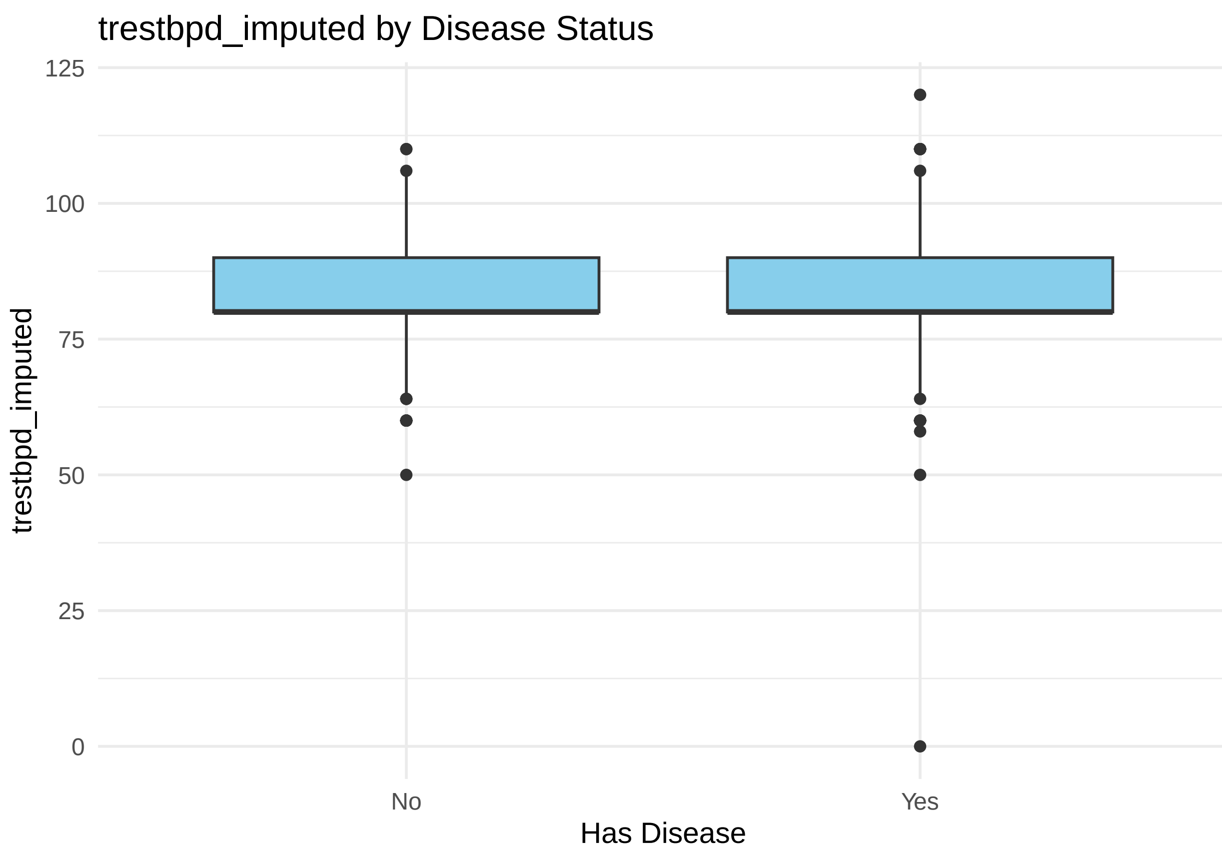

tpeakbpd_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
tpeakbpd_missing
No
Yes
Has Disease

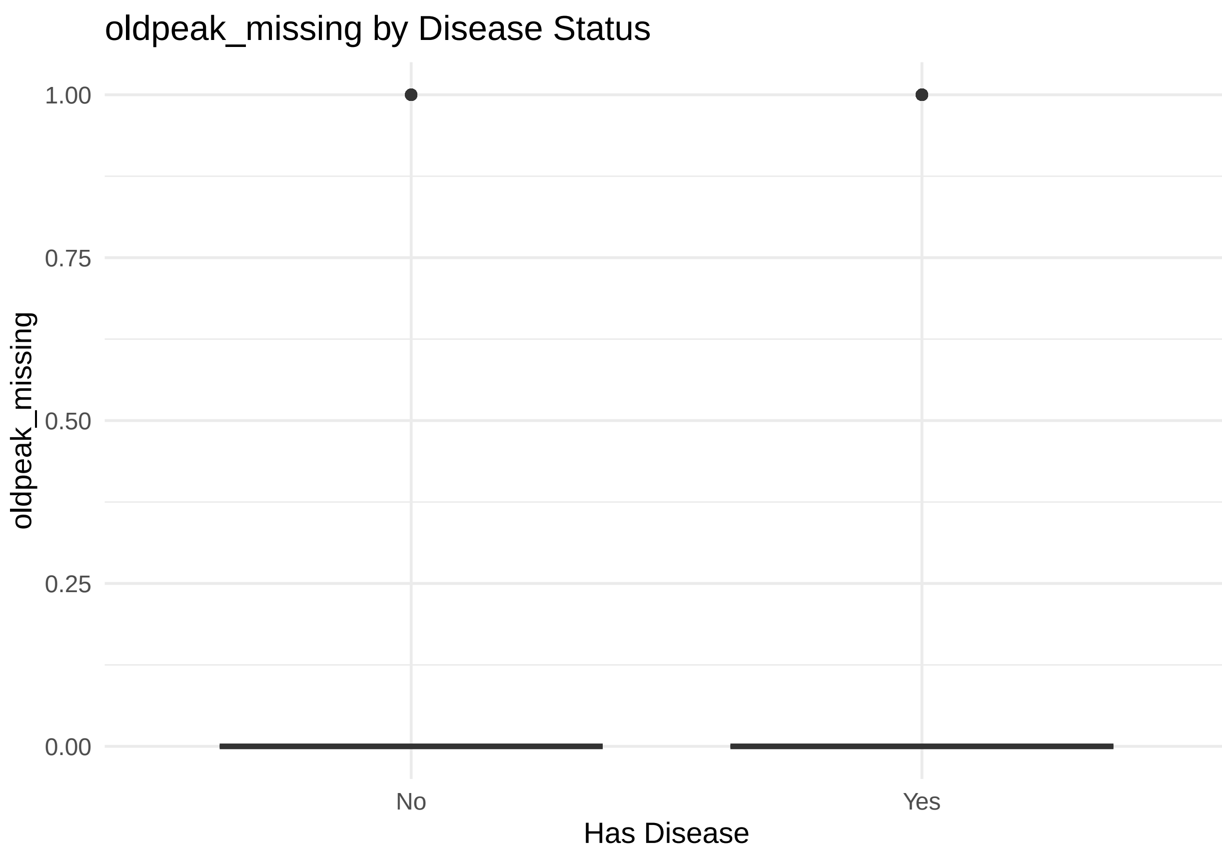


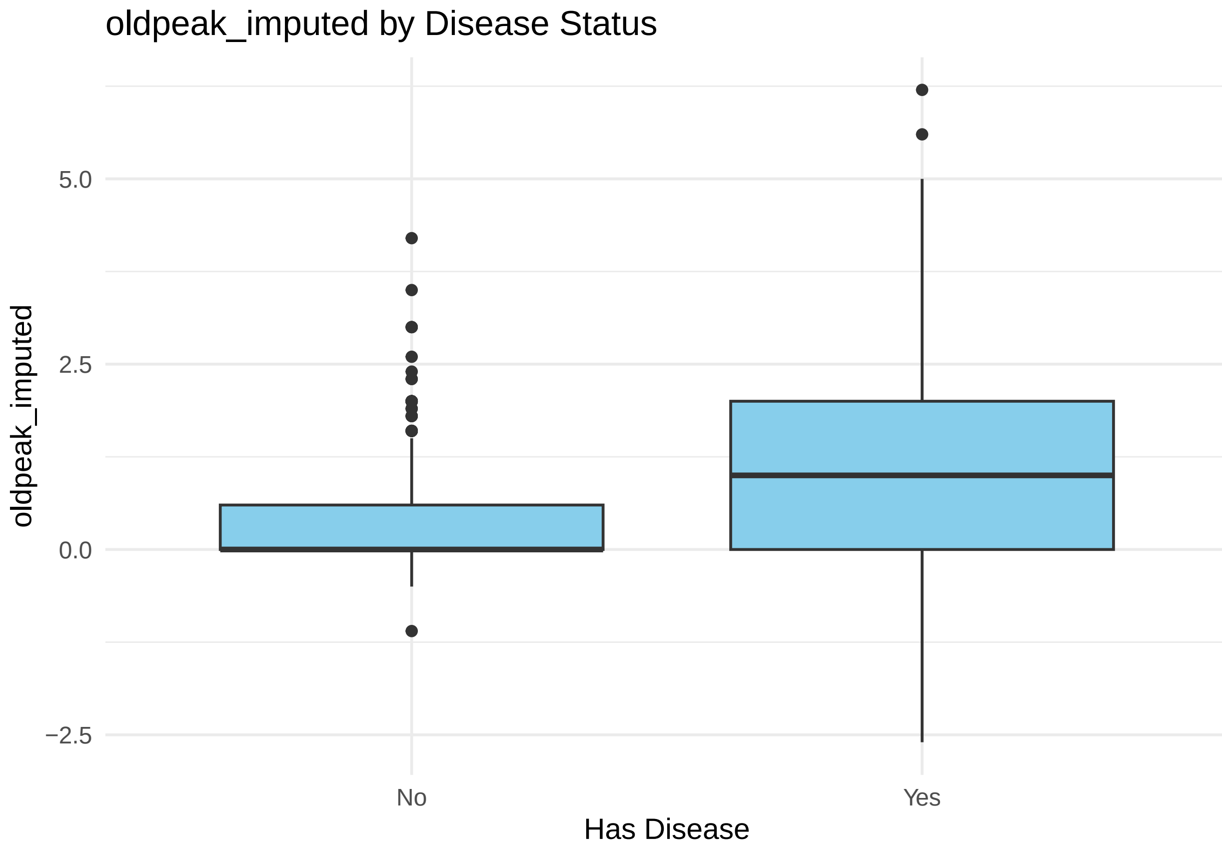
tpeakbpd_imputed by Disease Status
tpeakbpd_imputed
100
50
No
Yes
Has Disease

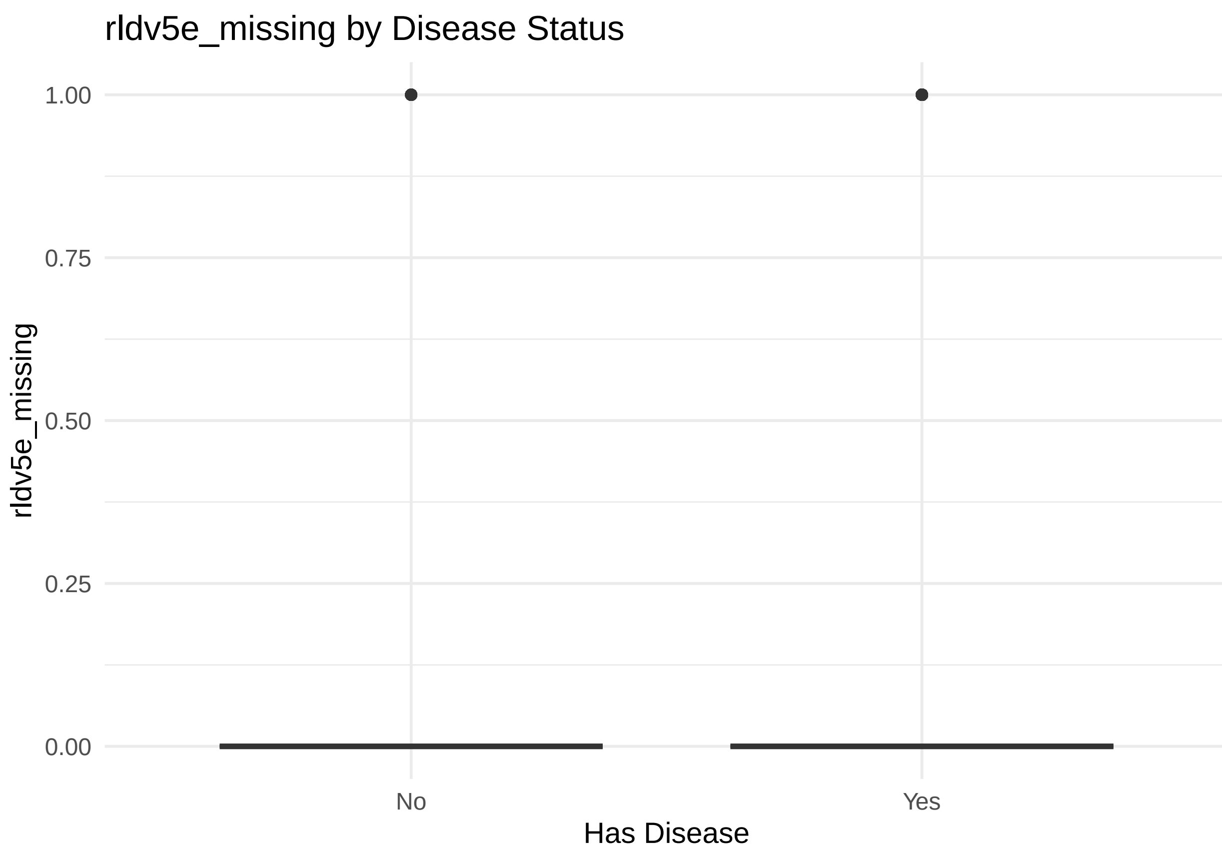


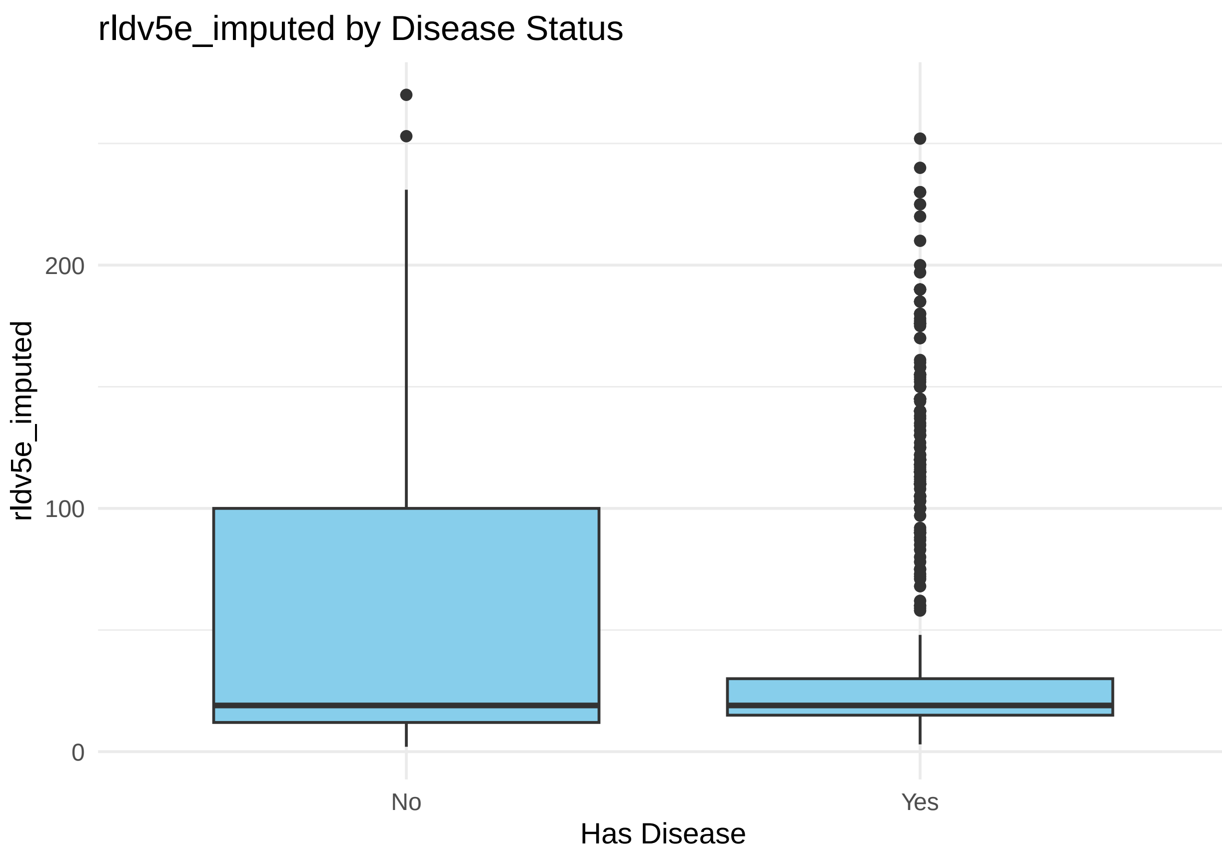
trestbpd_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
trestbpd_missing
No
Yes
Has Disease

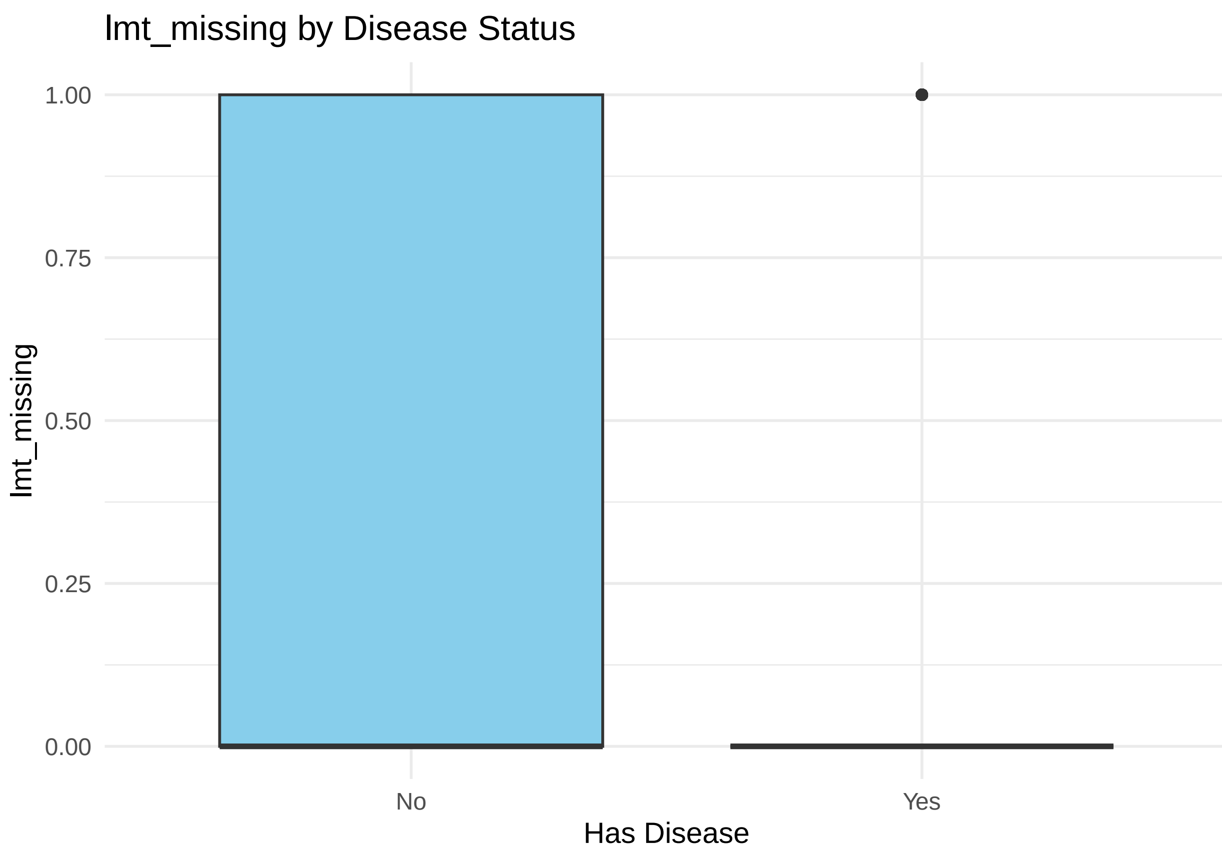

trestbpd_imputed by Disease Status
125
100
75
50
25
0
trestbpd_imputed
No
Yes
Has Disease

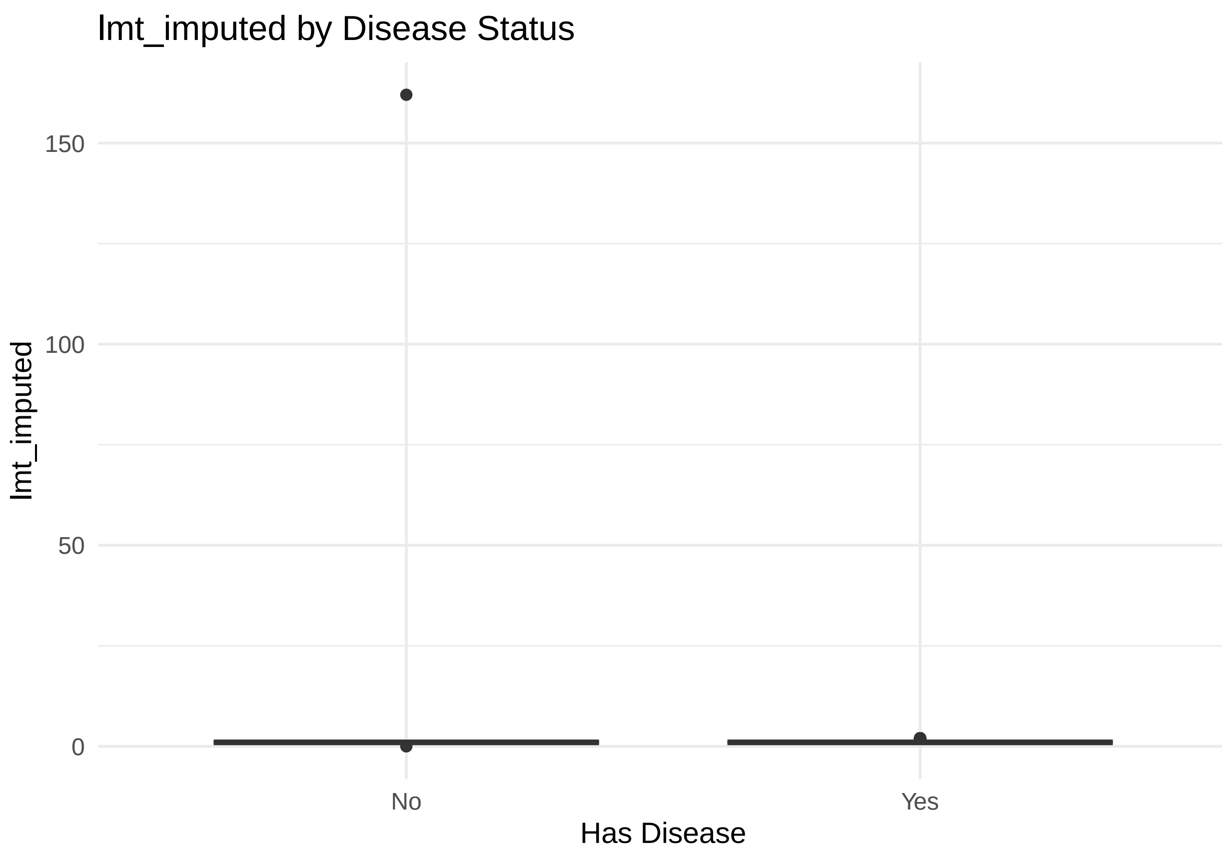

oldpeak_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
oldpeak_missing
No
Yes
Has Disease


oldpeak_imputed by Disease Status
oldpeak_imputed
5.0
2.5
0.0
−2.5
No
Yes
Has Disease


rldv5e_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
rldv5e_missing
No
Yes
Has Disease


rldv5e_imputed by Disease Status
rldv5e_imputed
200
100
0
No
Yes
Has Disease


Imt_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
Imt_missing
No
Yes
Has Disease



Welch's two-sample t-test

Statistical testing to determine if there are significant differences between the means of two features.

```
for (var in num_vars) {
  formula <- reformulate("has_disease", response = var)
  test_result <- t.test(formula, data = df)
  cat("=== T-test for", var, "===\n")
  print(test_result)
  cat("\n\n")
}
```

```
## === T-test for age ===
##
##  Welch Two Sample t-test
##
## data:  age by has_disease
## t = -8.5853, df = 833.36, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -6.462916 -4.057656
## sample estimates:
##  mean in group No mean in group Yes
##          50.58416          55.84444
##
##
##
```

```
## === T-test for ca_imputed ===
##
##  Welch Two Sample t-test
##
## data:  ca_imputed by has_disease
## t = -3.6443, df = 895.1, p-value = 0.0002836
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.25055510 -0.07514748
## sample estimates:
##  mean in group No mean in group Yes
##         0.1361386         0.2989899
##
##
##
## === T-test for thal_imputed ===
##
##  Welch Two Sample t-test
##
## data:  thal_imputed by has_disease
## t = -9.0385, df = 860.57, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -1.1314140 -0.7277019
## sample estimates:
##  mean in group No mean in group Yes
##          3.438119          4.367677
##
##
##
## === T-test for thaltime_imputed ===
##
##  Welch Two Sample t-test
##
## data:  thaltime_imputed by has_disease
## t = -0.82499, df = 890.94, p-value = 0.4096
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.5200975  0.2122537
## sample estimates:
##  mean in group No mean in group Yes
##          5.761634          5.915556
##
##
##
## === T-test for rldv5_imputed ===
##
##  Welch Two Sample t-test
##
## data:  rldv5_imputed by has_disease
## t = -1.7981, df = 828.98, p-value = 0.07252
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -1.05248435  0.04609371
```

```
## sample estimates:
##  mean in group No mean in group Yes
##          13.93317          14.43636
##
##
##
## === T-test for trestbps_missing ===
##
##  Welch Two Sample t-test
##
## data:  trestbps_missing by has_disease
## t = -1.8033, df = 896.85, p-value = 0.07167
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.061152495  0.002586638
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.07878788
##
##
##
## === T-test for trestbps_imputed ===
##
##  Welch Two Sample t-test
##
## data:  trestbps_imputed by has_disease
## t = -2.977, df = 896.52, p-value = 0.002989
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -5.998366 -1.231847
## sample estimates:
##  mean in group No mean in group Yes
##          129.9728          133.5879
##
##
##
## === T-test for chol_missing ===
##
##  Welch Two Sample t-test
##
## data:  chol_missing by has_disease
## t = 1.9913, df = 696, p-value = 0.04684
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  0.0003482239 0.0492667376
## sample estimates:
##  mean in group No mean in group Yes
##        0.04702970        0.02222222
##
##
##
## === T-test for chol_imputed ===
##
##  Welch Two Sample t-test
```

```
##
## data:  chol_imputed by has_disease
## t = 7.5093, df = 816.55, p-value = 1.559e-13
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  37.88176 64.69433
## sample estimates:
##  mean in group No mean in group Yes
##          227.8416          176.5535
##
##
##
## === T-test for thaldur_missing ===
##
##  Welch Two Sample t-test
##
## data:  thaldur_missing by has_disease
## t = -1.1721, df = 893.81, p-value = 0.2415
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.05008523  0.01263149
## sample estimates:
##  mean in group No mean in group Yes
##        0.05198020        0.07070707
##
##
##
## === T-test for thaldur_imputed ===
##
##  Welch Two Sample t-test
##
## data:  thaldur_imputed by has_disease
## t = 8.7171, df = 775.94, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  1.608559 2.543589
## sample estimates:
##  mean in group No mean in group Yes
##          9.764356          7.688283
##
##
##
## === T-test for met_missing ===
##
##  Welch Two Sample t-test
##
## data:  met_missing by has_disease
## t = -4.1789, df = 877.82, p-value = 3.222e-05
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.12675602 -0.04574123
## sample estimates:
##  mean in group No mean in group Yes
##        0.06930693        0.15555556
```

```
##
##
##
## === T-test for met_imputed ===
##
##  Welch Two Sample t-test
##
## data:  met_imputed by has_disease
## t = -7.6036, df = 544.71, p-value = 1.27e-13
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -16.563228  -9.762245
## sample estimates:
##  mean in group No mean in group Yes
##           8.12797          21.29071
##
##
##
## === T-test for thalach_missing ===
##
##  Welch Two Sample t-test
##
## data:  thalach_missing by has_disease
## t = -1.3416, df = 895.8, p-value = 0.1801
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.052219591  0.009815351
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.07070707
##
##
##
## === T-test for thalach_imputed ===
##
##  Welch Two Sample t-test
##
## data:  thalach_imputed by has_disease
## t = 12.322, df = 865.57, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  16.15533 22.27721
## sample estimates:
##  mean in group No mean in group Yes
##          148.0446          128.8283
##
##
##
## === T-test for thalrest_missing ===
##
##  Welch Two Sample t-test
##
## data:  thalrest_missing by has_disease
## t = -1.4592, df = 896.5, p-value = 0.1449
```

```
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.05445650  0.00801186
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.07272727
##
##
##
## === T-test for thalrest_imputed ===
##
##  Welch Two Sample t-test
##
## data:  thalrest_imputed by has_disease
## t = 6.3454, df = 773.7, p-value = 3.774e-10
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  4.183520 7.931421
## sample estimates:
##  mean in group No mean in group Yes
##          78.73020          72.67273
##
##
##
## === T-test for tpeakbps_missing ===
##
##  Welch Two Sample t-test
##
## data:  tpeakbps_missing by has_disease
## t = -2.2436, df = 894.11, p-value = 0.0251
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.070047808 -0.004679664
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.08686869
##
##
##
## === T-test for tpeakbps_imputed ===
##
##  Welch Two Sample t-test
##
## data:  tpeakbps_imputed by has_disease
## t = 3.1634, df = 860.4, p-value = 0.001614
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  1.988506 8.489841
## sample estimates:
##  mean in group No mean in group Yes
##          174.4109          169.1717
##
##
##
```

```
## === T-test for tpeakbpd_missing ===
##
##  Welch Two Sample t-test
##
## data:  tpeakbpd_missing by has_disease
## t = -2.2436, df = 894.11, p-value = 0.0251
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.070047808 -0.004679664
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.08686869
##
##
##
## === T-test for tpeakbpd_imputed ===
##
##  Welch Two Sample t-test
##
## data:  tpeakbpd_imputed by has_disease
## t = 0.38958, df = 814.22, p-value = 0.6969
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -1.517519  2.269054
## sample estimates:
##  mean in group No mean in group Yes
##          87.54950          87.17374
##
##
##
## === T-test for trestbpd_missing ===
##
##  Welch Two Sample t-test
##
## data:  trestbpd_missing by has_disease
## t = -1.8033, df = 896.85, p-value = 0.07167
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.061152495  0.002586638
## sample estimates:
##  mean in group No mean in group Yes
##        0.04950495        0.07878788
##
##
##
## === T-test for trestbpd_imputed ===
##
##  Welch Two Sample t-test
##
## data:  trestbpd_imputed by has_disease
## t = 0.20883, df = 879.12, p-value = 0.8346
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -1.163256  1.440273
```

```
## sample estimates:
##  mean in group No mean in group Yes
##          83.36881          83.23030
##
##
##
## === T-test for oldpeak_missing ===
##
##  Welch Two Sample t-test
##
## data:  oldpeak_missing by has_disease
## t = -1.8566, df = 896.85, p-value = 0.06369
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.063456965  0.001760795
## sample estimates:
##  mean in group No mean in group Yes
##        0.05198020        0.08282828
##
##
##
## === T-test for oldpeak_imputed ===
##
##  Welch Two Sample t-test
##
## data:  oldpeak_imputed by has_disease
## t = -12.186, df = 832.96, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.8839527 -0.6387035
## sample estimates:
##  mean in group No mean in group Yes
##         0.4257426         1.1870707
##
##
##
## === T-test for rldv5e_missing ===
##
##  Welch Two Sample t-test
##
## data:  rldv5e_missing by has_disease
## t = -8.4101, df = 779.1, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.2318428 -0.1440948
## sample estimates:
##  mean in group No mean in group Yes
##        0.05445545        0.24242424
##
##
##
## === T-test for rldv5e_imputed ===
##
##  Welch Two Sample t-test
```

```
##
## data:  rldv5e_imputed by has_disease
## t = 2.8953, df = 816.16, p-value = 0.003889
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##   3.581025 18.657808
## sample estimates:
##  mean in group No mean in group Yes
##          55.36386          44.24444
##
##
##
## === T-test for lmt_missing ===
##
##  Welch Two Sample t-test
##
## data:  lmt_missing by has_disease
## t = 9.4361, df = 742.88, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  0.2257806 0.3444065
## sample estimates:
##  mean in group No mean in group Yes
##         0.4628713         0.1777778
##
##
##
## === T-test for lmt_imputed ===
##
##  Welch Two Sample t-test
##
## data:  lmt_imputed by has_disease
## t = 0.78046, df = 403.8, p-value = 0.4356
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.4726466  1.0950288
## sample estimates:
##  mean in group No mean in group Yes
##          1.396040          1.084848
```

Summary for numerical variables

```
missing_flags <- colnames(df)[str_detect(colnames(df), "_missing$")]

# recommendation code generated with the help of ChatGPT.
# Create an empty data frame to store results
bivariate_summary <- data.frame(
  Variable = character(),
  p_value = numeric(),
  Missingness_Flag = logical(),
  Recommendation = character(),
  stringsAsFactors = FALSE
)
```

```r
# Loop through and compute t-tests
for (var in num_vars) {
  formula <- reformulate("has_disease", response = var)
  p_val <- tryCatch(t.test(formula, data = df)$p.value, error = function(e) NA)

  is_missing_flag <- paste0(var, "_missing") %in% colnames(df) | var %in% missing_flags

  recommendation <- if (!is.na(p_val) && p_val < 0.05) {
    "Include (predictive)"
  } else if (is_missing_flag) {
    "Include (missingness informative)"
  } else {
    "Optional (evaluate with model)"
  }

  bivariate_summary <- rbind(
    bivariate_summary,
    data.frame(
      Variable = var,
      p_value = round(p_val, 5),
      Missingness_Flag = is_missing_flag,
      Recommendation = recommendation
    )
  )
}

# View result
bivariate_summary
```

```
##             Variable p_value Missingness_Flag                   Recommendation
## 1                age 0.00000            FALSE             Include (predictive)
## 2         ca_imputed 0.00028            FALSE             Include (predictive)
## 3       thal_imputed 0.00000            FALSE             Include (predictive)
## 4   thaltime_imputed 0.40960            FALSE   Optional (evaluate with model)
## 5      rldv5_imputed 0.07252            FALSE   Optional (evaluate with model)
## 6   trestbps_missing 0.07167             TRUE Include (missingness informative)
## 7   trestbps_imputed 0.00299            FALSE             Include (predictive)
## 8       chol_missing 0.04684             TRUE             Include (predictive)
## 9       chol_imputed 0.00000            FALSE             Include (predictive)
## 10   thaldur_missing 0.24149             TRUE Include (missingness informative)
## 11   thaldur_imputed 0.00000            FALSE             Include (predictive)
## 12       met_missing 0.00003             TRUE             Include (predictive)
## 13       met_imputed 0.00000            FALSE             Include (predictive)
## 14   thalach_missing 0.18008             TRUE Include (missingness informative)
## 15   thalach_imputed 0.00000            FALSE             Include (predictive)
## 16 thalrest_missing 0.14486             TRUE Include (missingness informative)
## 17 thalrest_imputed 0.00000            FALSE             Include (predictive)
## 18 tpeakbps_missing 0.02510             TRUE             Include (predictive)
## 19 tpeakbps_imputed 0.00161            FALSE             Include (predictive)
## 20 tpeakbpd_missing 0.02510             TRUE             Include (predictive)
## 21 tpeakbpd_imputed 0.69695            FALSE   Optional (evaluate with model)
## 22  trestbpd_missing 0.07167             TRUE Include (missingness informative)
## 23  trestbpd_imputed 0.83463            FALSE   Optional (evaluate with model)
```

```
## 24  oldpeak_missing 0.06369             TRUE Include (missingness informative)
## 25  oldpeak_imputed 0.00000            FALSE             Include (predictive)
## 26   rldv5e_missing 0.00000             TRUE             Include (predictive)
## 27   rldv5e_imputed 0.00389            FALSE             Include (predictive)
## 28      lmt_missing 0.00000             TRUE             Include (predictive)
## 29      lmt_imputed 0.43558            FALSE    Optional (evaluate with model)
```

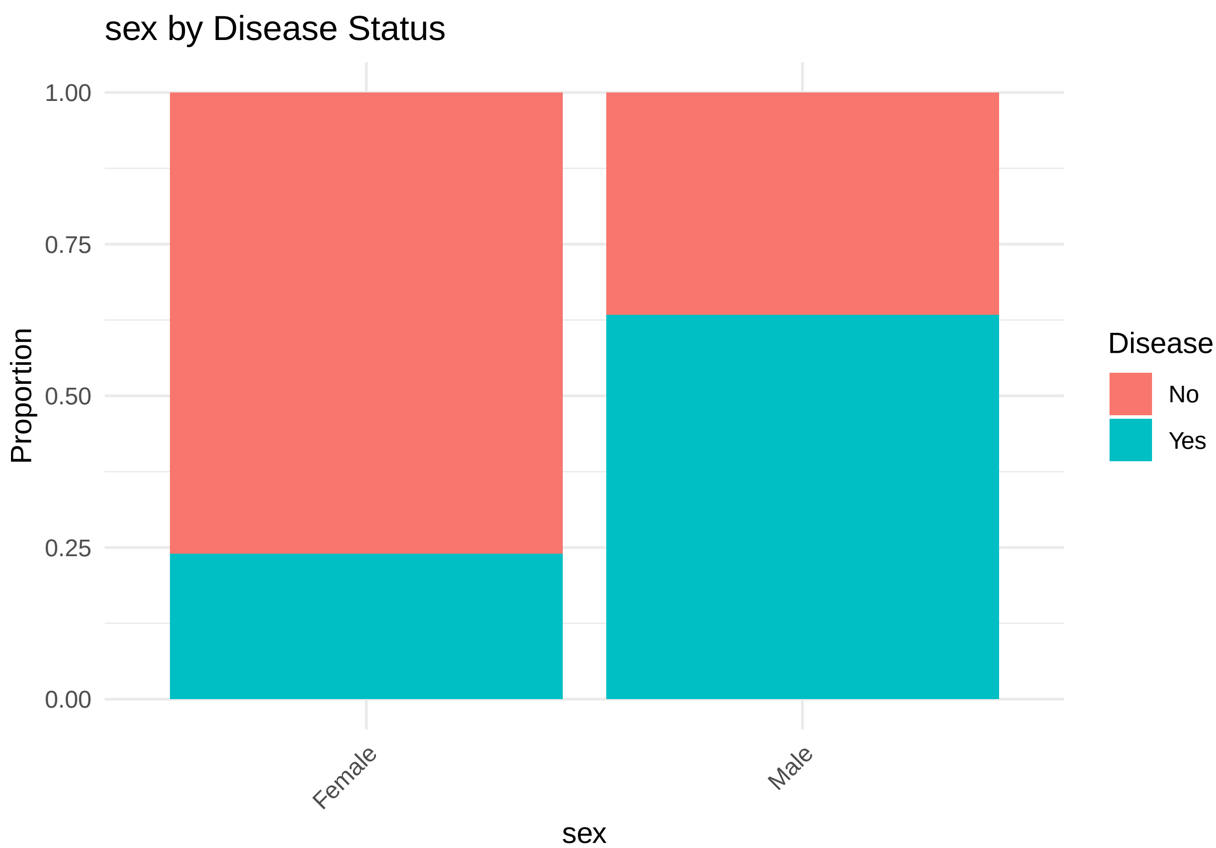
```
# create list of variables recommended as Include
selected_num_vars <- bivariate_summary %>%
  filter(str_detect(Recommendation, "Include")) %>%
  pull(Variable)
```

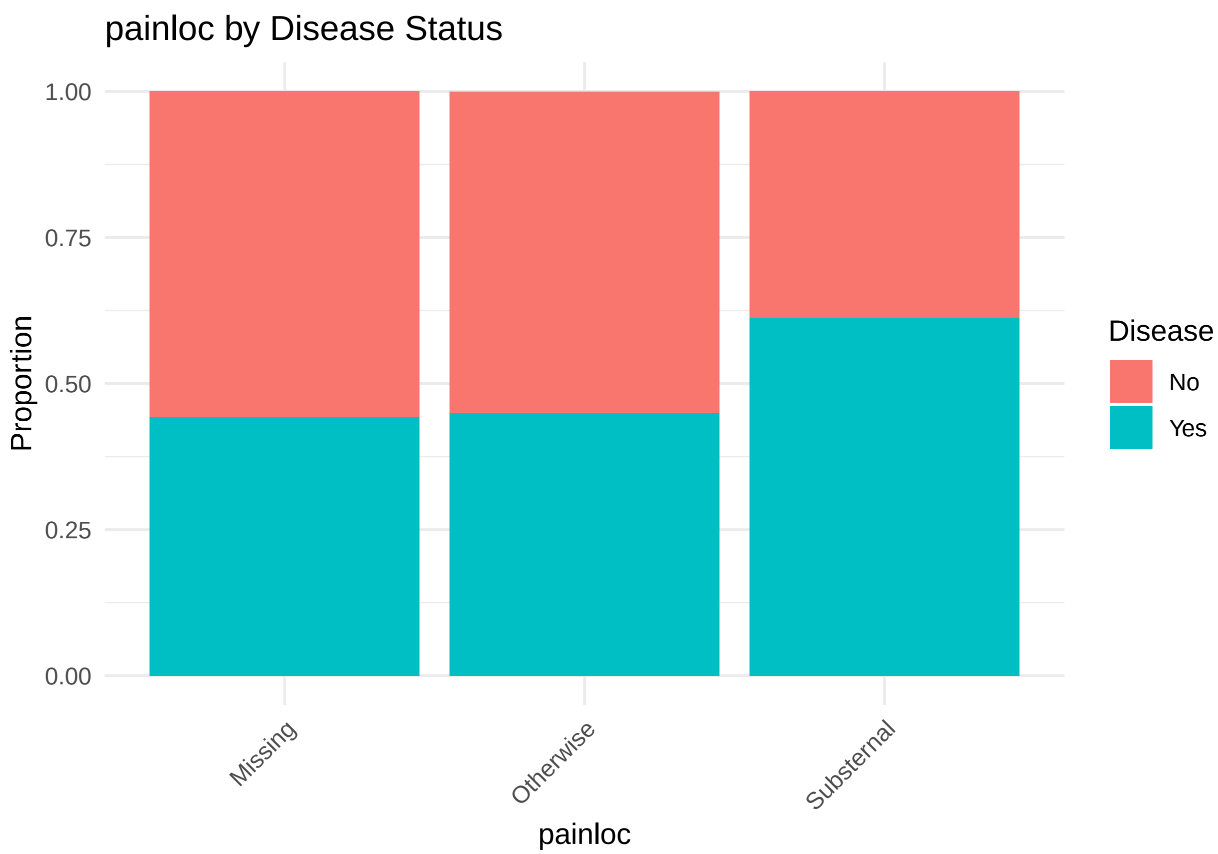
Based on recommendation, all variables recommended as 'Include' will be kept.

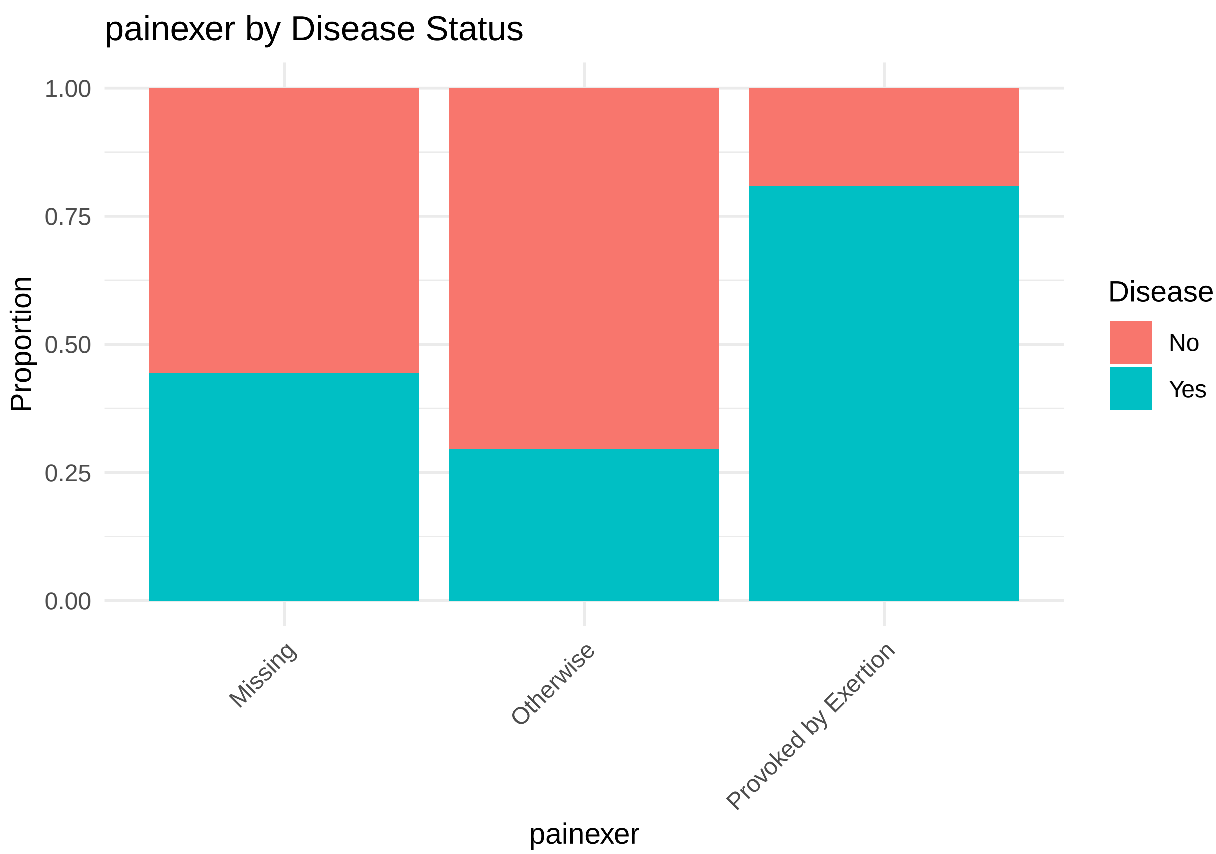
Categorical Vars

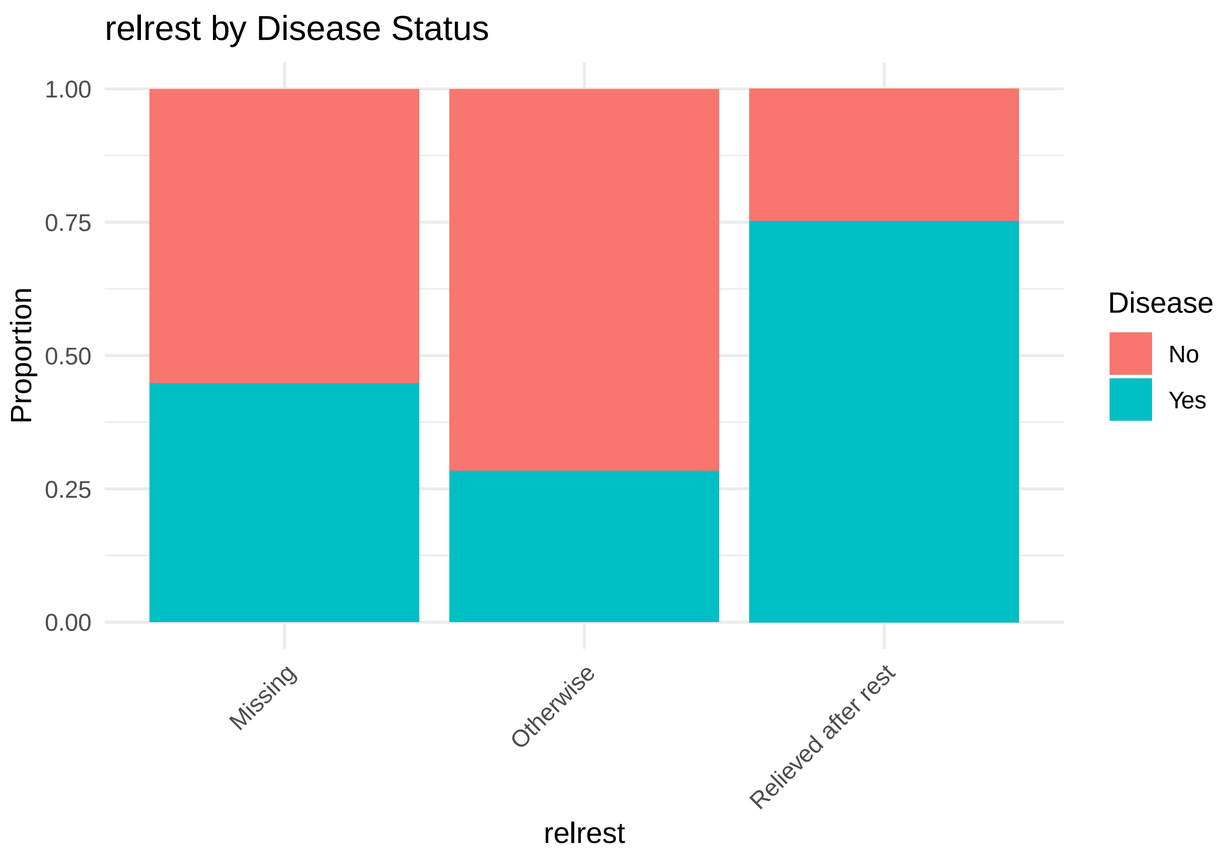
Bar Plots

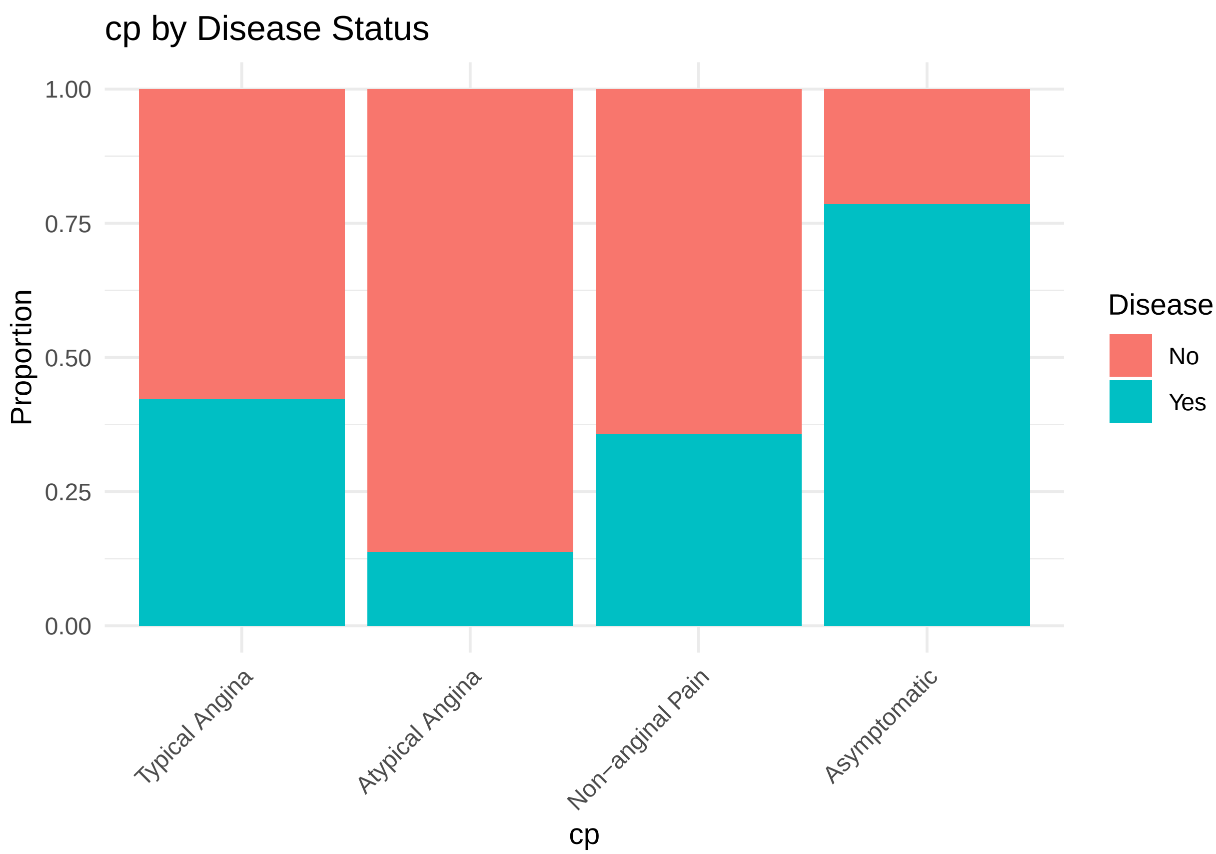
```
plot_bivariate_cat <- function(var) {
  ggplot(df, aes(x = .data[[var]], fill = factor(has_disease))) +
    geom_bar(position = "fill") +  # proportion stacked
    labs(title = paste(var, "by Disease Status"), y = "Proportion", fill = "Disease") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}

for (var in cat_vars) {
  print(plot_bivariate_cat(var))
}
```

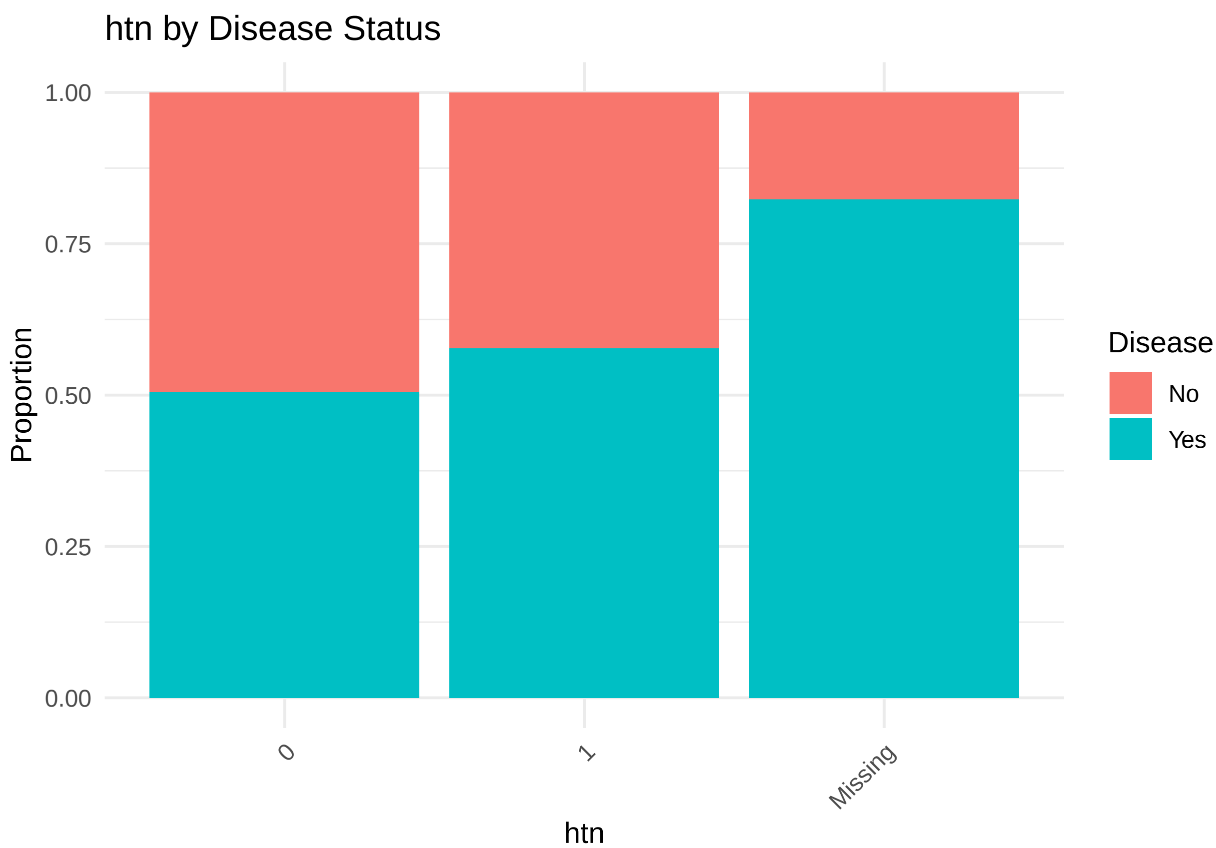

sex by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Female
Male
sex
Disease
No
Yes

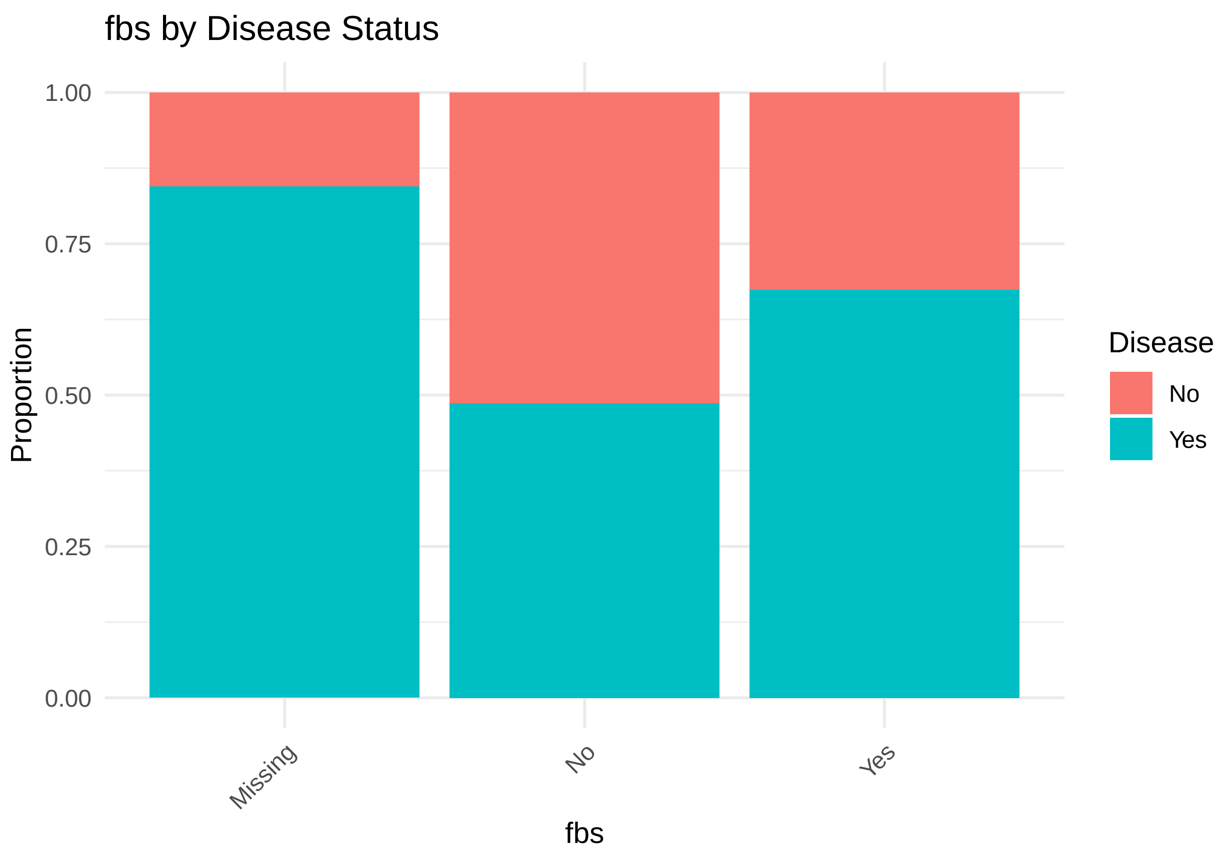

painloc by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
Otherwise
Substernal
painloc
Disease
No
Yes

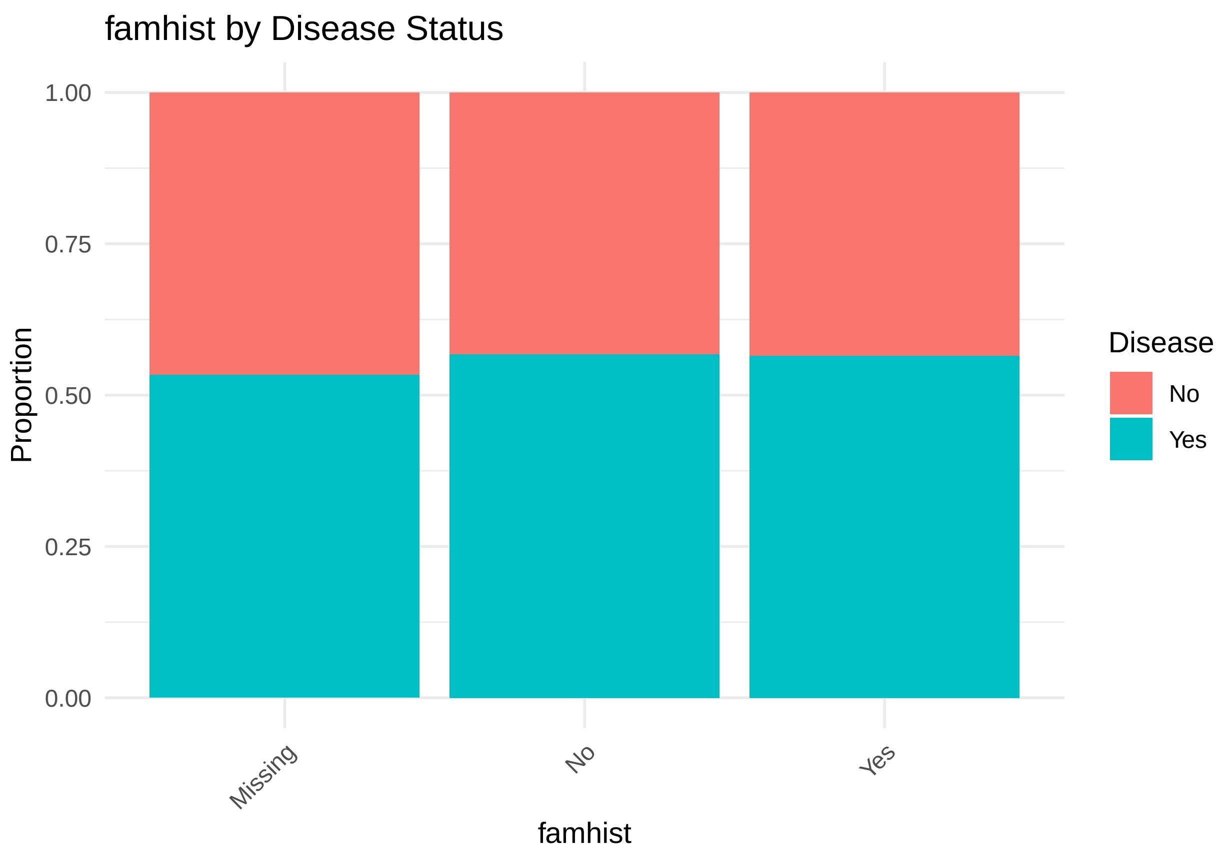

painexer by Disease Status
Proportion
1.00
0.75
0.50
0.25
0.00
Missing
Otherwise
Provoked by Exertion
painexer
Disease
No
Yes

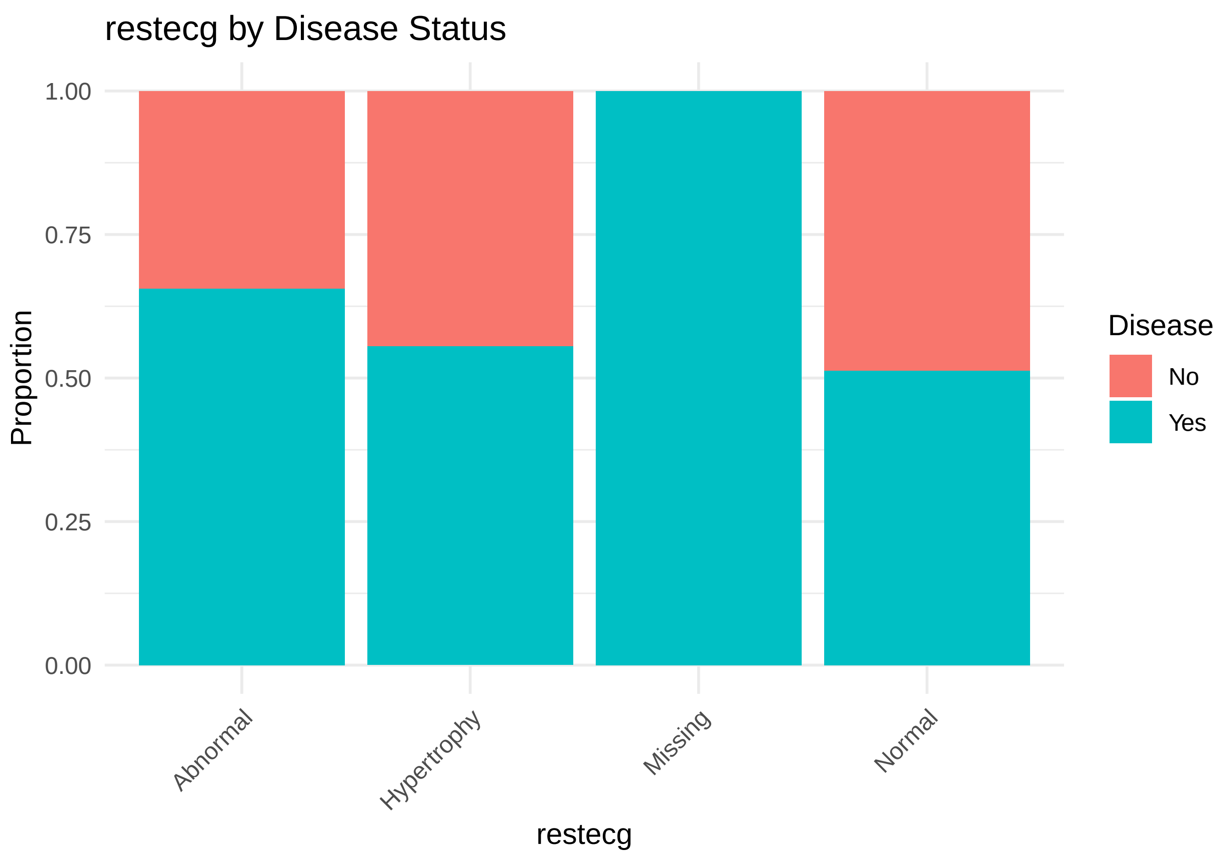

relrest by Disease Status
Proportion
1.00
0.75
0.50
0.25
0.00
Missing
Otherwise
Relieved after rest
relrest
Disease
No
Yes

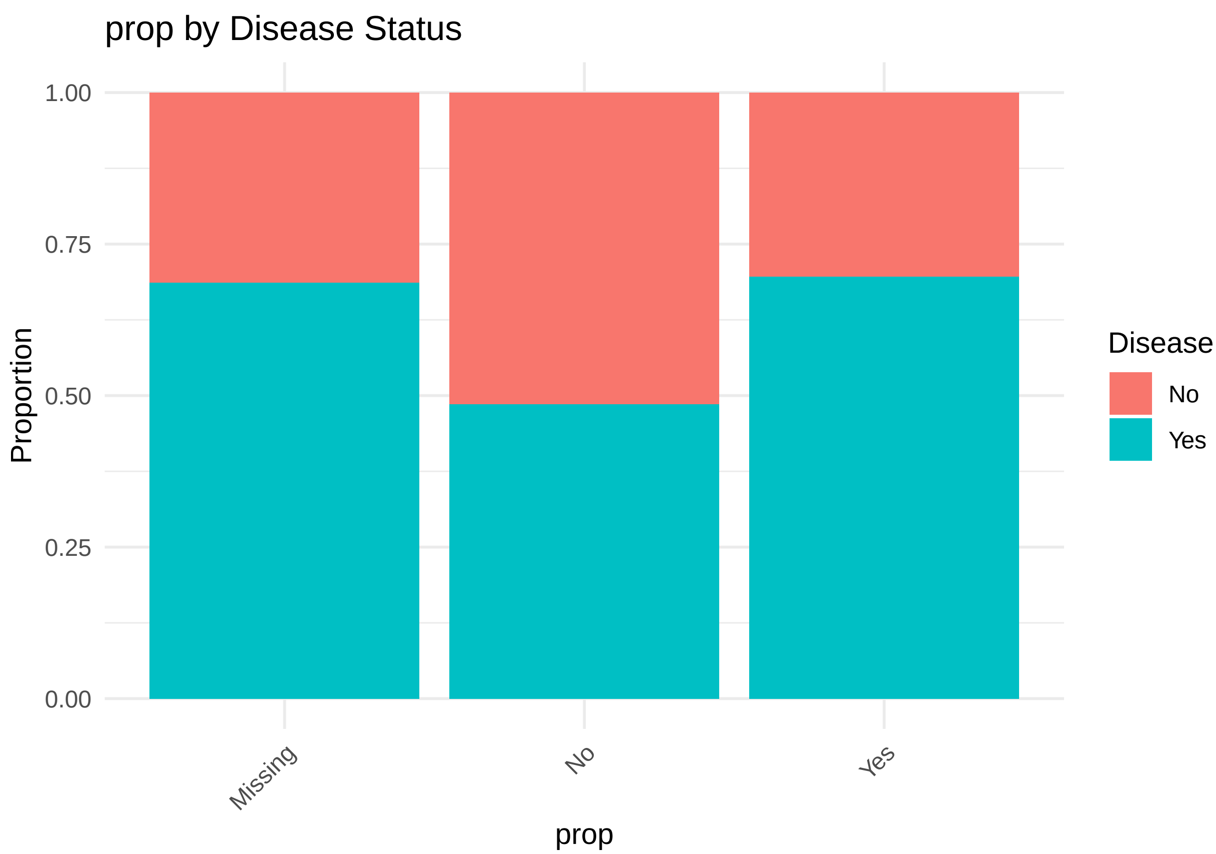

cp by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Typical Angina
Atypical Angina
Non–anginal Pain
Asymptomatic
cp
Disease
No
Yes

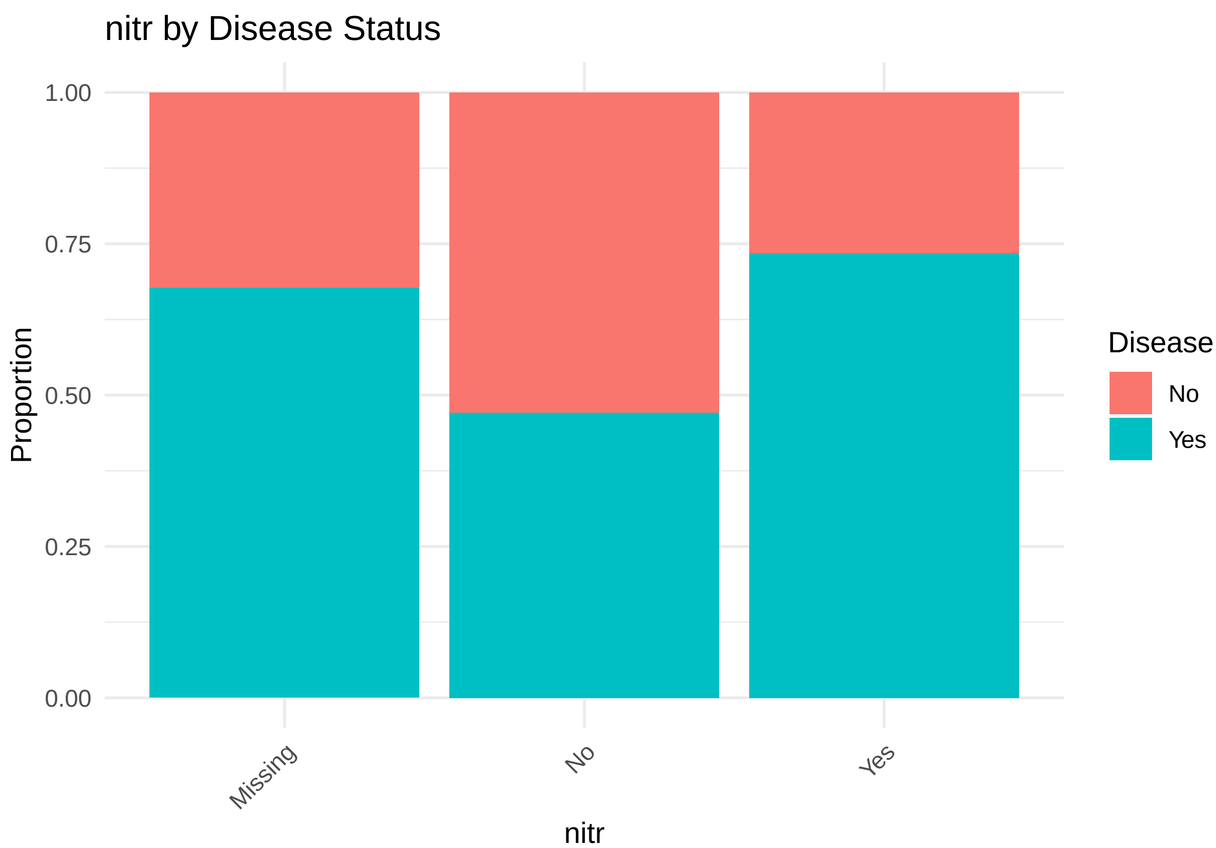

htn by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
0
1
Missing
htn
Disease
No
Yes

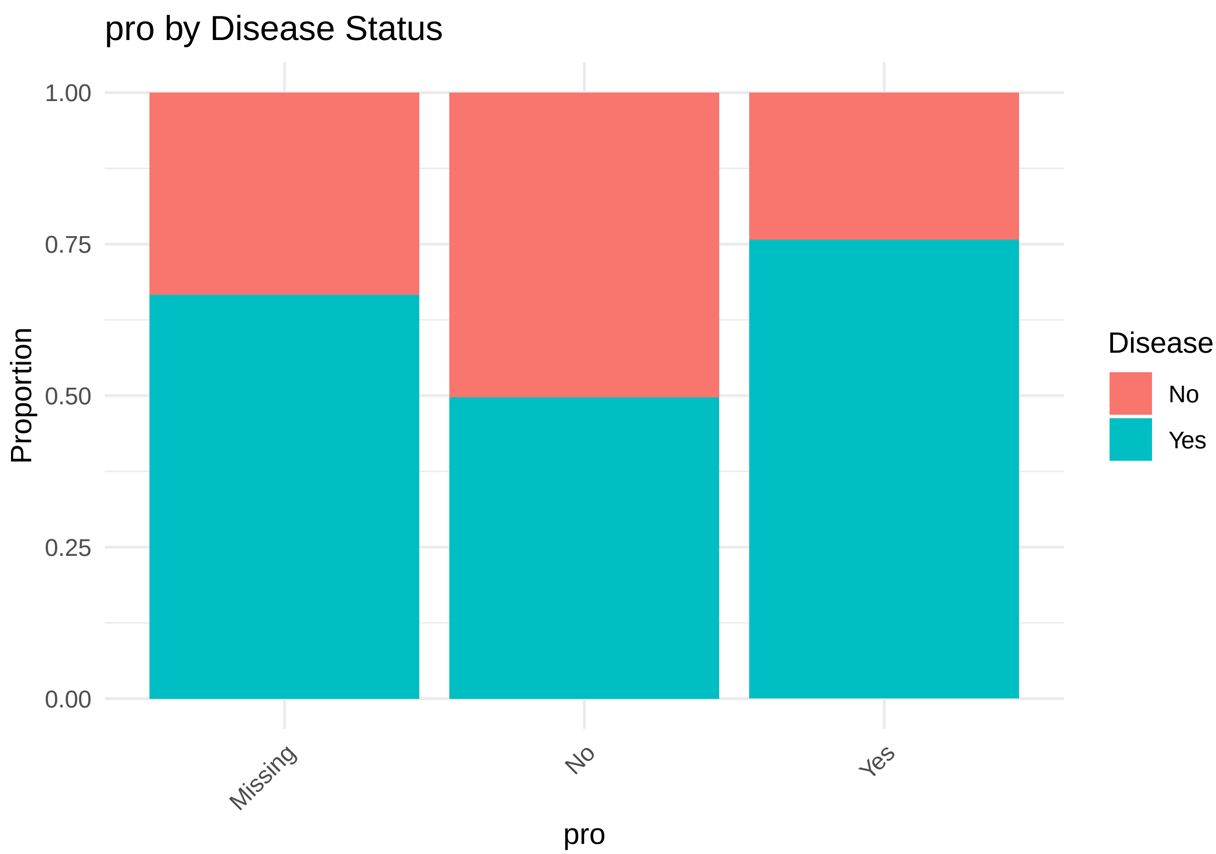

fbs by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
fbs
Disease
No
Yes

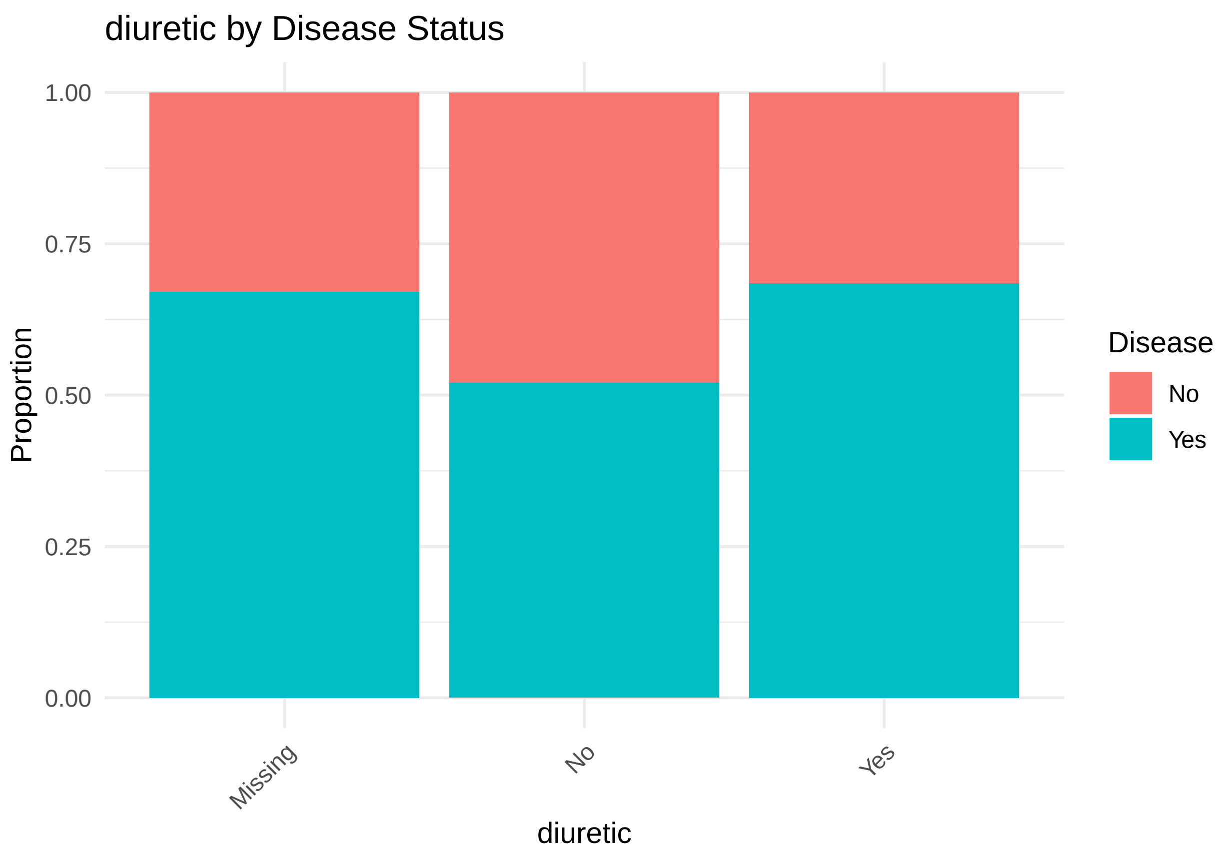

famhist by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
famhist
Disease
No
Yes

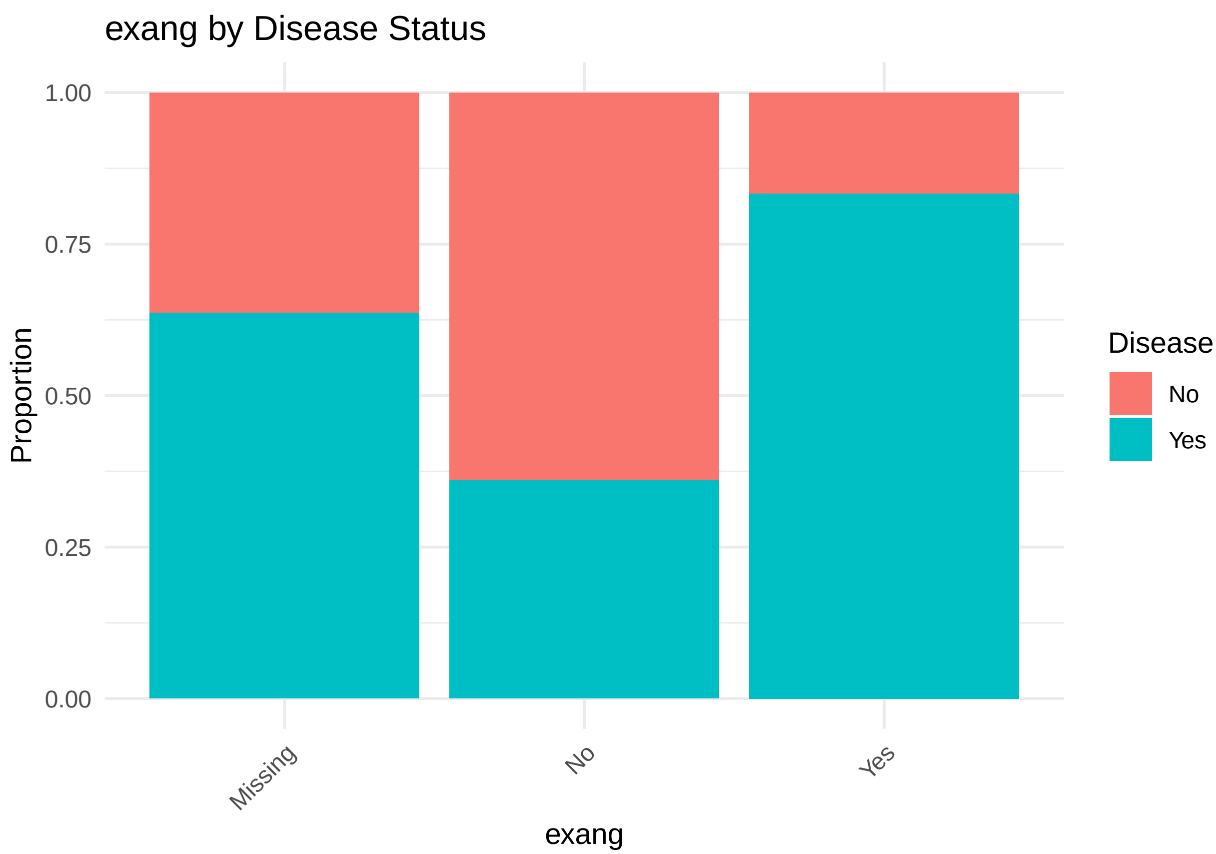

restecg by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Abnormal
Hypertrophy
Missing
Normal
restecg
Disease
No
Yes

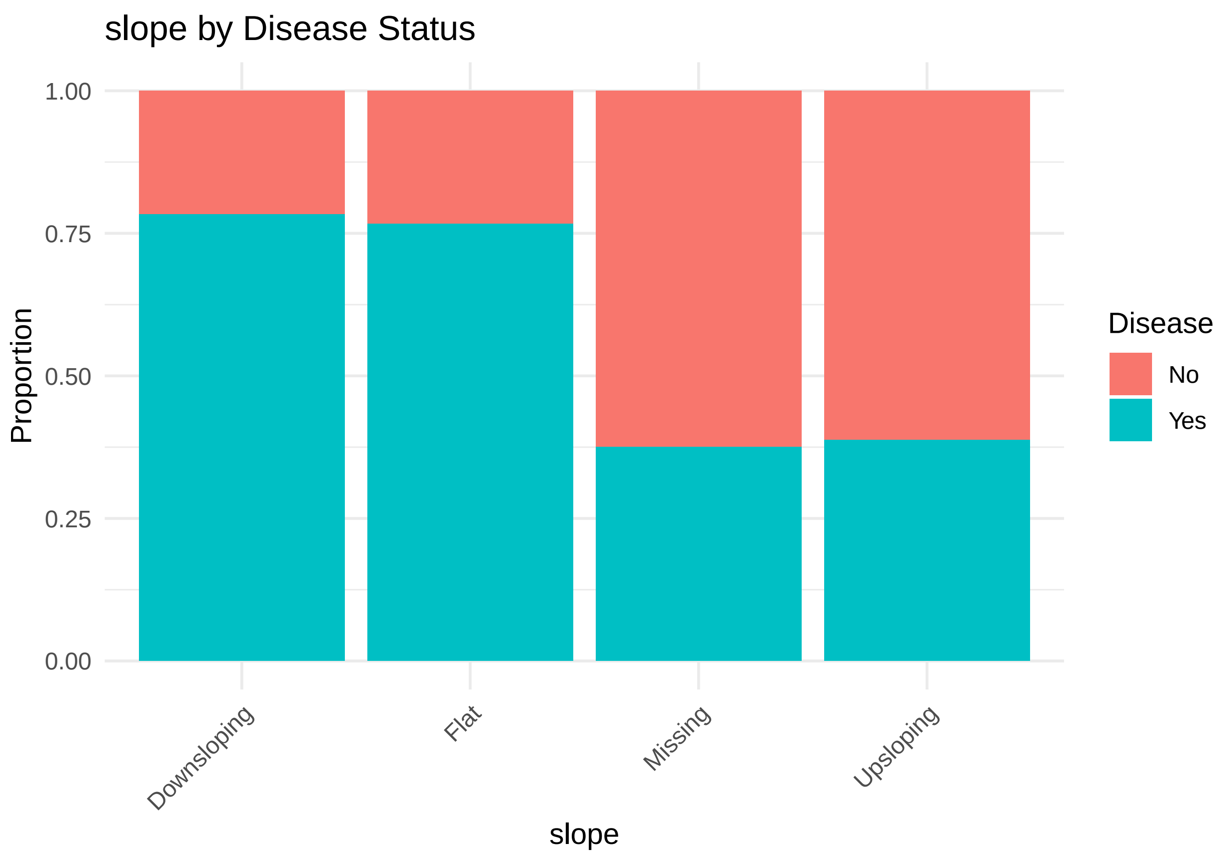

prop by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
prop
Disease
No
Yes

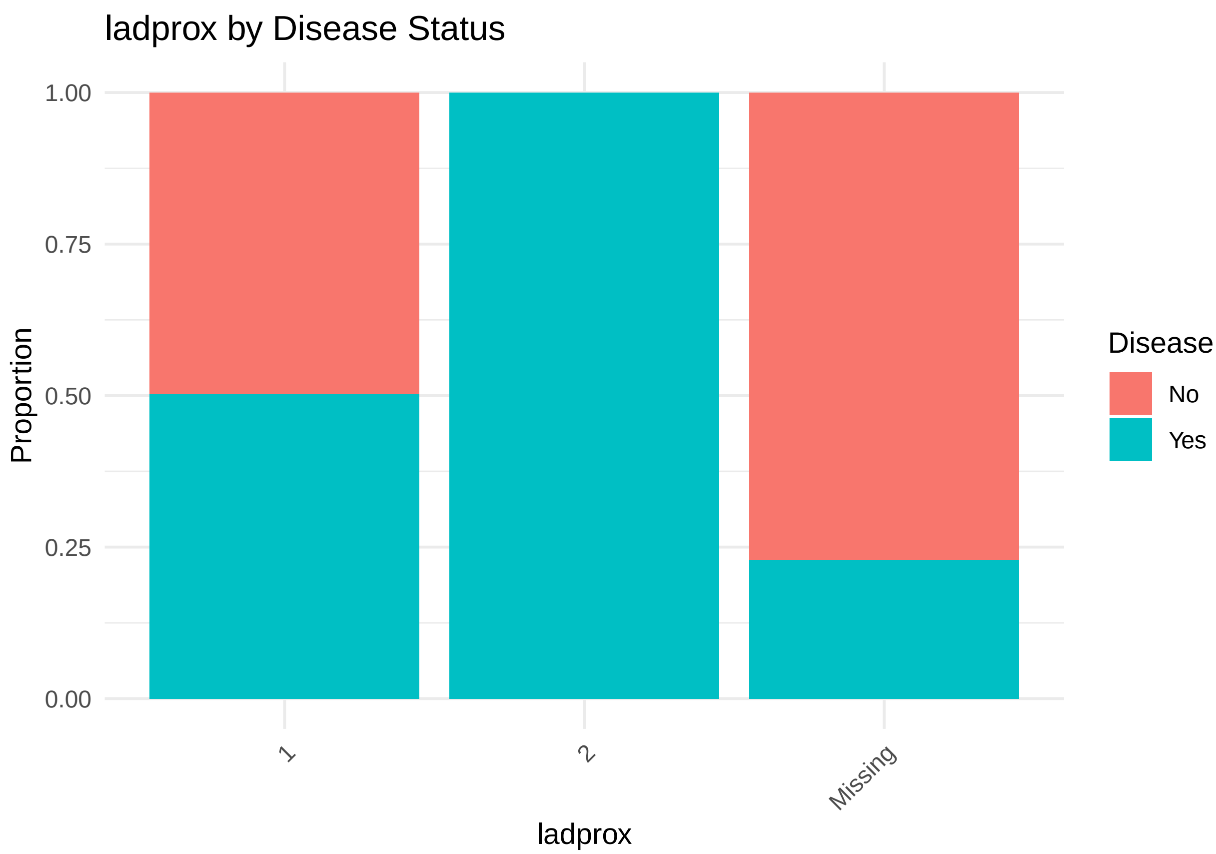

nitr by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
nitr
Disease
No
Yes

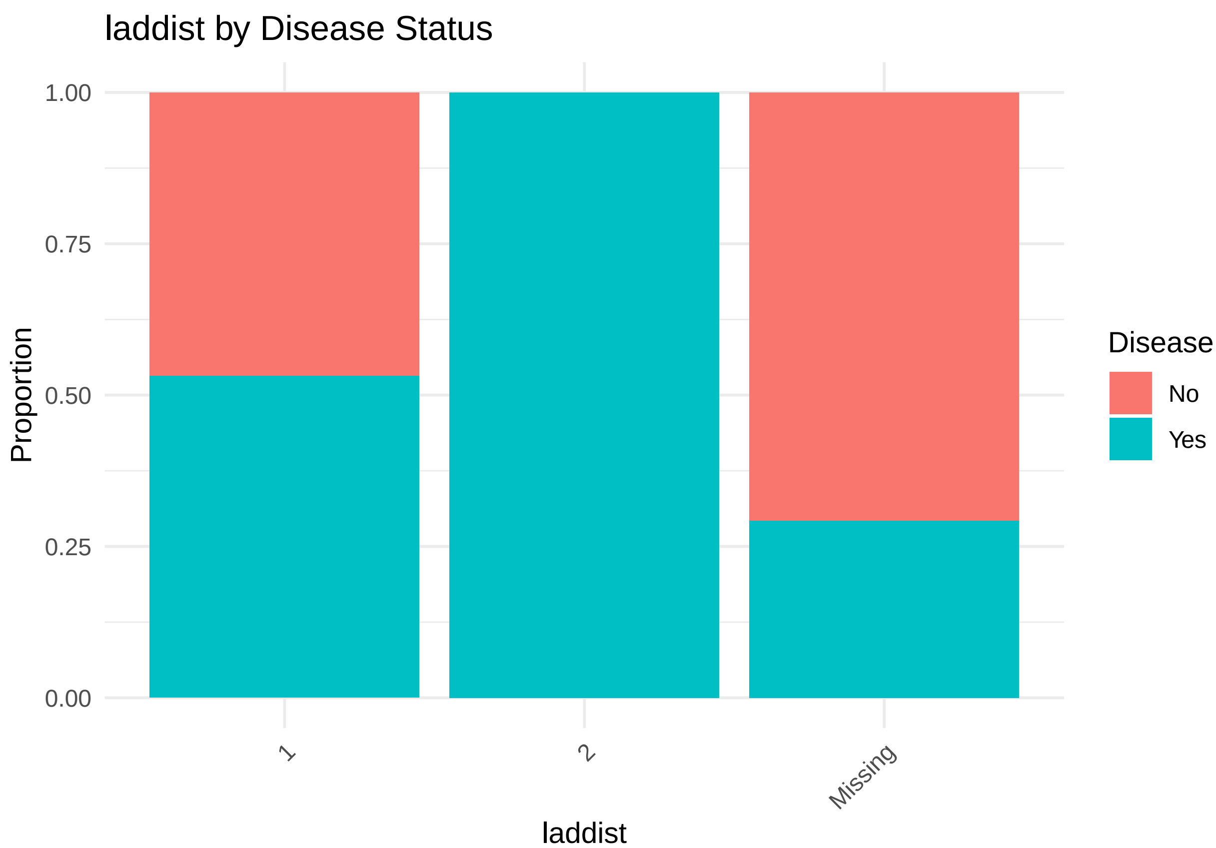

pro by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
pro
Disease
No
Yes

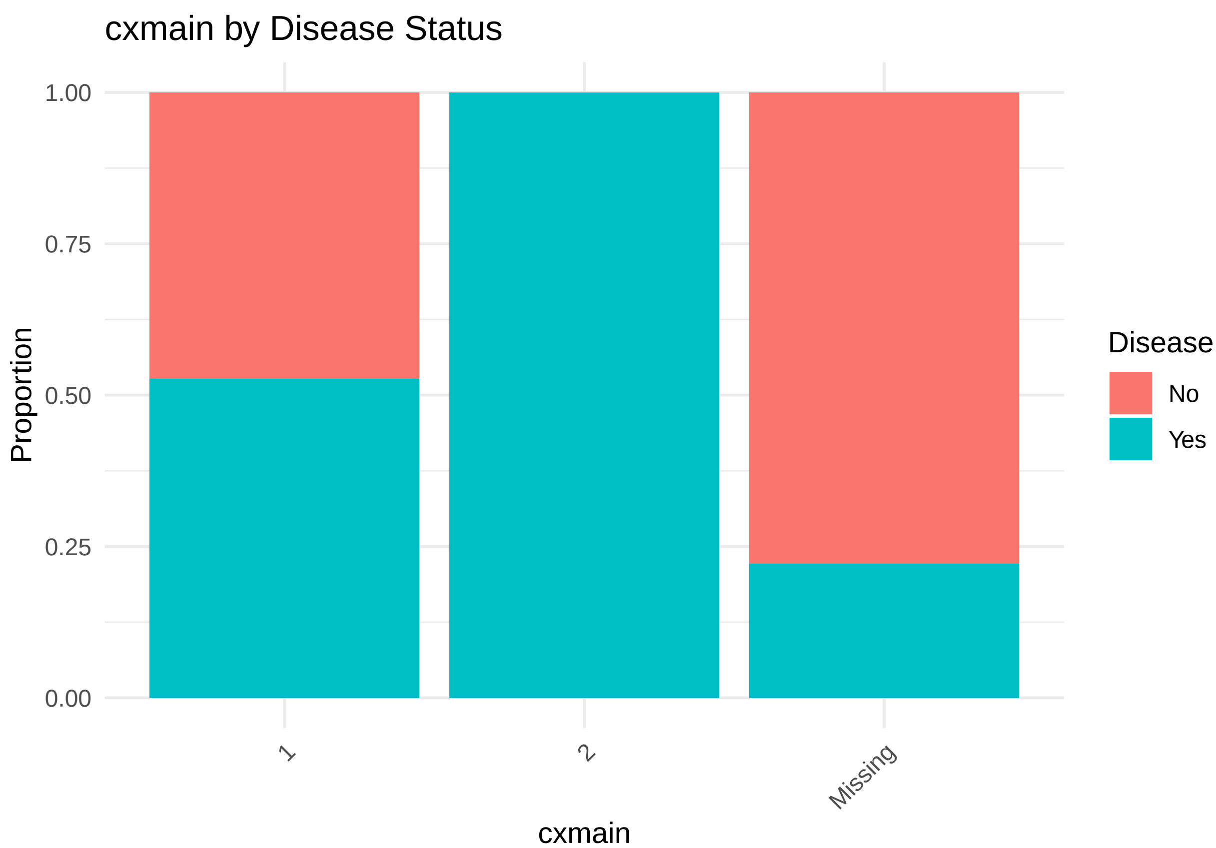

diuretic by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Missing
No
Yes
diuretic
Disease
No
Yes

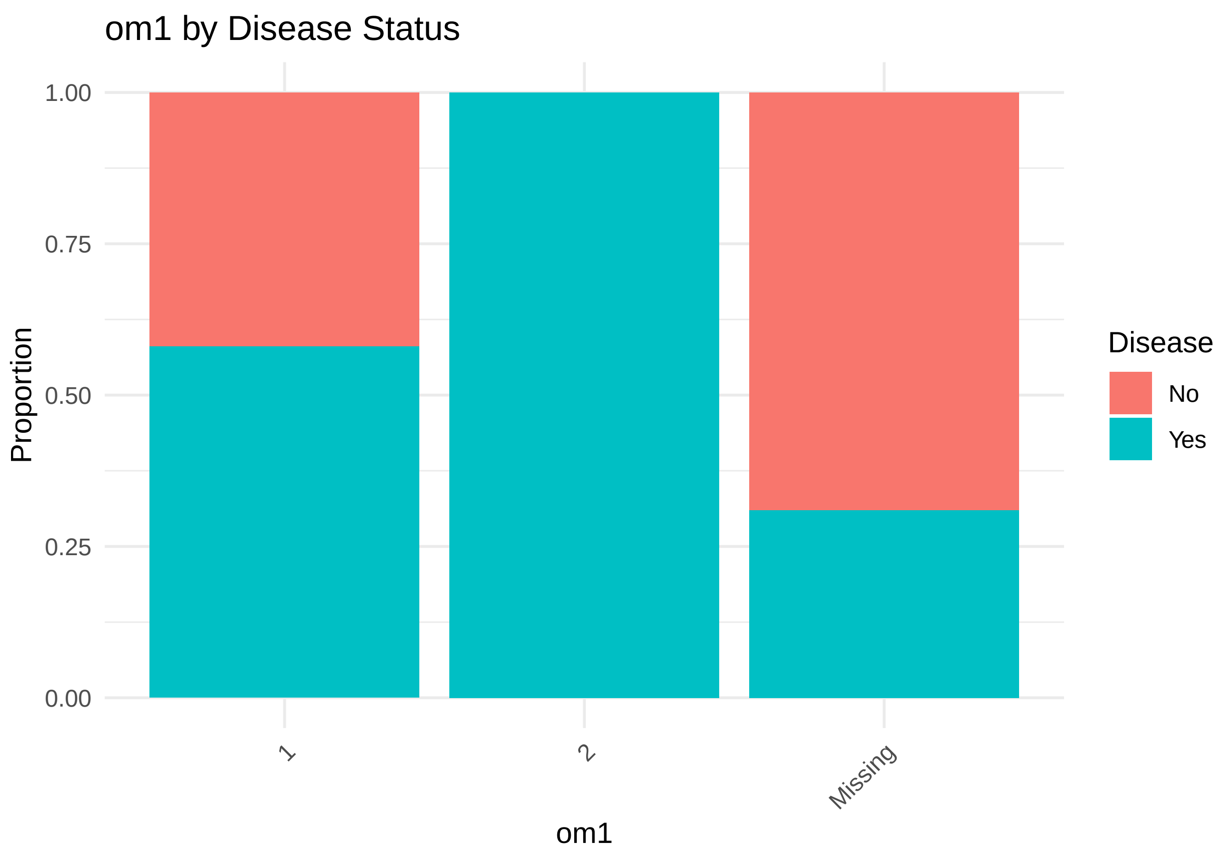

exang by Disease Status
Proportion
1.00
0.75
0.50
0.25
0.00
Missing
No
Yes
exang
Disease
No
Yes

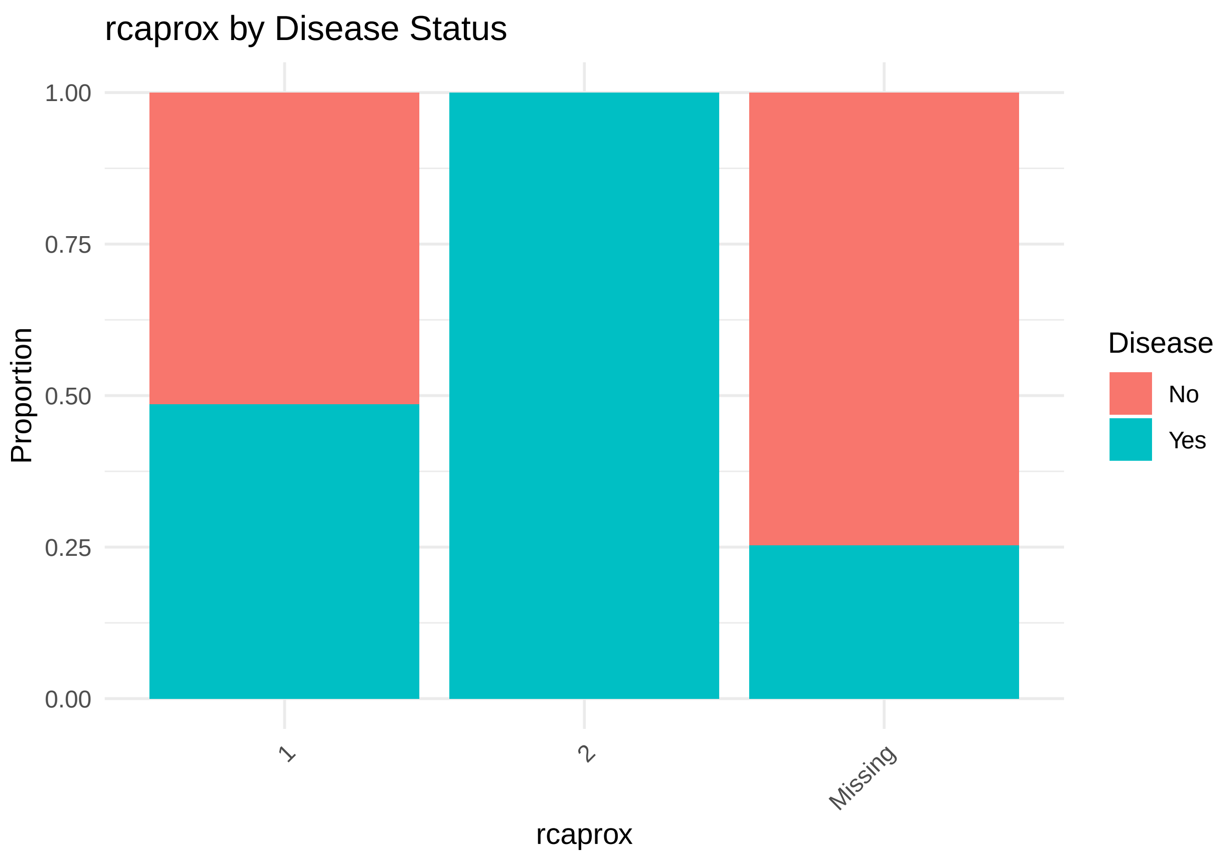

slope by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Downsloping
Flat
Missing
Upsloping
slope
Disease
No
Yes

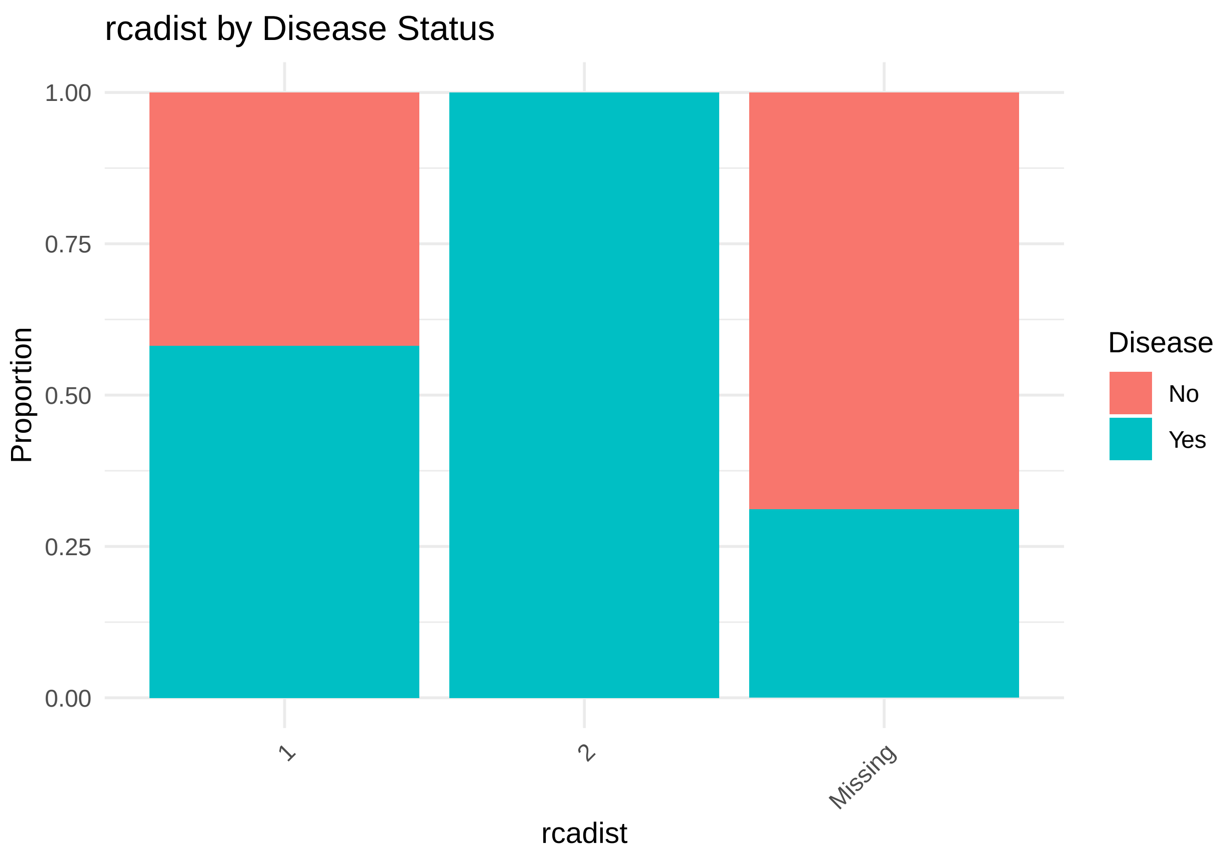

ladprox by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
1
2
Missing
ladprox
Disease
No
Yes


laddist by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
1
2
Missing
laddist
Disease
No
Yes

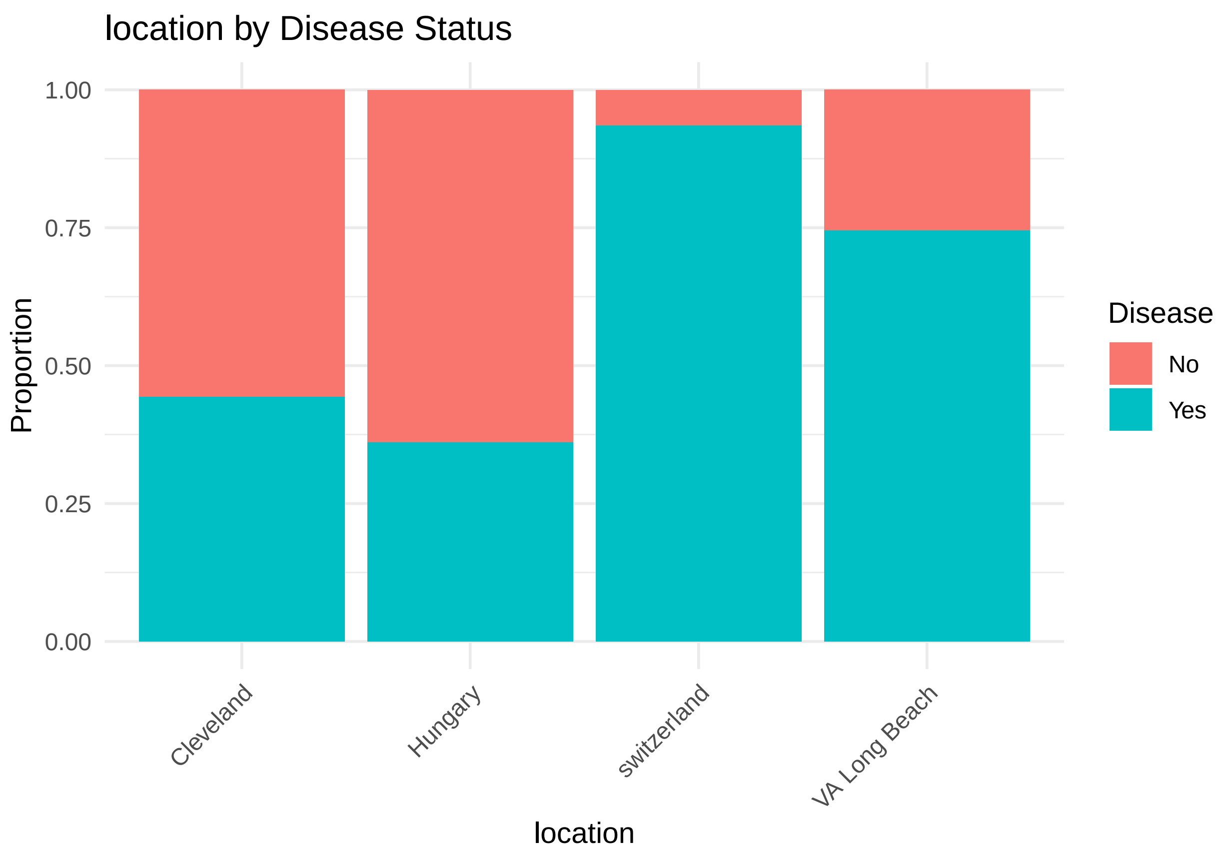

cxmain by Disease Status
Proportion
1.00
0.75
0.50
0.25
0.00
1
2
Missing
cxmain
Disease
No
Yes


om1 by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
1
2
Missing
om1
Disease
No
Yes


rcaprox by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
1
2
Missing
rcaprox
Disease
No
Yes



rcadist by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
1
2
Missing
rcadist
Disease
No
Yes

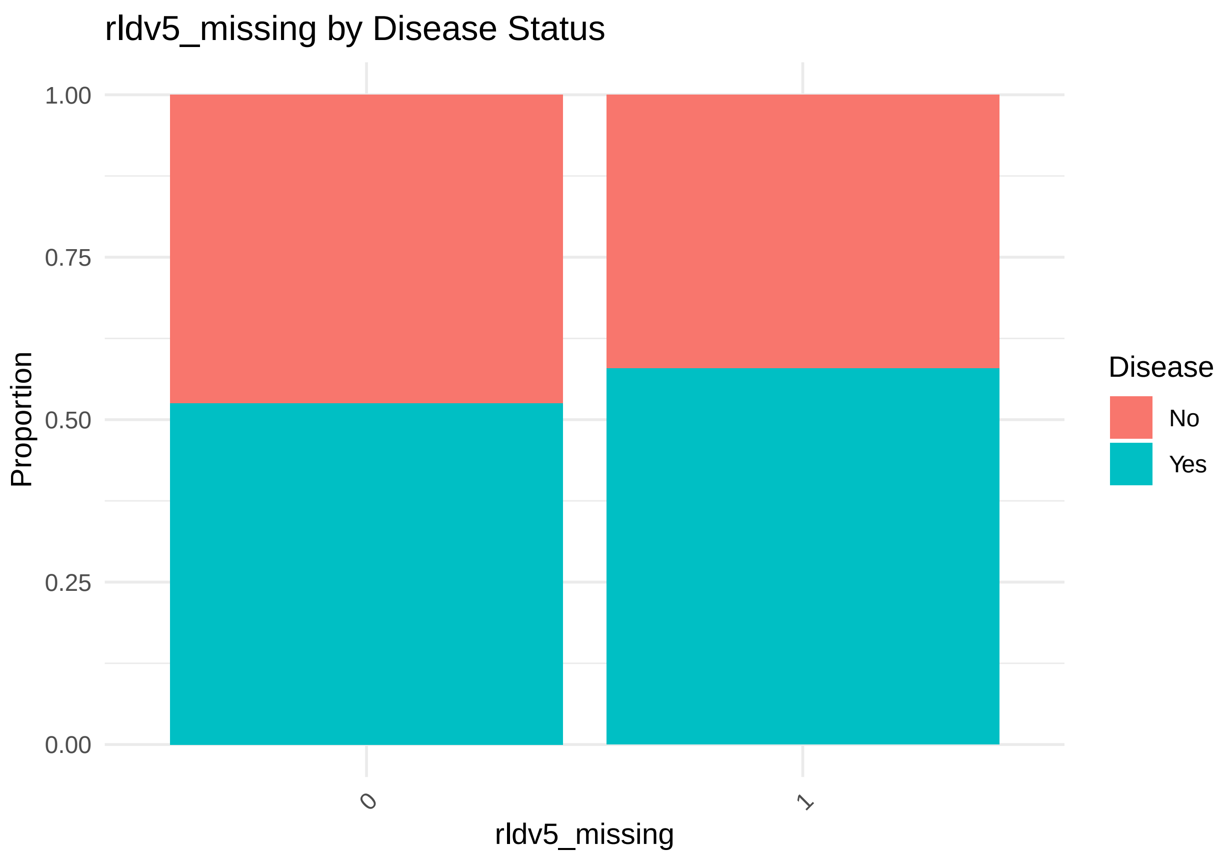

location by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
Cleveland
Hungary
switzerland
VA Long Beach
location
Disease
No
Yes


ca_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
0
1
ca_missing
Disease
No
Yes


thal_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
0
1
thal_missing
Disease
No
Yes


thaltime_missing by Disease Status
1.00
0.75
0.50
0.25
0.00
Proportion
0
1
thaltime_missing
Disease
No
Yes


smoke_profile by Disease Status
Proportion
1.00
0.75
0.50
0.25
0.00
Disease
No
Yes
heavy_long
heavy_medium
heavy_short
heavy_very_long
light_long
light_medium
light_short
light_very_long
missing
moderate_long
moderate_medium
moderate_short
moderate_very_long
non
very_heavy_long
very_heavy_medium
very_heavy_short
very_heavy_very_long
smoke_profile



The following groups of variables have similar distributions which means they are possibly correlated:

Group 1: “prop”, “nitr”, “pro”, “diuretic”, “exang” Group 2: “ladprox”, “laddist”, “cxmain”, “om1”, “rcaprox”, “rcadist”

```
cat_vars_to_keep <- c("sex", "painexer", "relrest", "cp", "htn", "fbs", "restecg", "exang", "slope", "la
```

```
# cramer v
cramerv <- function(dataframe, var_list){
  cramerV_matrix <- matrix(NA, nrow = length(var_list), ncol = length(var_list),
                           dimnames = list(var_list, var_list))

  for (i in seq_along(var_list)) {
    for (j in seq_along(var_list)) {
      tbl <- table(dataframe[[var_list[i]]], dataframe[[var_list[j]]])
      cramerV_matrix[i, j] <- suppressWarnings(cramerV(tbl))
    }
  }
  print(cramerV_matrix)
}

# Check correlation of variables that to have similar distribution
cramerv(df, c("prop", "nitr", "pro", "diuretic", "exang"))
```

Cramer’s V association test

```
##              prop    nitr     pro diuretic   exang
```

```
## prop     1.0000 0.7140 0.7072   0.6344 0.6277
## nitr     0.7140 1.0000 0.7479   0.6319 0.6450
## pro      0.7072 0.7479 1.0000   0.6259 0.6482
## diuretic 0.6344 0.6319 0.6259   1.0000 0.5616
## exang    0.6277 0.6450 0.6482   0.5616 1.0000
```

```
cramerv(df, c("ladprox", "laddist", "cxmain", "om1", "rcaprox", "rcadist"))
```

```
##          ladprox laddist cxmain    om1 rcaprox rcadist
## ladprox  1.0000  0.5490 0.5943 0.6370  0.6245  0.6277
## laddist  0.5490  1.0000 0.5914 0.6084  0.5994  0.6215
## cxmain   0.5943  0.5914 1.0000 0.6150  0.6324  0.6126
## om1      0.6370  0.6084 0.6150 1.0000  0.6385  0.6851
## rcaprox  0.6245  0.5994 0.6324 0.6385  1.0000  0.6128
## rcadist  0.6277  0.6215 0.6126 0.6851  0.6128  1.0000
```

```
drop_colinearity_cols <- c("prop", "nitr", "pro", "diuretic", "laddist",
                           "cxmain", "om1", "rcaprox", "rcadist")
```

Based on the results, each set of variables are moderate to highly correlated. To reduce multicollinearity, the following variables will be kept due to clinical significance: * `exang` - exercise-induced angina * `ladprox` - proximal LAD artery

Summary for categorical variables

```
# recommendation code generated with the help of ChatGPT.
cat_summary <- data.frame(
  Variable = character(),
  p_value = numeric(),
  Recommendation = character(),
  stringsAsFactors = FALSE
)

for (var in cat_vars) {
  tbl <- table(df[[var]], df$has_disease)
  p_val <- tryCatch(chisq.test(tbl)$p.value, error = function(e) NA)

  rec <- if (!is.na(p_val) && p_val < 0.05) {
    "Include (predictive)"
  } else {
    "Optional (evaluate with model)"
  }

  cat_summary <- rbind(
    cat_summary,
    data.frame(Variable = var, p_value = round(p_val, 5), Recommendation = rec)
  )
}
```

```
## Warning in chisq.test(tbl): Chi-squared approximation may be incorrect
## Warning in chisq.test(tbl): Chi-squared approximation may be incorrect
```

```
cat_summary
```

```
##             Variable p_value                 Recommendation
## 1                sex 0.00000           Include (predictive)
```

```
## 2           painloc 0.00001          Include (predictive)
## 3          painexer 0.00000          Include (predictive)
## 4           relrest 0.00000          Include (predictive)
## 5                cp 0.00000          Include (predictive)
## 6               htn 0.00051          Include (predictive)
## 7               fbs 0.00000          Include (predictive)
## 8           famhist 0.61273 Optional (evaluate with model)
## 9           restecg 0.00567          Include (predictive)
## 10             prop 0.00000          Include (predictive)
## 11             nitr 0.00000          Include (predictive)
## 12              pro 0.00000          Include (predictive)
## 13         diuretic 0.00082          Include (predictive)
## 14            exang 0.00000          Include (predictive)
## 15            slope 0.00000          Include (predictive)
## 16          ladprox 0.00000          Include (predictive)
## 17          laddist 0.00000          Include (predictive)
## 18           cxmain 0.00000          Include (predictive)
## 19              om1 0.00000          Include (predictive)
## 20          rcaprox 0.00000          Include (predictive)
## 21          rcadist 0.00000          Include (predictive)
## 22         location 0.00000          Include (predictive)
## 23       ca_missing 0.00002          Include (predictive)
## 24     thal_missing 0.49284 Optional (evaluate with model)
## 25 thaltime_missing 0.00000          Include (predictive)
## 26    smoke_profile 0.22461 Optional (evaluate with model)
## 27    rldv5_missing 0.12282 Optional (evaluate with model)
```

```r
# create list of all variables recommended as Include
selected_cat_vars <- cat_summary %>%
  filter(str_detect(Recommendation, "Include")) %>%
  pull(Variable)

selected_cat_vars
```

```
##  [1] "sex"              "painloc"          "painexer"         "relrest"
##  [5] "cp"               "htn"              "fbs"              "restecg"
##  [9] "prop"             "nitr"             "pro"              "diuretic"
## [13] "exang"            "slope"            "ladprox"          "laddist"
## [17] "cxmain"           "om1"              "rcaprox"          "rcadist"
## [21] "location"         "ca_missing"       "thaltime_missing"
```

Final Variable Sets

```r
# combine selected variables
included_vars <- c(selected_num_vars, selected_cat_vars)

model_df <- df %>%
  select(all_of(included_vars), has_disease) %>%
  select(-all_of(drop_colinearity_cols))

str(model_df)
```

```
## tibble [899 x 39] (S3: tbl_df/tbl/data.frame)
##  $ age              : num [1:899] 63 67 67 37 41 56 62 57 63 53 ...
##  $ ca_imputed       : num [1:899] 0 3 2 0 0 0 2 0 1 0 ...
```

```
##  $ thal_imputed    : num [1:899] 6 3 7 3 3 3 3 3 7 7 ...
##  $ trestbps_missing: num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ trestbps_imputed: num [1:899] 145 160 120 130 130 120 140 120 130 140 ...
##  $ chol_missing    : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ chol_imputed    : num [1:899] 233 286 229 250 204 236 268 354 254 203 ...
##  $ thaldur_missing : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ thaldur_imputed : num [1:899] 10.5 9.5 8.5 13 7 11.3 6 9 8 5.5 ...
##  $ met_missing     : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ met_imputed     : num [1:899] 13 13 10 17 9 16 7 10 9 7 ...
##  $ thalach_missing : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ thalach_imputed : num [1:899] 150 108 129 187 172 178 160 163 147 155 ...
##  $ thalrest_missing: num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ thalrest_imputed: num [1:899] 60 64 78 84 71 73 83 84 75 86 ...
##  $ tpeakbps_missing: num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ tpeakbps_imputed: num [1:899] 190 160 140 195 160 165 180 165 120 185 ...
##  $ tpeakbpd_missing: num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ trestbpd_missing: num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ oldpeak_missing : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ oldpeak_imputed : num [1:899] 2.3 1.5 2.6 3.5 1.4 0.8 3.6 0.6 1.4 3.1 ...
##  $ rldv5e_missing  : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ rldv5e_imputed  : num [1:899] 172 185 150 167 40 127 122 122 75 68 ...
##  $ lmt_missing     : num [1:899] 0 0 0 0 0 0 0 0 0 0 ...
##  $ sex             : Factor w/ 2 levels "Female","Male": 2 2 2 2 1 2 1 1 2 2 ...
##  $ painloc         : Factor w/ 3 levels "Missing","Otherwise",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ painexer        : Factor w/ 3 levels "Missing","Otherwise",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ relrest         : Factor w/ 3 levels "Missing","Otherwise",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ cp              : Ord.factor w/ 4 levels "Typical Angina"<..: 1 4 4 3 2 2 4 4 4 4 ...
##  $ htn             : Factor w/ 3 levels "0","1","Missing": 2 2 2 1 2 2 1 2 2 1 ...
##  $ fbs             : Factor w/ 3 levels "Missing","No",..: 3 2 2 2 2 2 2 2 2 3 ...
##  $ restecg         : Factor w/ 4 levels "Abnormal","Hypertrophy",..: 2 2 2 4 2 4 2 4 2 2 ...
##  $ exang           : Factor w/ 3 levels "Missing","No",..: 2 3 3 2 2 2 2 3 2 3 ...
##  $ slope           : Factor w/ 4 levels "Downsloping",..: 1 2 2 1 4 4 1 4 2 1 ...
##  $ ladprox         : Factor w/ 3 levels "1","2","Missing": 1 2 1 1 1 1 2 1 2 1 ...
##  $ location        : chr [1:899] "Cleveland" "Cleveland" "Cleveland" "Cleveland" ...
##  $ ca_missing      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ thaltime_missing: Factor w/ 2 levels "0","1": 1 1 1 1 2 2 1 1 1 1 ...
##  $ has_disease     : Factor w/ 2 levels "No","Yes": 1 2 2 1 1 1 2 1 2 2 ...
```

Export dataset for modeling

```r
# export modeling dataset to RDS file
saveRDS(model_df, here("Data/Clean", "heart_disease_modeling.rds"))

model_df %>% dim()
```

```
## [1] 899  39
```

The final modeling set has 899 observations and 39 features including the target variable.